B.Com. (Hons.) **III Semester**

PAPER : GENERIC ELECTIVE
Money and Banking
Reading Material



SCHOOL OF OPEN LEARNING
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Money and Banking

Generic Elective Paper for B. Com (Hons) Semester III

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Introduction

Life was very simple in the beginning of human existence. The basic human needs of food, clothing and shelter were fulfilled by man himself or the group in which he lived. Whatever simple production was there, was for self-consumption. There was no division of labour and no scope for exchange. But as time passed, human wants became varied and innumerable. It was simply not possible for any person to satisfy all his/her wants through own-production. Moreover, man realised the value of division of labour in enhancing production and making the process more efficient. This led to a complex division of labour and specialisation in production. Manufacturing of even a single commodity today is divided into many parts and production has become a joint venture in which large number of people participate. Every person gets his/her income through performing a very limited economic activity and spends this income on the commodities of his choice. Therefore, exchange has become a very important part of the economy. In the initial stages, the form of exchange was different and goods were exchanged for other goods. This was called barter. But this kind of exchange was possible in a small society where people had limited wants and knew of each other's wants. But barter was no longer practical in a big economy composed of innumerable people with innumerable wants. The necessary condition for barter to take place is double coincidence of wants, i.e., a person having a surplus of one commodity should be able to find another person who wants that very commodity and has something acceptable to offer in exchange at an agreed rate of exchange. But it is difficult to decide the terms of exchange as there is no common measure of value. Moreover, indivisibility of commodities and difficulty of storage make the barter system extremely difficult. In a complex economy, people invariably produce for others and cannot fulfil their wants except through a practical method of exchange, i.e. sale and purchase. This need resulted in the invention of money, something which is generally accepted in the process of exchange. Money may be any commodity chosen by common consent as a medium of exchange and all other commodities are expressed and valued in terms of this commodity.

It is not easy to go into the hoary origin of money and spell out precisely how and when it emerged in the pre-historic period. One theory is that the origin of money is not the result of man's conscious efforts, but it was discovered accidentally. According to Spalding, due to difficulties of the barter system (mentioned above), exchange must have become very difficult and some widely acceptable medium of exchange might have emerged e.g. articles of necessity or ornaments. The other theory contends that money was the result of man's rational efforts to find a common measure of value. According to G. Crowther, money "undoubtedly was an invention, it needed the conscious reasoning power of man to make the step from simple barter to money accounting." Adam Smith also believed that money resulted from the rational effort of man, but unlike Crowther, he thought that it was discovered as a medium of exchange and not as a unit of account. Whatever the origin of money, whether it emerged accidentally or was invented consciously, it is clear that it existed in societies which had no contact amongst themselves. Therefore, we can safely assume that it originated in different societies separately. Also, historically there is a set pattern of the evolution of money.

Carefully note that whether money was invented through conscious human effort or discovered accidently, what is important to know about money is that it removed the difficulties of barter and by lubricating the wheels of exchange, it facilitated division of labour and consequent increases in productivity.

Kinds of Money

It is agreed that the earliest form of money used in primitive societies was **commodity money**. Things which were commonly demanded (like salt, corn, utensils, furs, skins etc.) were used as money. In some regions, goats, cow or ox were chosen as medium of exchange. While elephant tusks, plumage of birds or tiger teeth served as money in tropical countries, it was shells in countries located on the sea-shore. Money took the form of tea in Tibet, rice in Japan and cattle in Vedic India. Even today, goats serve as money in some areas of Africa. Evidently, such commodities cannot perform efficiently as medium of exchange due to being indivisible, perishable and non-uniform.

With advancement of knowledge and civilisation, people started using **metals** as money. Metals had some merits and did not have the inadequacies of commodity money. To start with, pieces of brass, copper, iron, silver and gold were used as medium of exchange, but later on it was realised that iron, brass and copper were not good money materials as these were not scarce. Therefore, pieces of gold and silver and later on coins of gold and silver issued by rulers of various countries were used as money. Historical evidence shows that metal coins were circulating as money in our country around 400 B.C. Even today metal coins continue to be used as money though the techniques of coinage and quality of coins have improved tremendously.

Initially, the kings used to issue coins in their own image, certifying the weight and quality of the metal. Since intrinsic worth of these coins was no less than their face-value, these were called standard coins or **standard money**. Under gold standard, the banks and the government were supposed on demand to pay out gold in exchange for any form of money. **Token money**, on the other hand, is different from the full-bodied standard money as the face-value of the token coins is much higher than their metallic worth. It is not of any use to melt the token coins into metal. Today one, two and five rupee coins are an example of token money. Another important step in the development of money was the emergence of **paper money** which was slow and gradual. In the earlier times, goldsmiths used to issue receipts to people who deposited cash with them for safety reasons. In course of time, these paper receipts came to be accepted as money due to their credibility. Later on, paper money was issued by the state or by the central bank. It is believed that the Chinese were the first to use paper money which had the sanction of the state. Earlier, the currency notes issued were convertible into precious metals and in fact, represented gold reserves. But when paper money became widely acceptable, it became inconvertible. Today, the currency notes in circulation in different countries are not convertible, but they derive their acceptability from faith in the government.

The latest step in the development of money is **bank money** or **credit money**. This money came into being with the introduction and development of the banking system. Banks all over the world accept deposits which can be withdrawn or transferred through cheques. Banks promise to honour all the cheques issued by the depositors for amounts generally not exceeding the balance in their accounts. The cheque is an instrument through which the bank deposits become payable on

demand and can be transferred from one person or party to another. Though cheque in itself is not legal tender money (i.e. money which has the sanction of law), but it can be used to perform the same functions as money. One can pay for purchases through cheques and also receive the proceeds of sales through cheques. Bank money is generally more convenient than money, easier to carry and can be used to make or receive big payments. It is widely used in developed countries, but even in underdeveloped countries like India, it constitutes about 50 per cent of the money supply in the country. Bank money constitutes cheques, bank drafts and other credit instruments issued by the banks to transfer deposits.

Definition of Money

As far as the definition of money is concerned, some economists like Withers, Hicks and Bain define it with reference to its functions. According to these descriptive definitions, 'money is what money does' and what money does is described as working as a medium of exchange, measure of value, standard of deferred payments and store of value. Other economists like Hawtrey and Knapp adopt a legalistic view and argue that a commodity acquires the characteristic of general acceptability only when it is made legal tender by the state. According to this definition, a commodity cannot function as money unless it has the necessary backing of the state. This definition is allright in normal conditions, but in a situation of hyper-inflation (when the general price-level is rising very fast), even legal tender money loses the characteristic of general acceptability. Moreover, such money does not include demand deposits of banks. Most economists define money in terms of its general acceptability. Marshall, Cole and Keynes emphasize this aspect greatly in their writings. According to this definition, a commodity may or may not be legal tender, but it should possess general acceptability if it is to be used as money. Money may be classified on the basis of (1) legality and (2) liquidity. Legal tender money is backed by law and people are bound to accept it in exchange for goods and services or in discharge of debts. Non-legal tender money is generally accepted by people as a medium of exchange, but it is upto the person to accept it or not. Cheques, bank drafts, hundis, bills of exchange etc. are examples of non-legal tender or optional money. Money in the form of coins and currency notes is highly liquid and can be exchanged for any type of asset immediately. But there are other forms of money which are not as liquid as coins and notes, but can be converted into money involving some inconvenience, time and loss of value. Saving deposits with banks or post offices, time deposits, bills of exchange, debentures, bonds etc. fall in this category. These claims do not circulate as medium of exchange and are not actual money, but these can be called **quasi-money or near money**.

Self-Check Exercise

Note that the most important attribute of money is its general acceptability in exchange transactions and in settlement of debts. Money loses its utility the moment public loses confidence in its acceptability in transactions.

Functions of Money

Since the definition of money is also usually in terms of its functions, let us discuss the functions in some detail. The functions of money have been summed up in a couplet :

Money is a matter of functions four;

A medium, a measure, a standard, a store.

Medium of Exchange : The most important and primary function of money is that of acting

as a medium of exchange. This is, in fact, the most distinguishing characteristic of money which separates money from near-money and non-money assets. It also brings out the importance of the feature of general acceptability since no commodity can act as a medium of exchange unless it is acceptable to everyone. Money as a medium of exchange breaks up the act of exchange into two parts : sale and purchase. This removes the major difficulty of barter, i.e. double coincidence of wants. Hungry weavers who have surplus cloth do not have to search for naked farmers who have surplus food, but no cloth. This saves a lot of time, energy and resources. Thus money brings efficiency into exchange transactions. Moreover, it also promotes efficiency in allocation of resources by making it possible to exploit gains from specialisation in production and trade. People can engage in production, get income (in the form of money) from the proceeds of goods sold and spend that income or money on the goods and services they require. Thus money which is the general purchasing power, acts very efficiently as a medium of exchange.

Measure of Value : Money serves as a common unit of account or measure of value in terms of which the values of all goods and services are expressed. Thus money measures the value of economic goods and this value is expressed in terms of their money prices. In a money economy, it is possible to ascertain the relative exchange values of goods by comparing their market-prices. According to Crowther, "Money acts as a yardstick or standard measure of value to which all other things can be compared." There is a close relationship between the two primary functions of money. The commodity which is used as a medium of exchange in a society, is also used for measuring the values of various goods and services. In fact, the function of money as a unit of account is performed first, i.e. we first measure the value of goods and services to be exchanged and only then exchange various goods and services at pre-determined rates. Moreover, it is possible to measure the total value of different kinds of goods and services (measured in different units e.g. metres, kilograms, litres etc.) in terms of money. There is no other way to measure the national income of a country except in terms of money. Money as a measure of value is used for making all kinds of economic calculations. But sometimes there is a dichotomy and medium of exchange cannot be used as a unit of account. There is the famous case of German mark whose value declined so much due to hyper-inflation after the First World War that it was impossible to use it as a unit of account. Therefore American dollar or Swiss franc were used as a measure of value, but German mark remained in circulation as a medium of exchange. China faced a similar situation during the Second World War.

Standard of Deferred Payments : Credit plays a very important role in a modern capitalist economy. In most of the transactions, instant payments are not made. Debtors promise to pay on some future date and the debt as well as interest, if any, is settled in terms of money. This also applies to payments of rents, salaries, pensions, insurance premia etc. In an underdeveloped economy like India which is based on agriculture, there are instances of rent and wages being fixed and paid in kind. But in a money-using economy, most of the deferred payments and future obligations are stipulated in terms of money. This is for the simple reason that money can be expressed in definite and standardised units and its value (in terms of its purchasing power) remains generally stable over time. But in times of inflation or deflation, the value of money varies over time and money not only becomes a poor measure of value, but also a poor standard of deferred payments. All the same, in the absence of another universally acceptable standard, money continues to perform this function of settlement of debts. Therefore, just as money facilitates current transactions of goods and services through its function as a medium of exchange, it also facilitates credit transactions (i.e. exchange of present goods against future goods) through its function as a standard of deferred payments.

Store of Value : Another important function of money is serving as a store of value. People can hold their wealth in the form of money. This function is also derived from the use of money as a medium of exchange. As stated earlier, money breaks up the exchange transaction into two separate transactions of sale and purchase. Under barter, the two transactions are simultaneous, but the use of money separates them in time also. In a money-using economy, people get their incomes in the form of wages, salaries, rent, interest and profits at certain points of time. They may decide to spend the same immediately or at a later point of time. In the latter case, the full or at least a part of the income received is held in the form of money for varying periods. It is possible to do so because money has the unique feature of being a generalised purchasing power and is also the most liquid asset. This ensures that goods and services can be purchased by money at any time in future without delay or loss of value. Therefore money can be stored without loss in value (unless a situation of hyper-inflation prevails in the economy). Since man always felt the need of holding wealth, under the barter system he did so by storing commodities. But it involved substantial storage costs and loss in value of perishable commodities due to deterioration. Other assets can also serve as store of value, but money being the most liquid, is unique in this respect. According to Keynes, the role played by money as a store of value is no less important than its role as a medium of exchange. Money, in fact, acts as a bridge between present and future. But it is equally true that fluctuations in the value of money (due to inflation and deflation) affect its function as a store of value just as they affect its functions as a measure of value and a standard of deferred payments.

There are some other functions too that money performs. It helps us to transfer value from one person to another, from one place to another and over time due to its general acceptability. Many a time, such functions are performed by near-money such as cheques and bank drafts. Money also helps consumers in maximising their utility by equalising the ratio of marginal utilities to that of respective prices (expressed in terms of money). It also helps producers to maximise production by equalising the marginal productivities of various factors of production. Money also facilitates the distribution of national income among the various factors of production on the basis of their marginal productivities. Banks and firms keep sufficient money reserves for meeting their liabilities lest they become insolvent and lose their goodwill. Money, on account of its perfect liquidity, can be converted into any type of asset according to its profitability at a point of time. Money is also the basis of credit since circulation of credit instruments is not possible in the absence of sufficient case balances.

Self-Check Exercise

Note that people accept money in exchange and hold it for short or long duration (as a store of value) in the belief that other people will similarly accept it whenever the need arises.

Money is also used as a standard measure (a yardstick) of value; current and future (deferred payments).

Role of Money

The importance of money in an economy cannot be over-emphasized. Money may not produce anything, but not much can be produced without the help of money today. Money renders invaluable services in various economic processes going on in a modern capitalist economy. According to Dey, "Money is one of the most fundamental of all Man's inventions..... in the whole commercial side of Man's social existence, money is the essential invention on which all the rest is based." The important position of money is largely due to its two main characteristics. Money being a medium of exchange, is used for making all transactions and settling most of the debts. Purchases of consumer

goods and services, factors of production and claims such as bonds, bills, payment of taxes, are all made through money. As a result of this, money is used for distributing national income to workers, traders, government employees, shareholders and all others. Secondly, since money is general purchasing power and can be conveniently used to lay claims on goods and services, people generally prefer to hold their wealth in the form of money (which includes both currency and bank deposits).

In a capitalist economy, all the three central problems of what, how and for whom to produce, are solved through the price-mechanism which consists of prices of all goods and factors of production expressed in money. This economy is basically unplanned where a large number of consumers and producers take their decisions individually and it is only money prices of various goods and factors determined in more or less free markets, that bring order into the system. Since money splits exchange transaction into purchase and sale, two classes of buyers and sellers come into existence and their desires find their expression in the form of demand and supply respectively and influence various prices. Equilibrium between demand for and supply of a commodity/factor of production determines its price in the market. The set of goods prices helps consumers in choosing the basket of commodities according to their preferences and enables them to maximise their utility within the constraint of their income. The sets of goods and factor prices help the producer in choosing the most profitable line and technique of production. Profit is nothing but the difference between the price and cost of a commodity where the latter depends upon the prices and quantities of various factors of production used in the process of producing this commodity. The share of each person in the national income also depends upon the quantities of various factors of production supplied by him and their prices. To what extent that share in income (expressed in money) enables the person to lay claim on goods and services depends upon the prices of the latter. Thus, all the three problems of allocation of resources, choice of technique of production and distribution of income are solved in a capitalist economy through the price mechanism which is nothing but the value of various commodities and factors of production expressed in terms of money.

Money is essential for the development of an organised credit market. Such a credit market is not possible in a barter economy. Money also removes all trade barriers at national as well as international level by making specialisation and exchange of goods and services possible. Money is also important for the government in the sense that all taxes, fees, fines and other public revenues are realised in money only. The government also organises its public expenditure on activities that ensure maximum social advantage. Moreover, money is not only a technical device serving as a medium of exchange, measure of value, standard of deferred payments and store of value, but also influences the behaviour of such vital economic variables as level of output and employment in the economy. In a situation of unemployment in a modern capitalist economy, monetary adjustments are found useful to solve this problem. By increasing the supply of money (which we shall study shortly), rate of interest is sought to be reduced so that investment is encouraged, increasing output and employment. Also by increasing the monetary expenditure on consumption and investment through reduction in taxes and undertaking government projects, output and employment are sought to be increased. In a situation of inflation, the opposite kind of monetary policy (contractionary in nature) is adopted. Therefore, we can say that money is not wanted merely because it performs some useful functions, but it has become indispensable for the efficient management of a modern economy.

The role of money is not restricted to the capitalist economy. In a socialist economy also, where all the economic activities are coordinated by the central planning authority, money has to be used mainly as a medium of exchange and a unit of account. Resources are allocated on the basis of shadow prices which are also expressed in money. Wages and salaries are paid in terms of money.

People are free to organise their consumption on the basis of prevailing prices which may be partly or wholly controlled by the government. Therefore, even in a socialist society based essentially on economic planning, money and the price mechanism play a major role in the allocation of resources, distribution of income and expenditure on consumption. Lenin himself pointed out that a socialist economy could not be a moneyless economy.

Despite the crucial role that money plays in any modern economy, it cannot be considered as an unmixed blessing. It does promote employment, economic growth and welfare if it is properly managed. But if mismanaged, it can cause economic recession or inflation and result in untold miseries for the common people. As stated before, money acts as a measure of value, but its performance depends upon the extent of stability in its value in terms of goods and services or general purchasing power. We are able to measure wheat in quintals and cloth in metres simply because the weight of a quintal as well as the length of a metre is fixed. But if the yardstick itself is variable, all measurements lose their precision. Because of this, there emerges a difference between real values and monetary values of economic variables like national income, wages and other factor prices. Keynes talked of money illusion that workers suffer from because they cannot distinguish between money wages and real wages. But these days workers and their trade unions are aware of the effect of rising prices on their real wages and try to link their wages to the rate of inflation (through dearness allowance). But the problem of economic fluctuations caused by imperfections in various techniques of monetary control is quite real and causes hardships to the people.

To conclude, we can say that money is a tool and a very useful tool at that, but like all tools, it has to be used properly and efficiently so that it does not cause any harm to the user.

Self-Check Exercise

In a money economy money does many things. However, its most basic function is its use in exchange transactions and as a store of value. Imagine what will happen to productivity if money was not available.

MEASURES OF MONEY SUPPLY

Having defined money, described its functions and importance in a modern economy, we now intend to discuss the supply of money and its various measures. But before we talk about the supply of money, let us say something briefly about the demand for money. It is obvious that unlike other consumer goods, money is not demanded for its own sake, nor does it possess any utility to satisfy human wants. It represents general purchasing power and is demanded because it helps people to gain command over goods and services which possess utility. It is also true that money is a barren and unproductive asset and does not yield anything. Other assets like stocks and shares, houses etc. yield returns in terms of dividends and rent. Money given on loan also yields interest. But even then people (households as well as firms) do hold their wealth in the form of money for three purposes.

According to classical economists, money is held by people for (1) transaction and (2) precautionary motives. Derived from the principal function of money as a medium of exchange is the transaction demand for money. Consumers require money to purchase goods and services while producers need money to obtain factors of production and intermediate goods which are required in the process of production. Therefore, the transaction demand for money depends upon the total volume of transactions in an economy. In addition to the money required for meeting certain and foreseen expenditures, people also keep money to cover unexpected expenditures resulting from uncertain and unforeseen circumstances e.g. accident, sudden illness, loss of job etc. Thus the motive to hold money to guard against future uncertainties is called precautionary motive. The demand for money on this account depends upon the level of income and also the access to credit market and the degree of liquidity of other assets. If there is a developed and organised money market, people can easily borrow or convert their assets into money to meet their unexpected needs. Precautionary motive is also a kind of transaction motive where the transactions are unforeseen and uncertain.

Keynes' innovation is the speculative motive for holding money. Both transaction and precautionary demand for money are derived from its function as a medium of exchange. But speculative demand for money is totally alien to the classical economics. In Keynesian theory, however, this demand for money occupies a strategic position. According to Keynes, speculative demand for money results from people's desire to make capital gains by buying bonds and other financial assets when their prices are low and selling them when their prices rise. In other words, people speculate about the future level of prices of various securities. As rational individuals who try to maximise their gains, they would hold those securities whose prices they expect to rise and try to dispose off those securities whose prices they anticipate to fall. Keynes defined the speculative motive as **"the desire of earning profit by knowing better than the market what the future will bring forth."** Obviously, speculative demand for money makes use of the function of money as a store of value. People tend to hold money (which is the most liquid asset) so that they can convert it into securities the moment it becomes profitable. Keynes related the speculative demand for money to the rate of interest. Since the total monetary return on bonds is fixed, their prices are inversely related to the rate of interest. When the bond prices rise, the rate of interest falls and vice-versa. People buy bonds only when their prices are low and the rate of interest is high. Otherwise, if the bond prices are high and the rate of interest is low, they prefer to hold money so that they can buy bonds as soon as their prices fall. Also, if they expect the interest rate to rise, they will convert their bonds into money and if they expect the rate of interest to fall, they will buy bonds with the stock of money they have. Thus, the speculative demand for money varies inversely with the rate of interest

and expectations about the rate of interest and bond prices also play a role.

Let us now discuss the supply of money and its various measures. Supply of money is the total volume of money held by the public where public includes the private individuals and business firms operating in the economy and **excludes the producers of money which are government, central bank and commercial banks.** Thus the money held by the government and the currency lying with the central bank and commercial banks is not included in the money supply. Therefore, the money supply of a country at any point of time (**this means it is a stock**) is the total amount of money in circulation. This may be held by individuals, households, business firms, institutions, local authorities, non-bank financial institutions and non-departmental public sector undertakings (like Indian Airlines, Hindustan Steel, etc.) and even foreign banks, governments and International Monetary Fund. The reason for measuring the stock of money in this way is to separate the producers or suppliers of money from the holders or demanders of money.

Since money is measurable, we can calculate the total stock of money at a particular point of time. Measuring money at different points of time, we can construct a whole time series of money supply which will show the behaviour of money supply over time. This information can be used to analyse the effects of changes in the money-supply on several important economic variables like level of income, prices, employment, rate of interest, investment, balance of payments etc., and to control the supply of money to attain certain policy goals. We shall discuss this aspect later, but let us now concentrate on various components of the money-supply.

The simplest measure of money-supply (denoted by M or M_1) consists of currency with the public (notes and coins), demand deposits (DD) at commercial banks and other deposits of the Reserve Bank of India (RBI). Currency includes coins and notes out of which coins and one rupee notes are issued by the government of India while notes of denomination of rupees two and above are issued by the RBI. All this is legal tender money. Demand deposits are defined as bank deposits payable on demand through cheques or otherwise. These deposits can serve as a medium of exchange if these are acceptable to the other party. If not acceptable, these can be immediately converted into cash. Thus, demand deposits are as liquid as currency. Other deposits of the RBI are its deposits other than those held by the government and banks. They include demand deposits of quasi-govt. institutions (like IDBI), foreign central banks and governments, IMF and the World Bank, etc. However, these other deposits (OD) of the RBI constitute a very small proportion (say less than one per cent) of the total money supply and hence can be ignored. Till 1967-68 in our country, the RBI used to publish only a single measure of money supply (M) defined as the sum of currency and demand deposits, both held by the public. From 1967-68, the RBI started publishing additionally a broader measure of money supply called 'aggregate monetary resources' (AMR). It was defined as M or M_1 plus the time-deposits of banks held by the public. From April 1977, another change was introduced and since then the RBI has been publishing data on four alternative measurers of money-supply, M_1, M_2, M_3 and M_4.

$M_1 = C + DD + OD$ (where C denotes currency, DD denotes demand deposits and OD denotes Other deposits of the RBI).

$M_2 = M_1$ + Savings deposits with post office savings banks.

$M_3 = M_1$ + net time deposits of banks.

$M_4 = M_3$ + total deposits with the Post Office savings organisations (excluding National Savings Certificates).

We have already explained M_1. Currency consists of paper currency as well as coins. Demand

deposits are the **net** demand deposits of banks and **not** their total deposits. Total deposits include both deposits from the public and inter-bank deposits while the latter are excluded from the definition of money. We have already discussed what other deposits of the RBI include and also pointed out their quantitative insignificance. M_3 is the same as AMR which, apart from M_1, includes time deposits of all banks (net of inter-bank deposits). M_2 and M_4 include Post Office deposits in addition to M_1 and M_3 respectively. It should be remembered that these deposits are not withdrawable by cheque as are demand deposits of banks. While M_2 includes only savings deposits of post offices, M_4 includes all post office deposits, whether savings or time deposits.

In all the different concepts of money supply described above, currency is the most liquid asset, followed by demand deposits of banks which can be easily converted into cash if the need arises. Saving deposits with post offices fall next in terms of liquidity and can be converted into money at a short notice. Time-deposits, whether of banks or post offices, come last in terms of liquidity and cannot be redeemed into money before the stipulated maturity period without loss of time and money. The RBI also views the four measures of money-supply to represent different degrees of liquidity, M_1 being the most liquid and M_4 being the least liquid. When it comes to the choice of the measure of money supply to be used, it all depends upon the context. It is necessary to know the reasons for which money is demanded and the sources from which this demand can be met. The most common measure of money supply is that provided by M_1 which is considered appropriate by most of the economists. M_3 is another important and broader measure of money supply which encompasses time-deposits of banks also.

But whatever the measure of money supply used, one thing that stands out clearly is that the quantity of money in circulation has increased over time and even its rate of growth has accelerated over time. For example, the rate of growth of M_1 increased from 3.6% during the fifties to more than 13% during the eighties. You may ask who controls the supply of money ? Basically it is the central bank, commercial banks and government who can influence or control the supply of money in the economy. The central bank not only issues currency, but also influences deposit money through its monetary policy. It can increase the supply of money by following a cheap money policy and decrease it through a tight money policy. Commercial banks can create credit in terms of demand deposits on the basis of money deposited with them by the people. The government can influence the supply of money through its fiscal policy (relating to revenue and expenditure) and public borrowing. An expansionary fiscal policy increases the supply of money by reducing taxes and increasing government expenditure (which may be partly financed by deficit financing). The opposite happens in case of a contractionary fiscal policy.

Self-Check Exercise

Carefully note that :

1. Supply of Money in an economy means the stock of money in the hands of the users of money (i.e. individuals, households, business firms, institutions, local authorities, non-bank financial institutions etc.) and not the amounts lying in the vaults of producers of money (i.e. The RBI, Commercial Banks and government)
2. Various types of money are distinguished on the basis of their liquidity i.e., the ease with which these can be used in exchange transactions. Currency is the most liquid form of money; it can be used in exchange, as and when needed without loss of time and value. On the other hand, Savings Bank Deposits are less liquid form of money since their use involves a cost (loss of interest) and loss of time.

MONEY AND PRICES

As you know, money does not have any utility of its own and does not satisfy human wants directly. However, people exchange goods and services for money and it functions as a measure of value. As mentioned earlier, this measure of value is not constant itself and its own value keeps on changing. What is the value of money ? Since money helps us to gain command over goods and services which satisfy our wants, its value is determined by what a unit of money will buy in terms of a representative assortment of goods and services. In other words, the value of money is nothing, but its purchasing power which varies inversely with the general price-level. By general price-level, we mean prices of all the goods and services as distinct from prices of individual goods relative to those of other goods (which are called relative prices). Change in relative prices performs the function of allocation of resources in an economy. For example, a rise in the price of X-good relative to the price of Y-good, raises profits in the production of X and induces producers to shift resources from the production of Y to that of X. But when all prices in the economy rise or fall together, relative prices remain unchanged so that no transfer or reallocation of resources takes place. What happens when the general price level rises ? When that happens, the value of money declines, as a unit of money now commands a smaller amount of goods and services. On the contrary, a fall in general price-level raises the value of money since a unit of money can now buy more of goods and services.

The Quantity Theory of Money

We have related the value of money to the general price-level. It will be interesting to go into the reasons for changes in the general price level/value of money. Why do the variations in general price level take place ? The classical theory in this regard is known as the 'Quantity Theory of Money' or 'Fisher's Equation' (since it was formulated by Irving Fisher). According to this theory, changes in the general price-level are direct result of changes in the quantity of money in circulation. The equation formulated by Fisher is written as :

$$MV = PT$$

where M stands for the quantity of money in circulation in the economy and V stands for the velocity of circulation i.e. the number of times a unit of money changes hands on the average during a given period of time. (If a 100 rupees note changes hands 10 times during a month, it performs the function of 1000 rupees, not 100). Thus MV stands for the effective supply of money over a given period of time. On the right hand side of the equation, P stands for the average price-level and T for the volume of real transactions. Thus PT represents the money value of all transactions in the economy. What does the equation $MV \equiv PT$ really signify ? It refers to the simple fact, which has to be true under all circumstances, that the stock of money multiplied by its velocity (i.e. the total supply of money) is always equal to the total value of all the transactions (i.e. the total transaction demand for money). In other words, this equation is an identity and conveys only this that all transactions have to be carried out through money. A change in any of the four variables (M,V, P and T) has to be compensated by equal change in one or more remaining variables. For example, if the amount of real transactions (T) increases, but the money supply (M) is fixed, then either each unit of money will be used a greater number of times to carry out the larger volume of transactions (i.e. V will increase) or the average price-level P must fall. Suppose T increases from 500 to 1000 and M remains constant at 500, then either V must double or P must fall to one-half of its previou value. On the other hand, if the product MV on the left side of the equation remains constant (either

because of constant M and V or because of compensating opposite changes in M and V) and T increases, then P will have to fall proportionately to make the product PT equal to the constant MV. As an identity, the equation MV = PT conveys nothing more than the simple fact that the total value of all transactions in the economy (PT) must be executed through the effective money-supply in the system, i.e., MV.

But according to the classical economists, the equation MV = PT was not merely a definitional identity, it was a theory which relates changes in the general price level (P) directly to the changes in the quantity of money (M). Their main hypothesis was that changes in the general price level are directly proportional to changes in the quantity of money in circulation *because* volume of real transactions (T) and velocity of circulation of money (V) are assumed constant. If V and T are assumed constant, it is quite obvious that changes in P will naturally be proportional to the changes in M. If M doubles, P will also double and if M decreases, P will also be pulled down proportionately. Thus so long as we accept the assumption of constant T and V, the conclusion of the quantity theory of money that changes in the quantity of money cause proportionate changes in the general price-level, holds true. But if a rise in the quantity of money is offset either by a decline in V or an increase in T, it will not lead to a proportional rise in the general price level.

Self-Check Exercises

1. Construct examples to show that if V and T are held constant, P will rise and fall proportionately to rise and fall in the quantity of M.
2. Construct another example to show that P may change in the opposite direction to changes in the quantity of M if V and T are allowed to vary.

Evaluation of the Quantity Theory

This equation was criticised by some economists for ignoring credit money which is a very important component of money supply in a modern economy. Therefore, Fisher extended this original equation by incorporating the volume of bank deposits (M') and its velocity of circulation (V'). The extended version stands as follows :

$MV + M'V' = PT$

If $M = 200$, $V = 6$, $M' = 500$, $V' = 4$ and $T = 1600$

then $P = \frac{MV + M'V'}{T} = \frac{200 \times 6 + 500 \times 4}{1600} = \frac{3200}{1600} = 2$. The value of money can be determined by $\frac{1}{P}$ or $\frac{T}{MV+M'V'}$. In the above case, value of money is $\frac{1}{2}$ or 0.5. Now suppose V, V' and T remain constant, but M rises from 200 to 400 and M' rises from 500 to 1000, then P will rise to $\frac{MV + M'V'}{T} = \frac{400 \times 6 + 1000 \times 4}{1600} = \frac{6400}{1600} = 4$. The value of money will fall to $\frac{1}{4} = 0.25$.

Let us examine this theory critically. The net effect of a change in the quantity of money on the general price level will depend upon the following factors :

(a) how the change in the quantity of money affects aggregate demand ?

(b) how the change in aggregate demand affects the level of output ?

(c) does velocity of circulation of money remain constant or varies ?

Let us examine these questions in some detail. As far as the first question is concerned, the assumption of the classical economists that an increase in the quantity of money will result in a proportional increase in aggregate demand, is based on the notion that rational individuals have no use for idle cash balances or in other words, there is no speculative demand for money. Their argument was that if people have more money than they need for their day-to-day transactions (i.e. if they are able to save), they will lend out the surplus money to the enterpreneurs at some interest rather than keeping it idle with them. The enterpreneurs, in turn, will use the borrowed funds by investing them in capital goods because the funds have a cost in terms of the interest to be paid. Thus the savings of the people will automatically generate an equal amount of investment demand in the economy and aggregate demand will increase exactly in proportion to the increase in the quantity of money. But in our discussion of the Keynesian theory of the rate of interest, we stated that rational individuals do hold some idle cash balances for speculative purpose. Therefore, it is not necessary that an increase in the quantity of money will always be lent out to enterpreneurs for investment, thus resulting in a proportional increase in aggregate demand. It is quite possible that it may wholly or mainly be held as idle cash balances, thus not resulting in any increase in aggregate demand.

The second question whether a change in the level of aggregate demand will or will not affect the level of output in the economy, will depend on whether the economy is working at the full employment level or whether there are unemployed resources in the economy. If there is already full employment of resources in the economy, then an increase in aggregate demand resulting from a change in the quantity of money cannot lead to an increase in output and will, therefore, simply raise the general price-level. If, however, there are unemployed resources available in the economy, the increase in aggregate demand will definitely raise the level of output rather than general price-level. To what extent the increase in aggregate demand will raise output and to what extent it will raise the price-level, will depend upon the extent of unemployment in the economy. Thus when there is considerable unemployment of resources in the economy, there will be the possibility of increase in aggregate demand leading to an increase in output rather than a proportional increase in the general price level. Therefore, we can conclude that the quantity theory of money will not hold if (a) an increase in the quantity of money does not lead to an increase in aggregate demand, but only results in a greater amount of idle cash balances and (b) even if an increase in the quantity of money does raise aggregate demand, but due to unemployed resources in the economy, this increase in aggregate demand results in higher output, rather than higher price-level.

Finally, the question whether velocity of circulation of money is constant or not, is an empirical issue. Some economists consider it a fairly stable and predictable variable, while others think it is quite volatile. Famous economist Kalecki has shown that V is not constant and over short periods of time, it varies with the rate of interest. If the rate of interest is high, holding idle cash becomes costly and people try to manage their transactions with smaller cash balances by using each unit of money more often (i.e. by increasing V). Therefore, the higher the rate of interest, the higher tends to be the velocity of circulation, V. If V is not constant, the conclusion of the quantity theory of money does not hold good.

The classical economists assumed T to be constant because they believed that the economy would always have full employment and aggregate demand would always equal aggregate supply. They assumed that rational individuals would have no use for idle cash balances (i.e. there would be no speculative demand for money) and whatever they save will generate an equal amount of

investment demand in the economy.

Thus we may say that conclusion of the Quantity Theory of Money will hold provided (a) consequent to a change in the quantity of money, aggregate demand changes proportionately i.e. no idle cash is held by people for speculative purpose; (b) when the aggregate demand changes, level of output does not change at all i.e. there is already full employment in the economy and (c) there is no change in V (velocity of circulation of money). On the other hand, it is equally possible that changes in the quantity of money may not affect the price-level at all. This will happen if (i) the whole of the newly pumped money is held as idle cash for speculative purposes i.e. aggregate demand does not increase at all; (ii) in case there is some increase in aggregate demand (proportionate or less than proportionate), this is completely neutralised by increase in output because there are sufficient unemployed resources available in the economy and (iii) the impact of an increase in the quantity of money is neutralised by a compensating change in V. However, these two are extreme possibilities and normally we can expect an increase in the quantity of money leading partly to an increase in the general price-level and partly to an increase in the level of output. Therefore, normally the conclusion of the quantity theory of money will hold, but partly.

There are many weaknesses of the Quantity Theory of Money. It takes M, V and T as independent variables (out of which it assumes V and T to be constant and this may not be correct in reality) and considers only P as the dependent variable. In fact, changes in P may also affect V, T and M. Also, it emphasizes only one function of money i.e. the medium of exchange function and ignores its function as a store of value. Moreover, the rate of interest is completely ignored by this theory. Perhaps the biggest weakness of this theory is that it does not explain how changes in the quantity of money work their way into the economic system and what is the chain of events that follows.

Self-Check Exercises

Note the following points :

1. Because of the speculations motive, aggregate demand may not rise at all consequent to an increase in the quantity of M.
2. Even if aggregate demand rises in proportions to a change in the quantity of M, the price level may remain unchanged if enough unemployed resources are available in the economy.
3. The biggest defect of the Quantity Theory of Money is its failure to explain the channel by which in increase in M bids up the price level.

The Income and Expenditure Approach

It was Keynes who explained the Income and Expenditure Approach according to which changes in the quantity of money work their way into the economic system through a certain process and the net effect of such changes on the price-level depends on a number of 'ifs' and 'buts'. We can describe the sequence of events roughly as follows. An initial change in the quantity of money will affect bond-prices (by increasing the demand for bonds) and the rate of interest since bond-prices and rate of interest are inversely related to each other and are two sides of the same coin. Normally, an increase in the quantity of money may be expected to raise bond-prices and thus lower the rate of interest. A reduction in the quantity of money may be expected to produce the opposite effects viz. a fall in bond-prices and a rise in the rate of interest. A lowering of the rate of interest (caused by an increase in the quantity of money) normally induces producers to make larger investment

because the cost of investment (in terms of the rate of interest) is now lower than before. Here we assume that other factors affecting producers' profit-expectations remain constant. Now, even if consequent upon an increase in the quantity of money, the rate of interest does fall and the investment is stimulated, the extent of rise in aggregate demand will depend upon that in aggregate income which in turn depends on the value of the multiplier. Finally, the increase in aggregate demand will raise the general level of prices provided the economy is working at the full employment level and there are no unemployed resources available in the economy. Thus, we can see that it is not possible to relate changes in the general price-level directly to changes in the quantity of money as the Quantity Theory of Money does. In fact there is many a slip between the cup and the lip.

First of all, an increase in the quantity of money may be neutralised by an increase in idle cash balances and may fail to lower the rate of interest. This will happen if the rate of interest is already so low that people expect it to rise and hence postpone the purchase of bonds. Secondly, even if the rate of interest does fall, it may not stimulate investment if the marginal efficiency of capital also falls due to technological, social, political or economic factors. Thirdly, even if a fall in the rate of interest does stimulate investment, it may not lead to much increase in the level of income if the value of the multiplier is low. Finally, even if an increase in investment does result in a greater increase in income-level via the multiplier and hence an increase in aggregate demand, it may not affect the general level of prices if the level of total output increases in proportion to that of aggregate demand due to availability of unemployed resources in the economy. Therefore, the Keynesian Income and Expenditure Approach describes all the conditions which should be fulfilled if the changes in the quantity of money are to result in proportionate changes in the general price level. Increase in the quantity of money should raise bond-prices and lower the rate of interest. A decline in the rate of interest should induce more investment which should raise the income-level through the multiplier. A rise in the income-level should raise aggregate demand, but not the level of output in the economy so that the general price-level rises proportionately. If any one of these conditions is not fulfilled, the direct relationship between the quantity of money and general price-level breaks. Thus, the Income and expenditure Approach explains how changes in the quantity of money work their way through the economic system step by step and eventually result in changes in the general price-level.

Self-Check Exercises

Carefully note the 'ifs' and 'buts' in the way changes in the quantity of M work their way through the economic system.

1. A change in the quantity of M is expected to change the rate of interest. How ? When it may not ?
2. A change in the rate of interest is expected to affect investment. How ? When it may not ?
3. A change in investment is expected to affect the levels of output and employment through the multiplier process.
4. It is possible that in one situation the change affects only output and in another situation it affects only the price level.

UNIT 2

Financial Markets and Institutions

This Unit covers the following topics:

- Meaning, Types, Role and Functions of Financial Markets
- Meaning and Types of Financial Institutions
- Problem of Adverse Selection in Financial Market
- Role and Functions of Derivatives
- Features, Structure and Constituents of Money Market in India
- Money Market Reforms in India
- Structure of Capital Market in India
- Capital Market Reforms in India

Financial Market

Definition: Financial Market refers to a marketplace, where creation and trading of financial assets, such as shares, debentures, bonds, derivatives, currencies, etc. take place. It plays a crucial role in allocating limited resources, in the country's economy. It acts as an intermediary between the savers and investors by mobilising funds between them.

The financial market provides a platform to the buyers and sellers, to meet, for trading assets at a price determined by the demand and supply forces.

Functions of Financial Market

The functions of the financial market are explained with the help of points below:

- It facilitates mobilisation of savings and puts it to the most productive uses.
- It helps in determining the price of the securities. The frequent interaction between investors helps in fixing the price of securities, on the basis of their demand and supply in the market.
- It provides liquidity to tradable assets, by facilitating the exchange, as the investors can readily sell their securities and convert assets into cash.
- It saves the time, money and efforts of the parties, as they don't have to waste resources to find probable buyers or sellers of securities. Further, it reduces cost by providing valuable information, regarding the securities traded in the financial market.
- The financial market may or may not have a physical location, i.e. the exchange of asset between the parties can also take place over the internet or phone also.

Classification of Financial Market

1. **By Nature of Claim**

Debt Market: The market where fixed claims or debt instruments, such as debentures or bonds are bought and sold between investors.

Equity Market: Equity market is a market wherein the investors deal in equity instruments. It is the market for residual claims.

2. **By Maturity of Claim**

Money Market: The market where monetary assets such as commercial paper, certificate of deposits, treasury bills, etc. which mature within a year, are traded is called money market. It is the market for short-term funds. No such market exist physically; the transactions are performed over a virtual network, i.e. fax, internet or phone.

Capital Market: The market where medium and long term financial assets are traded in the capital market. It is divided into two types:

Primary Market: A financial market, wherein the company listed on an exchange, for the first time, issues new security or already listed company brings the fresh issue.

Secondary Market: Alternately known as the Stock market, a secondary market is an organised marketplace, wherein already issued securities are traded between investors, such as individuals, merchant bankers, stockbrokers and mutual funds.

3. **By Timing of Delivery**

Cash Market: The market where the transaction between buyers and sellers are settled in real-time.

Futures Market: Futures market is one where the delivery or settlement of commodities takes place at a future specified date.

4. **By Organizational Structure**

Exchange-Traded Market: A financial market, which has a centralised organisation with the standardised procedure.

Over-the-Counter Market: An OTC is characterised by a decentralised organisation, having customised procedures.

Since last few years, the role of the financial market has taken a drastic change, due to a number of factors such as low cost of transactions, high liquidity, investor protection, transparency in pricing information, adequate legal procedures for settling disputes, etc.

The Role of Financial Markets

Financial markets play many important economic roles.

They enable individuals to achieve a better balance between current and future consumption. For example, entrepreneurs with good investment projects may be in need of financing while individuals wanting to provide for their retirement may be looking for avenues in which to invest their savings.

Financial markets bring the borrowers in contact with the lenders and in the process make both better off.

Financial markets also allow efficient risk sharing among investors. As we will see later, risks are of two types: diversifiable and non-diversifiable. Diversifiable risk can be eliminated by holding assets the returns of which are not perfectly correlated. Financial markets not only help

investors in diversifying some of the risk, but also offer a wide array of financial instruments with very different risk-return relationships. This enables individuals to choose the risk profile of their investments according to their risk-tolerance levels. Investors who are extremely risk averse, for example, may choose to invest a large fraction of their wealth in risk-free securities (such as gilts), whereas more risk-tolerant investors may elect to invest in speculative stocks.

Financial markets also make possible the separation of ownership and management that is a practical necessity for running large organisations. Many corporations have hundreds of thousands of shareholders with very different tastes, wealth, risk tolerances and personal investment opportunities. Despite this, they can all agree on one thing they should continue to invest in real assets as long as the marginal return on the investment equals the rate of return on similar investments in capital markets. Since shareholders are unanimous about the investment criterion, they can delegate the operations of an enterprise to a professional management team. Managers do not need to know anything about the preferences of their shareholders; neither do they need to consult their own tastes. Managers need to follow only one objective: to invest in projects that yield a higher return compared with that offered by equivalent investments in capital markets (the opportunity cost of capital). Put another way, the managers objective becomes that of investing in projects that in present value terms cost less than the benefits they bring in, i.e. investing in positive net present value projects. This objective maximises the market value of each stockholders stake in the concern and therefore turns out to be in the best interest of all the shareholders.

The separation of ownership and management is a fundamental condition for the successful operation of a capitalist economy. It means that individuals can decide how much to consume now and how much to invest for the future. Once they have decided how much to invest, by using the wide array of financial instruments available in capital markets, they can choose the time pattern of consumption plan and the risk characteristic of the consumption plans that suit them.

The managers of large private corporations, on the other hand, can borrow money from capital markets to buy real assets. The real assets may be tangible, such as machinery, factories and offices, or they may be intangible, such as technical expertise, trademarks and patents. When the managers maximise net present value they make everyone better off. Thus, well functioning financial markets ensure that individual maximisation leads to a socially optimal outcome.

Financial markets lead to the efficient allocation of resources via information conveyed through market prices. Consider the case of a farmer who has land that can be used to grow wheat, corn or oatmeal. He is reasonably certain about how much it will cost him to grow any of these crops and how much output his land will yield. There is, though, considerable uncertainty about the price his crop will fetch after harvesting. This price uncertainty depends not only on factors such as weather conditions but also on the levels of demand and supply that may prevail in future

However, the farmer can look at the futures prices of wheat, corn and oatmeal and, knowing his cost structure, decide which is the most profitable crop for him. He can also use the futures markets to assure a guaranteed price for the crop and then go ahead with planting the crop. In this way financial markets ensure that the land is put to its most efficient use.

The stock market aggregates diverse opinions of market participants and conveys how much the equity of a company is worth under current management. Suppose the shares of Company A are trading at a given price and suppose that another organisation, Company B, can use the assets of Company A more efficiently. Then Company B may decide to acquire Company A. If it does so, the assets of Company A will be put to more efficient and productive use under the management of Company B. If there were no stock market, then it might be difficult for Company B to notice that the assets of Company A were not being put to best use. Even if Company B noticed this, it might not be possible to acquire Company A and thereby transfer the assets. Thus, the presence of a well- functioning stock market leads to the more efficient utilisation of assets and enables poor management to be disciplined through a market for corporate control.

When a company announces a plan of future actions, such as starting a new project or the takeover of another company, the stock price may respond in a positive or a negative way. The management of the company can observe the stock price reaction and learn what the market participants collectively think of its proposed plan. If the stock price reaction is negative, the management may wish to re-examine its own calculations and reconsider its decision. Thus, the stock market helps management to get a second opinion about its investment decisions. Moreover, since stock prices reflect the value of the assets under current management, the market provides a measure of how well management is doing its job and therefore helps the process of evaluating managerial performance.

Banks play important economic roles as well. In addition to bringing borrowers and lenders together, banks also act as monitors of companies. If finance is provided entirely through the diverse ownership of shareholders, no one alone has an incentive to spend resources to monitor the management and ensure that it is acting in the best interest of those shareholders. Monitoring is best carried out by only one party since duplication may not result in improved monitoring and would waste resources. But shareholders cannot combine to hire somebody to monitor because of a free rider problem; each would want others to bear the costs of monitoring the monitor. When a bank lends to a corporation, it has an incentive to be the single monitor.

To summarise, well- functioning financial markets bring borrowers and lenders together, improve risk sharing, lead to the efficient allocation of resources, provide information to market participants, allow separation of ownership and management and help the monitoring of management. Together they improve the quality of investment decisions and the welfare of all market participants.

Finance, it can be argued, is the very centre of business and therefore of management. It is certainly an area of which any general manager must have a very secure grasp. For that reason, it is given considerable coverage in Mastering Management. The Finance Module begins with four sections looking at the finance function within a company, financial markets and investment decisions, risk and return, and the efficiency of markets. The module will be completed over eight subsequent parts: planned to be Parts 2, 4, 5, 6, 9, 13, 15 and 17. The topics scheduled to be covered include: discounted cash flow and other criteria for assessing the viability of projects; cost of capital; choice of capital structure; the changing nature of finance; dividend policy; options and derivatives; warrants and convertibles; interest rates and currency swaps; financial regulations and bank capital requirements; managerial remuneration and company performance.

Summary

Financial markets serve a number of useful purposes. These include bringing borrowers and lenders together; sharing risk between investors; separating ownership and management; achieving a better balance for individuals between current and future consumption; efficiently allocating resources, effectively utilising assets, and helping the process of evaluating managerial performance through the signals contained in market prices.

Six key roles of financial markets

To facilitate saving by businesses and households: Offering a secure place to store money and earn interest

To lend to businesses and individuals: Financial markets provide an intermediary between savers and borrowers

To allocate funds to productive uses: Financial markets allocate capital to where the risk-adjusted rate of return is highest

To facilitate the final exchange of goods and services such as contactless payments systems, foreign exchange etc.

To provide forward markets in currencies and commodities: Forward markets allow agents to insure against price volatility

To provide a market for equities: Allowing businesses to raise fresh equity to fund their capital investment and expansion.

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Financial Institution

What Is a Financial Institution (FI)?

A financial institution (FI) is a company engaged in the business of dealing with financial and monetary transactions such as deposits, loans, investments, and currency exchange. Financial institutions encompass a broad range of business operations within the financial services sector including banks, trust companies, insurance companies, brokerage firms, and investment dealers. Virtually everyone living in a developed economy has an ongoing or at least periodic need for the services of financial institutions.

How Financial Institutions Work

Financial institutions serve most people in some way, as financial operations are a critical part of any economy, with individuals and companies relying on financial institutions for transactions and investing. Governments consider it imperative to oversee and regulate banks and financial institutions because they do play such an integral part of the economy. Historically, bankruptcies of financial institutions can create panic.

In the United States, the Federal Deposit Insurance Corporation (FDIC) insures regular deposit accounts to reassure individuals and businesses regarding the safety of their finances with financial institutions. The health of a nation's banking system is a linchpin of economic stability. Loss of confidence in a financial institution can easily lead to a bank run.

Types of Financial Institutions

Financial institutions offer a wide range of products and services for individual and commercial clients. The specific services offered vary widely between different types of financial institutions.

Commercial Banks

A commercial bank is a type of financial institution that accepts deposits, offers checking account services, makes business, personal, and mortgage loans, and offers basic financial products like certificates of deposit (CDs) and savings accounts to individuals and small businesses. A commercial bank is where most people do their banking, as opposed to an investment bank.

Banks and similar business entities, such as thrifts or credit unions, offer the most commonly recognized and frequently used financial services: checking and savings accounts, home mortgages, and other types of loans for retail and commercial customers. Banks also act as payment agents via credit cards, wire transfers, and currency exchange.

Investment Banks

Investment banks specialize in providing services designed to facilitate business operations, such as capital expenditure financing and equity offerings, including initial public offerings (IPOs). They also commonly offer brokerage services for investors, act as market makers for trading exchanges, and manage mergers, acquisitions, and other corporate restructurings.

Insurance Companies

Among the most familiar non-bank financial institutions are insurance companies. Providing insurance, whether for individuals or corporations, is one of the oldest financial services. Protection of assets and protection against financial risk, secured through insurance products, is an essential service that facilitates individual and corporate investments that fuel economic growth.

Brokerage Firms

Investment companies and brokerages, such as mutual fund and exchange-traded fund (ETF) provider Fidelity Investments, specialize in providing investment services that include wealth management and financial advisory services. They also provide access to investment products that may range from stocks and bonds all the way to lesser-known alternative investments, such as hedge funds and private equity investments.

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Asymmetric Information in Financial Market

Financial markets exhibit asymmetric information in any transaction in which one of the two parties involved has more information than the other and thus has the ability to make a more informed decision.

Economists say that asymmetric information leads to market failure. That is, the law of supply and demand that regulates the pricing of goods and services is skewed.

The 2007–2008 subprime loan crisis was a classic example of the way asymmetric information can skew a market and cause market failure.

Understanding Asymmetric Information

Asymmetric information in the financial markets can occur whenever either the buyer or seller has more information on the past, present, or future performance of an investment. One party can make an informed decision, but the other party cannot.

The buyer may know that the asset is under-priced, or the seller may know that it is under - priced. In either case, one party has the opportunity to profit from the transaction at the expense of the other.

Example: The Subprime Meltdown and Asymmetric Information

The 2007-2008 subprime mortgage crisis could serve as a textbook illustration of the effects of asymmetric information. The products behind the crisis were mortgage-backed securities. Banks had extended the mortgages to consumers and then sold them to third parties. Those third parties packaged them together in batches and sold them on to investors. The securities were rated high-quality and were sold as such.

But many or most of the individual mortgages included in those products had been extended to borrowers buying bubble-priced homes that were beyond their means. When prices stalled the borrowers were stuck, as were the secondary buyers of their mortgages.

Unless nobody did their homework at any stage of this complicated process, the sellers had information that the end buyers did not. That is, they knew that risky mortgages were being passed off as high-quality debt. They were profiting from asymmetric information.

Other Examples of Asymmetric Information

Asymmetric information can occur in any situation involving a borrower and a lender when the borrower fails to disclose negative information about his or her real financial state. Or the borrower may simply fail to anticipate a worst-case scenario such as a job loss or an unanticipated expense.

This is why unsecured loans can be so costly. The lender can review the borrower's credit history and salary level but cannot foresee bad luck. The lender will charge a risk premium to compensate for the disparity in information.

Ignoring Risks

Economists who study asymmetric information suggest that such situations can pose a moral hazard to one party in a transaction. Such a moral hazard can occur when the seller or buyer knows or reasonably suspects that a real but undisclosed risk is involved in the transaction.

As an example, consider again the sale of those mortgage-backed securities. The sellers may have done their homework and therefore have known they were selling low-quality mortgages packaged as top-rated investments. Or they may have seen early warning signs of an imminent collapse in home prices.

Did the buyers have the same information? If they did, they presumably were engaged in the same game of pass-the-trash and were counting on reselling the securities at a profit before the end came.

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Role and Functions of Derivatives

Derivatives usually perform the following functions.

1) Price discovery of the underlying asset

Price discovery is a method of determining the price for a specific commodity or security through basic supply and demand factors related to the market. Price discovery is the general process used in determining the spot price. These prices are dependent upon market conditions affecting supply and demand. For example, if the demand for a particular commodity is higher than its supply, the price will typically increase and vice versa.

Futures market prices depend on a continuous flow of information from around the world and require a high degree of transparency. A broad range of factors (climatic conditions, political situations, debt default, refugee displacement, land reclamation and environmental health, for example) impact supply and demand of assets (commodities in particular) - and thus the current and future prices of the underlying asset on which the derivative contract is based. This kind of information and the way people absorb it constantly changes the price of a commodity. This process is known as price discovery.

2) Techniques of risk management

Financial derivatives are useful for dealing with various types of risks, mainly market, credit and operational risks. The importance of derivatives has been increasing since the instrument has been used to hedge against price movements. The financial tool assists with the transfer of risks associated with a specific portfolio without requiring selling the portfolio itself. Essentially, derivatives allow investors to manage their risks and so reach the desired risk profile and allocation more efficiently.

The relationship between derivatives and risk management is relatively simple. Derivatives are seen as the tool that enables banks and other financial institutions to break down risks into smaller elements. From this, the elements can be bought or sold to align with the risk management objectives. So, the original purpose of derivatives was to hedge and spread risks. The main motive of the financial tool has aided with the great development and expansion of derivatives.

3) Operational advantages

Derivative markets entail lower transaction costs. They have greater liquidity compared to spot markets. Derivative markets allow short selling of underlying securities more easily.

4) Market efficiency

Spot markets for securities probably would be efficient even if there were no derivative markets. A few profitable arbitrage opportunities exist, however, even in markets that are usually efficient. The presence of these opportunities means that the price of some assets is temporarily out of line. Investors can earn return s that exceed what the market deems fair for the given risk level. There are important linkages between spot and derivative prices .The ease and low cost of transacting in these markets facilitate the arbitrage trading and rapid

price adjustments that quickly eradicate these profit opportunities .Society benefits because the prices of underlying goods more accurately reflect the good's true economic values.

Why Derivatives are important?

The uses of derivative instruments are generally attributed to:

- Risk Sharing

Derivatives are mainly used to hedge risk associated with the underlying asset to the willing parties to take risk. The risk comes from several sources and is unavoidable. Derivatives are mainly intended to reduce the risks through transferring, spreading, etc. to the third parties who are risk seekers. The reducible risks include business risk, market risk, interest rate risk, inflation risk, currency risk/exchange rate risk, political risk, credit risk, weather risk, legal and regulatory risks, operational risks, valuation risks, etc. These risks can be reduced in different ways such as,

- By selling the source of it
- By diversification
- By buying insurance against losses

- Implementation of Asset allocation Decisions

Derivatives are useful in implementing the asset allocation strategies on account of their property of low cost of diversification and leverage.

- Information gathering

Derivative markets affect the information structure of the financial system. The economic benefit of the information is that the potential imbalances can be visualized more easily by the higher implied volatilities.

- Price discovery and Liquidity

Derivative markets offer liquidity in their transactions. Futures and forwards markets are the important source of price information.

Money Market in India: Features, Structure, Constituents, Participants and Defects

(@ https://www.economicsdiscussion.net/india/money-market/money-market-in-india-features-structure-constituents-participants-and-defects/31348)

Meaning of Money Market:

Money market is a market for short-term funds. We define the short-term as a period of 364 days or less. In other words, the borrowing and repayment take place in 364 days or less. The manufacturers need two types of finance: finance to meet daily expenses like purchase of raw material, payment of wages, excise duty, electricity charges etc., and finance to meet capital expenditure like purchase of machinery, installation of pollution control equipment etc.

The first category of finance is invested in the production process for a short-period of time. The market where such short-time finance is borrowed and lent is called 'money market'. Almost every concern in the financial system, be it a financial institution, business firm, a corporation or a government body, has a recurring problem of liquidity management, mainly because the timing of the expenditures rarely synchronize with that of the receipts.

The most important function of the money market is to bridge this liquidity gap. Thus, business and finance firms can tide over the mismatches of cash receipts and cash expenditures by purchasing (or selling) the shortfall (or surplus) of funds in the money market.

In simple words, the money market is an avenue for borrowing and lending for the short-term. While on one hand the money market helps in shifting vast sums of money between banks, on the other hand, it provides a means by which the surplus of funds of the cash rich corporations and other institutions can be used (at a cost) by banks, corporations and other institutions which need short-term money.

A supplier of funds to the money market can be virtually anyone with a temporary excess of funds. The government bonds, corporate bonds and bonds issued by banks are examples of money market instruments, where the instrument has a ready market like the equity shares of a listed company. The money markets refer to the market for short-term securities (one year or less in original maturity) such as treasury bills, certificates of deposits, commercial paper etc. Money market instruments are more liquid in nature.

The money market is a market where money and highly liquid marketable securities are bought and sold. It is not a place like the stock market but an activity and all the trading is done through telephones. One of the important features of the money market is honor of commitment and creditworthiness.

The money market form an important part of the financial system by providing an avenue for bringing equilibrium of the surplus funds of lenders and the requirements of borrowers for short periods ranging from overnight up to a year. Money market provides a non-inflationary way to finance government deficits and allow governments to implement monetary policy through open market operations and provide a market based reference point for setting interest rate.

Features and Objectives of Money Market:

Features of Money Market:

Following are the features of money market:

1. Money market has no geographical constraints as that of a stock exchange. The financial institutions dealing in monetary assets may be spread over a wide geographical area.

2. Even though there are various centers of money market such as Mumbai, Calcutta, Chennai, etc., they are not separate independent markets but are inter-linked and interrelated.

3. It relates to all dealings in money or monetary assets.

4. It is a market purely for short-term funds.

5. It is not a single homogeneous market. There are various sub-markets such as Call money market, Bill market, etc.

6. Money market establishes a link between RBI and banks and provides information of monetary policy and management.

7. Transactions can be conducted without the help of brokers.

8. Variety of instruments are traded in money market.

Objectives of Money Market:

Following are the objectives of money market:

1. To cater to the requirements of borrowers for short term funds, and provide liquidity to the lenders of these funds.

2. To provide parking place for temporary employment of surplus fund.

3. To provide facility to overcome short term deficits.

4. To enable the central bank to influence and regulate liquidity in the economy.

5. To help the government to implement its monetary policy through open market operation.

Structure of Indian Money Market:

(i) Broadly speaking, the money market in India comprises two sectors- (a) Organised sector, and (b) Unorganised sector.

(ii) The organised sector consists of the Reserve Bank of India, the State Bank of India with its seven associates, twenty nationalised commercial banks, other scheduled and non-scheduled commercial banks, foreign banks, and Regional Rural Banks. It is called organised because its part is systematically coordinated by the RBI.

(iii) Non-bank financial institutions such as the LIC, the GIC and subsidiaries, the UTI also operate in this market, but only indirectly through banks, and not directly.

(iv) Quasi-government bodies and large companies also make their short-term surplus funds available to the organised market through banks.

(v) Cooperative credit institutions occupy the intermediary position between organised and unorganised parts of the Indian money market. These institutions have a three-tier structure. At the top, there are state cooperative banks. At the local level, there are primary credit societies and urban cooperative banks. Considering the size, methods of operations, and dealings with the RBI and commercial banks, only state and central, cooperative banks should be included in the organised sector. The cooperative societies at the local level are loosely linked with it.

(vi) The unorganised sector consists of indigenous banks and money lenders. It is unorganised because activities of its parts are not systematically coordinated by the RBI.

(vii) The money lenders operate throughout the country, but without any link among themselves.

(viii) Indigenous banks are somewhat better organised because they enjoy rediscount facilities from the commercial banks which, in turn, have link with the RBI. But this type of organisation represents only a loose link with the RBI.

Constituents of Indian Money Market:

Money market is a centre where short-term funds are supplied and demanded. Thus, the main constituents of money market are the lenders who supply and the borrowers who demand short-term credit.

I. Supply of Funds:

There are two main sources of supply of short-term funds in the Indian money market:

(a) Unorganised indigenous sector, and

(b) Organised modern sector.

(i) Unorganized Sector:

The unorganised sector comprises numerous indigenous bankers and village money lenders. It is unorganized because its activities are not controlled and coordinated by the Reserve Bank of India.

(ii) Organized Sector:

The organized modern sector of Indian money market comprises:

(a) The Reserve Bank of India;

(b) The State Bank of India and its associate banks;

(c) The Indian joint stock commercial banks (scheduled and non-scheduled) of which 20 scheduled banks have been nationalised;

(d) The exchange banks which mainly finance Indian foreign trade;

(e) Cooperative banks;

(f) Other special institutions, such as, Industrial Development Bank of India, State Finance Corporations, National Bank for Agriculture and Rural Development, Export-Import Bank, etc., which operate in the money market indirectly through banks; and

(g) Quasi-government bodies and large companies also make their funds available to the money market through banks.

II. Demand for Funds:

In the Indian money market, the main borrowers of short-term funds are: (a) Central Government, (b) State Governments, (c) Local bodies, such as, municipalities, village panchayats, etc., (d) traders, industrialists, farmers, exporters and importers, and (e) general public.

Sub-Markets of Organised Money Market:

The organised sector of Indian money market can be further classified into the following sub-markets:

A. Call Money Market:

The most important component of organised money market is the call money market. It deals in call loans or call money granted for one day. Since the participants in the call money market are mostly banks, it is also called interbank call money market.

The banks with temporary deficit of funds form the demand side and the banks with temporary excess of funds form the supply side of the call money market.

The main features of Indian call money market are as follows:

(i) Call money market provides the institutional arrangement for making the temporary surplus of some banks available to other banks which are temporary in short of funds.

(ii) Mainly the banks participate in the call money market. The State Bank of India is always on the lenders' side of the market.

(iii) The call money market operates through brokers who always keep in touch with banks and establish a link between the borrowing and lending banks.

(iv) The call money market is highly sensitive and competitive market. As such, it acts as the best indicator of the liquidity position of the organised money market.

(v) The rate of interest in the call money market is highly unstable. It quickly rises under the pressures of excess demand for funds and quickly falls under the pressures of excess supply of funds.

(vi) The call money market plays a vital role in removing the day-to-day fluctuations in the reserve position of the individual banks and improving the functioning of the banking system in the country.

B. Treasury Bill Market:

The treasury bill market deals in treasury bills which are the short-term (i.e., 91, 182 and 364 days) liability of the Government of India. Theoretically these bills are issued to meet the short-term financial requirements of the government.

But, in reality, they have become a permanent source of funds to the government. Every year, a portion of treasury bills are converted into long-term bonds. Treasury bills are of two types: ad hoc and regular.

Ad hoc treasury bills are issued to the state governments, semi- government departments and foreign central banks. They are not sold to the banks and the general public, and are not marketable.

The regular treasury bills are sold to the banks and public and are freely marketable. Both types of ad hoc and regular treasury bills are sold by Reserve Bank of India on behalf of the Central Government.

The treasury bill market in India is underdeveloped as compared to the treasury bill markets in the U.S.A. and the U.K.

In the U.S.A. and the U.K., the treasury bills are the most important money market instrument:

(a) Treasury bills provide a risk-free, profitable and highly liquid investment outlet for short-term, surpluses of various financial institutions;

(b) Treasury bills from an important source of raising fund for the government; and

(c) For the central bank the treasury bills are the main instrument of open market operations.

On the contrary, the Indian Treasury bill market has no dealers expect the Reserve Bank of India. Besides the Reserve Bank, some treasury bills are held by commercial banks, state government and semi-government bodies. But, these treasury bills are not popular with the non-bank financial institutions, corporations, and individuals mainly because of absence of a developed treasury bill market.

C. Commercial Bill Market:

Commercial bill market deals in commercial bills issued by the firms engaged in business. These bills are generally of three months maturity. A commercial bill is a promise to pay a specified amount in a specified period by the buyer of goods to the seller of the goods. The seller, who has sold his goods on credit draws the bill and sends it to the buyer for acceptance. After the buyer or his bank writes the word ‘accepted’ on the bill, it becomes a marketable instrument and is sent to the seller.

The seller can now sell the bill (i.e., get it discounted) to his bank for cash. In times of financial crisis, the bank can sell the bills to other banks or get them rediscounted from the Reserved Bank. In India, the bill market is undeveloped as compared to the same in advanced countries like the U.K. There is absence of specialised institutions like acceptance houses and discount houses, particularly dealing in acceptance and discounting business.

D. Collateral Loan Market:

Collateral loan market deals with collateral loans i.e., loans backed by security. In the Indian collateral loan market, the commercial banks provide short- term loans against government securities, shares and debentures of the government, etc.

E. Certificate of Deposit and Commercial Paper Markets:

Certificate of Deposit (CD) and Commercial Paper (CP) markets deal with certificates of deposit and commercial papers. These two instruments (CD and CP) were introduced by Reserve Bank of India in March 1989 in order to widen the range of money market instruments and give investors greater flexibility in the deployment of their short-term surplus funds.

Participants in Money Market:

A large number of borrowers and lenders make up the money market.

Some of the important players are listed below:

1. Central Government:

Central Government is a borrower in the money market through the issue of Treasury Bills (T-Bills). The T-Bills are issued through the RBI. The T-Bills represent zero risk instruments. They are issued with tenure of 91 days (3 months), 182 days (6 months) and 364 days (1 year). Due to its risk free nature, banks, corporates and many such institutions buy the T-Bills and lend to the government as a part of it short- term borrowing programme.

2. Public Sector Undertakings:

Many government companies have their shares listed on stock exchanges. As listed companies, they can issue commercial paper in order to obtain its working capital finance. The PSUs are only borrowers in the money market. They seldom lend their surplus due to the bureaucratic mindset. The treasury operations of the PSUs are very inefficient with huge cash surplus remaining idle for a long period of time.

3. Insurance Companies:

Both general and life insurance companies are usual lenders in the money market. Being cash surplus entities, they do not borrow in the money market. With the introduction of CBLO (Collateralized Borrowing and Lending Obligations), they have become big investors. In between capital market instruments and money market instruments, insurance companies invest more in capital market instruments. As their lending programmes are for very long periods, their role in the money market is a little less.

4. Mutual Funds:

Mutual funds offer varieties of schemes for the different investment objectives of the public. There are many schemes known as Money Market Mutual Fund Schemes or Liquid Schemes. These schemes have the investment objective of investing in money market instruments.

They ensure highest liquidity to the investors by offering withdrawal by way of a day’s notice or encashment of units through Bank ATMs. Naturally, mutual funds invest the corpus of such schemes only in money market. They do not borrow, but only lend or invest in the money market.

5. Banks:

Scheduled commercial banks are very big borrowers and lenders in the money market. They borrow and lend in call money market, short-notice market, repo and reverse repo market. They borrow in rediscounting market from the RBI and IDBI. They lend in commercial paper market by way of buying the commercial papers issued by corporates and listed public sector units. They also borrow through issue of Certificate of Deposits to the corporates.

6. Corporates:

Corporates borrow by issuing commercial papers which are nothing but short-term promissory notes. They are issued by listed companies after obtaining the necessary credit rating for the CP. They also lend in the CBLO market their temporary surplus, when the interest rate rules very high in the market. They are the lender to the banks when they buy the Certificate of Deposit issued by the banks. In addition, they are the lenders through purchase of Treasury bills.

There are many other small players like non-banking finance companies, primary dealers, provident funds and pension funds. They mainly invest and borrow in the CBLO market in a small way.

Defects of Indian Money Market: Need for Reforms

A well-developed money market is a necessary pre-condition for the effective implementation of monetary policy. The central bank controls and -regulates the money supply in the country through the money market. But, unfortunately, the Indian money market is inadequately developed, loosely organised and suffers from many weaknesses.

Major defects are discussed below:

I. Dichotomy between Organised and Unorganised Sectors:

The most important defect of the Indian money market is its division into two sectors- (a) the organised sector and (b) the unorganised sector. There is little contact, coordination and cooperation between the two sectors. In such conditions it is difficult for the Reserve Bank to ensure uniform and effective implementations of its monetary policy in both the sectors.

II. Predominance of Unorganised Sector:

Another important defect of the Indian money market is its predominance of unorganised sector. The indigenous bankers occupy a significant position in the money- lending business in the rural areas. In this unorganised sector, no clear-cut distinction is made between short- term and long-term and between the purposes of loans.

These indigenous bankers, which constitute a large portion of the money market, remain outside the organised sector. Therefore, they seriously restrict the Reserve Bank’s control over the money market.

III. Wasteful Competition:

Wasteful competition exists not only between the organised and unorganised sectors, but also among the members of the two sectors. The relation between various segments of the money

market is not cordial; they are loosely connected with each other and generally follow separatist tendencies.

For example, even today, the State Bank of Indian and other commercial banks look down upon each other as rivals. Similarly, competition exists between the Indian commercial banks and foreign banks.

IV. Absence of All-India Money Market:

Indian money market has not been organised into a single integrated all-Indian market. It is divided into small segments mostly catering to the local financial needs. For example, there is little contact between the money markets in the bigger cities, like, Bombay, Madras, and Calcutta and those in smaller towns.

V. Inadequate Banking Facilities:

Indian money market is inadequate to meet the financial need of the economy. Although there has been rapid expansion of bank branches in recent years particularly after the nationalisation of banks, yet vast rural areas still exist without banking facilities. As compared to the size and population of the country, the banking institutions are not enough.

VI. Shortage of Capital:

Indian money market generally suffers from the shortage of capital funds. The availability of capital in the money market is insufficient to meet the needs of industry and trade in the country. The main reasons for the shortage of capital are- (a) low saving capacity of the people; (b) inadequate banking facilities, particularly in the rural areas; and (c) undeveloped banking habits among the people.

VII. Seasonal Shortage of Funds:

A Major drawback of the Indian money market is the seasonal stringency of credit and higher interest rates during a part of the year. Such a shortage invariably appears during the busy months from November to June when there is excess demand for credit for carrying on the harvesting and marketing operations in agriculture. As a result, the interest rates rise in this period. On the contrary, during the slack season, from July to October, the demand for credit and the rate of interest decline sharply.

VIII. Diversity of Interest Rates:

Another defect of Indian money market is the multiplicity and disparity of interest rates. In 1931, the Central Banking Enquiry Committee wrote- “The fact that a call rate of 3/4 per cent, a hundi rate of 3 per cent, a bank rate of 4 per cent, a bazar rate of small traders of 6.25 per cent and a Calcutta bazar rate for bills of small trader of 10 per cent can exist simultaneously indicates an extraordinary sluggishness of the movement of credit between various markets.”

The interest rates also differ in various centres like Bombay, Calcutta, etc. Variations in the interest rate structure are largely due to the credit immobility because of inadequate, costly and time-consuming means of transferring money. Disparities in the interest rates adversely affect the smooth and effective functioning of the money market.

IX. Absence of Bill Market:

The existence of a well-organised bill market is essential for the proper and efficient working of money market. Unfortunately, in spite of the serious efforts made by the Reserve Bank of India, the bill market in India has not yet been fully developed.

The short-term bills form a much smaller proportion of the bank finance in India as compared to that in the advanced countries.

Many factors are responsible for the underdeveloped bill market in India:

(i) Most of the commercial transactions are made in terms of cash.

(ii) Cash credit is the main form of borrowing from the banks. Cash credit is given by the banks against the security of commodities. No bills are involved in this type of credit,

(iii) The practice of advancing loans by the sellers also limits the use of bills,

(iv) There is lack of uniformity in drawing bills (hundies) in different parts of the country,

(v) Heavy stamp duty discourages the use of exchange bills.

(vi) Absence of acceptance houses is another factor responsible for the underdevelopment of bill market in India.

(vii) In their desire to ensure greater liquidity and public confidence, the Indian banks prefer to invest their funds in first class government securities than in exchange bills,

(viii) The Reserve Bank of India also prefers to extend rediscounting facility to the commercial banks against approved securities.

Undeveloped Nature of Indian Money Market:

An insight into the various defects and inadequacies of the Indian money market reveals that as compared to the advanced international money markets like the London Money Market, the New York Money Market, etc., Indian money market is still an undeveloped money market. It is “a money market of a sort where banks and other financial institutions lend or borrow funds for short periods.”

The following characteristics of Indian money market highlight its undeveloped nature:

(i) The Indian money market does not possess highly developed and adequately developed banking system.

(ii) It lacks sufficient and regular supply of short-term assets such as bills of exchange, treasury bills, short-term government bonds, etc.

(iii) There is no uniformity in the interest rates which vary considerably among different financial institutions as well as centres,

(iv) In the Indian money market, there are no dealers in short-term assets who can function as intermediaries between the government and the banking system,

(v) No doubt, a well-developed call money market exists in India, there is absence of other necessary sub-markets such as the acceptance market, commercial bill market, etc.

(vi) There is no proper coordination between the different sectors of the money market,

(vii) The Indian money market does not attract foreign funds and thus lacks international status.

Measures to Improve Indian Money Market:

Suggestions to Remove Defects:

In a view of the various defects in the Indian money market, the following suggestions have been made for its proper development:

(i) The activities of the indigenous banks should be brought under the effective control of the Reserve Bank of India.

(ii) Hundies used in the money market should be standardised and written in the uniform manner in order to develop an all-India money market,

(iii) Banking facilities should be expanded especially in the unbanked and neglected areas,

(iv) Discounting and rediscounting facilities should be expanded in a big way to develop the bill market in the country.

(v) For raising the efficiency of the money market, the number of the clearing houses in the country should be increased and their working improved.

(vi) Adequate and less costly remittance facilities should be provided to the businessmen to increase the mobility of capital.

(vii) Variations in the interest rates should be reduced.

Reserve Bank and Indian Money Market:

The Reserve Bank of Indian has taken various measures to improve the existing defects and to develop a sound money market in the country.

Important among them are:

(i) Through the introduction of two schemes, one in 1952 and the other in 1970, the Reserve Bank has been making efforts to develop a sound bill market and to encourage the use of bills in the banking system. The variety of bills eligible for use has also been enlarged.

(ii) A number of measures have been taken to improve the functioning of the indigenous banks. These measures include- (a) their registration; (b) keeping and auditing of accounts; (e) providing financial accommodation through banks; etc.

(iii) The reserve bank is fully effective in the organised sector of the money market and has evolved procedures and conventions to integrate and coordinate the different components of money market.

Due to the efforts of the Reserve Bank, there is now much more coordination in the organised sector than that in the unorganised sector or that between organised and unorganised sectors.

(iv) The difference between various sections of the money market has been considerably reduced. With the enactment of the Banking Regulation Act, 1949, all banks in the country have been given equal treatment by the Reserve Bank as regards licensing, opening of branches, share capital, the type of loans to be given, etc.

(v) In order to develop a sound money market, the Reserve Bank of Indian has taken measures to amalgamate and merge banks into a few strong banks and given encouragement to the expansion of banking facilities in the country,

(vi) The Reserve Bank of India has been able to reduce considerably the differences in the interest rates between different sections as well as different centres of the money market.

Now the interest rate structure of the country is much more sensitive to changes in the bank rate. Thus, the Reserve Bank of India has succeeded to a great extent in improving the Indian money market and removing some of its serious defects.

But, there are certain difficulties faced by the Reserve Bank in controlling the money market:

(i) The absence of bill market restricts the Reserve Bank's ability to withdraw surplus funds from the money market by disposing of bills.

(ii) The existence of indigenous bankers is the major hurdle in the way of integrating the money market.

(iii) Inadequate development of call money market is another difficulty in controlling the money market. The banks do not maintain fixed ratios between their cash reserves and deposits and the Reserve Bank has to undertake large open market operations to influence the policy of the banks.

Working Group on Money Market:

In, 1986, the Reserve Bank of India set up a Working Group under the chairmanship of Mr. N. Vaghul to examine the possibilities of enlarging the scope of money market and to recommend specific measures for evolving other suitable money market instruments.

The Working Group submitted its Report in January, 1987. It has made a number of recommendations for activating and developing the Indian money market.

Some Important recommendations are as follows:

(i) Measures should be taken to improve the operation of the call money market,

(ii) Rediscounting market should be developed with a view to facilitating the emergence of genuine bill culture in the country.

(iii) A short-term commercial paper should be introduced.

(iv) An active secondary market for Government paper, especially a '182 days Treasury Bill' Refinance facility, should be developed.

(v) A Finance House should be set up to deal in short-term money market instruments.

(vi) Banks and private non-bank financial institutions should be encouraged to provide factoring services.

(vii) There should be continuing development and refinement of money market instruments, and every new instrument must be approved by the Reserve Bank.

Recent Measures Taken by RBI:

The Reserve Bank of India has taken the following measures to implement the recommendation of the Working Group since 1987:

(i) With a view to make bill financing attractive to the borrowers, from April 1987, the effective interest rate on bill discounting for categories subject to the maximum lending rate has been fixed at a rate one percentage point lower than the maximum lending rate.

(ii) In order to attract additional funds into rediscount market, the ceiling on the bill rediscounting rate has been raised from 11.5% to 12:5%

(iii) Access to bill rediscounting market has been increased by selectively increasing the number of participants in the market.

(iv) 182 Day Treasury Bills have been introduced in 1987. In 1992-93, 364 Day Treasury Bills were introduced and the auction of 182 Day Bill has been discontinued. Like 182-Day Treasury bills, 364 Day Bills can be held by commercial banks for meeting Statutory Ratio.

(v) In August 1989, the government remitted the duty on usance bills. This step removed a major administrative constraint in the use of bill system.

(vi) Total deregulation of money market interest rates with effect from May 1, 1989 is a significant step taken by RBI towards the activation of money market. Removing the interest ceiling on money rates would make them flexible and lend transparency to transactions in the money market.

(vii) Certificates of Deposits (CDs) were introduced in June 1989 to give investors greater flexibility in employment of their short-term funds.

(viii) Another money market instrument, Commercial Paper (CP), was introduced in 1990-91 to provide flexibility to the borrowers rather than additionally of funds over and above the eligible credit limit.

(ix) Since July 1987, the Credit Authorisation Scheme (CAS) has been liberalised to allow for greater access to credit to meet genuine demand in production sectors without the prior sanction of the Reserve Bank.

(x) In April, the Discount and Finance House of Indian Limited (DFHI) was established with a view to increasing the liquidity of money market instruments.

(xi) In 1991, the scheduled commercial banks and their subsidiaries were permitted to set up Money Market Mutual Fund (MMMF) which would provide additional short-term avenue to investors and bring money market instruments within the reach of individuals and small bodies.

As a result of various measures taken by the RBI, the Indian money market has shown signs of notable development in many ways:

(i) It is becoming more and more organised and diversified.

(ii) The government trading in various instruments, like 364 Day treasury Bills, commercial bills and commercial paper, has increased considerably.

(iii) The volume of inter-bank call money, short notice money and term money transactions have grown significantly.

(iv) At present, scheduled commercial banks, cooperative banks, Discount and Finance House of India (DFHI) are participating in the money market both as lenders and borrowers of short-term funds, while Life Insurance Corporation of India (LIC), Unit Trust of India (UTI), General Insurance Corporation of India (GIC), Industrial Development Bank of India (IDBI) and National Bank for Agriculture and Rural Development (NABARD) are participating as lenders.

Discount and Finance House of India (PFHI):

The Working Group of Money Market, in its Report submitted in 1987, recommended, among other things, that a Finance House should be set up to deal in short-term money market instruments.

As a follow- up on the recommendations of the Working Group, the Reserve Bank in India, in collaboration with the public sector banks and financial institutions, set up the Discount and Finance House of India Limited (DFHI) in April 1988.

DFHI is the apex body in the Indian money market and its establishment is a major step towards developing a secondary market for money instruments. DFHI, which commend its operations from April 25, 1988 deals in short-term money market instruments.

As a matter of policy, the aim of the DFHI is to increase the volume of turnover rather than to becomes the repository of money market instruments. The initial paid up capital of DFHI is Rs. 150 crores. Apart from this, it has lines of refinance from RBI and a line of credit from the consortium of public sector banks.

As the apex agency in the Indian money market, the DFHI has been playing an important role ever since its inception. It has been promoting the active participation of the scheduled commercial banks and their subsidiaries, state and urban cooperative banks and all-Indian financial institutions in the money market.

The objective is to ensure that short-term surplus and deficits of these institutions are equilibrated at market-related rates through inter-bank transactions and various money market instruments. In 1990-91 the DFHI opened its branches at Delhi, Calcutta, Madras, Ahmedabad and Bangalore in order to decentralise its operations and provide money market facilities at the major money market centres in the country.

DFHI has been providing secondary market for money instruments and Government of India Treasury Bills.

Certificate of Deposit (CD) and Commercial Paper (CP):

In March 1989, Reserve Bank of India decided to introduced Certificates of Deposit (CD) and Commercial Paper (CP) in order to widen the range of money market instruments and give investors greater flexibility in the deployment of their short-term surplus funds.

I. Certificates of Deposit (CD):

The Certificates of Deposit (CD) can be issued only by the scheduled commercial banks in multiple of Rs. 25 lakhs subject to the minimum size of an issue being Rs. 1 crore. Their maturity will vary between three months and one year. CDs will be issued at discount to face value and the discount rate will be freely determined. They will be further freely transferable by endorsement and delivery. CDs will, however, be subject to reserve requirements. Banks will neither be allowed to grant loans against CDs, nor can they buy their own CDs.

II. Commercial Paper (CP):

Commercial Paper (CP) can be issued by a listed company which has a net worth of at least Rs. 10 crores and a working capital limit of not less than Rs. 25 crore. CPs will be issued in multiples of Rs. 25 lakhs subject to the minimum size of an issue being Rs. 1 crore. Their maturity ranges from three months to six months. They will be freely transferable by endorsement and delivery.

The company issuing CP will have to obtain every six months a specified rating from an agency approved by the Reserve Bank. The company can raise money through CP upto a maximum amount equivalent to 20% of its working capital limits. Banks will not be permitted to either underwrite or co- accept the issue of CP.

On January 3, 1990, the Reserve Bank issued guidelines, for issue of CP, according to which a company will have to obtain P1 + rating from Credit Rating Information Service of India Ltd. and also classification under Health Code Number from its financing banks and it has also to maintain the current ratio of 1.33 : 1 to be eligible to issue CP.

Money Market Reforms in India

Reserve Bank of India is the biggest regulator of the Indian markets. It controls the monetary policy of India. Its control is however limited to the organised part of economy and the unorganised sector which has a significant presence is largely unregulated. RBI frequently introduces many reforms to bolster the Indian economy which is in a state of constant flux and is continuously evolving. The major money market reforms came after the recommendations of S. Chakravarty Committee and Narsimham Committee. These were major changes which helped unfold the banking potential of India and shape our financial institutions to world class standards. It was soundness of these reforms which helped our economy to easily tide over the economic crisis which had gripped the world in 2008. These are discussed below:

1. Deregulation of Interest Rates

Interest rates are now subject to market conditions as the ceiling limit on them have been removed by RBI after 1989.The important interest rates in India are-Bank rate, Medium-term lending rate, Prime Lending rate, Bank Deposit rate, Call rate, Certificate of Deposit rate, Commercial paper rate etc. This deregulation got a major push after the economic liberalisation of 1991. Chakravarty Committee was a strong proponent of free and flexible interest rates to promote savings, investments, government financial system and stability. RBI removed the upper ceiling of 16.5% and instead fixed a minimum of 16% per annum. The rates were further relaxed after the Narasimhan Committee report in 1991.

2. Reforms in Call and Term money market

The reforms in call and term money market were done to infuse more liquidity into the system and enable price discovery. RBI undertook several important steps to check the constraints and remove them systematically. It was in October 1998, RBI announced that non-banking financial institutions should not participate in call/term money market operations and it should purely be an interbank operating segment and encouraged other participants to migrate to collateralised segments to improve stability. Also, reporting of all call/notice money market transactions through negotiated dealing system within 15 minutes of conclusion of transaction was made mandatory. The volume of operations in this segment was not increased much even after the reforms.

3. Introduction of new money market instruments

The Reserve Bank of India has played an important role in the introduction of new money market instruments to diversify the market. These new instruments are 182 days treasury bills, longer maturity bills, dated Government securities, certificates of deposits and commercial papers, 3—4 days repos and 1 day repos from 1998-99, commercial papers in 1990 and interbank participation certificates with/without risk in 1988.

Traditionally, the 91 days treasury bills have been the main instrument used by Government of India for raising short term funds. The investments came from commercial banks. In January 1993, the Government of India introduced the system of weekly out time, which has become quite popular. The 182 days bills, which were discontinued in 1992, have been reintroduced

from 1998-99. Now Indian money market has 14 days, 91 days, 182 days and 364 days treasury bills.

Demand for Treasury bill is no longer exclusively linked with statutory liquidity rates considerations. The secondary market transactions aiming at effective management of short term liquidity are on the increase.

The Government has been raising nearly Rs. 16,000 crores through this measurement. The interest rate variations in these bills have been between 7.15 to 11 per cent. Indian money market is following the unique practice of converting treasury bills into dated securities of 2 years or 5 years, normally carrying interest rate of 12 per cent.

4. Introducing Liquidity Adjustment Facility

RBI introduced a Liquidity Adjustment Facility in June 2000 which was operated through fixed repo and reverse repo rates. This helped establishment of interest rate as an important monetary instrument and granted greater flexibility to RBI to respond to market needs and suitably adjust liquidity in the market. Repo and Reverse Repo rates were introduced in 1992 and 1996 respectively.

5. Refinance by RBI

This is a potent tool by RBI to meet the any liquidity shortages and for credit control to select sectors. The export credit refinance facility to banks is provided under Section 17(3) of RBI Act 1934. It is available to all scheduled commercial banks who are authorised to deal in foreign exchange and have extended export credit. The SCBs are prvided export credit to the tune of 50% of the outstanding export credit. The concept of directed credit was also changed as the Narasimhan Committee recommended reduction of directed credit from 40 to 10%. It also suggested narrowing of priority sector and realigning focus to small farmers and low income target groups. The refinance rate is linked to bank rate.

6. Regulation of Non-Banking Financial Companies

RBI Act was amended in 1997 to bring the NBFCs under its regulatory framework. A NBFC is a company registered under Companies Act, 1956 and is involved in making loans and advances, acquisition of shares, stocks, bonds, securities issued by government etc. They are similar to banks but are different from the latter as they cannot accept demand deposits and cannot issue cheques. They have to be registered with RBI to operate within India. There are a host of regulations which NBFCs have to follow to smoothly operate within India like accept deposit for a minimum period, cannot accept interest rate beyond the prescribed rate given by RBI.

7. Debt Recovery

RBI has set up special Recovery Tribunals which provide legal assistance to banks for recovery of dues.

8. Institutional Development:

The post reforms period saw significant institutional development and procedural reforms aimed at developing a strong secondary market in government securities.

Discount and Finance House of India Ltd:

Has been set up as a part of the package of reforms of the money market. It buys bills and other short term papers from banks and financial institutions. It provides short term investment opportunity to banks.

To develop a secondary market in Government securities, it started buying and selling securities to a limited extent in 1992. To enable Discount and Finance House of India Ltd. (DFHI), to deal in Government securities, the Reserve Bank of India provides necessary refinance.

The institutional infrastructure in government securities has been strengthened with the system of Primary Dealers (PDs) announced in March 1995 and that of Satellite Dealers (SDs) in December 1996.

Similarly, Securities Trading Corporation of India was established in 1994, to provide better market and liquidity for dated securities, and to hold short term money market assets like treasury bills. The National Stock Exchange (NSE), has an exclusive trading floor for transparent and screen based trading in all types of debt instruments.

9. Money Market Mutual Funds:

In 1992 setting up of Money Market Mutual Funds was announced to bring it within in the reach of individuals. These funds have been introduced by financial institutions and banks.

With these reforms the money market is becoming vibrant. There is further scope of introducing new market players and extending refinance from Reserve Bank of India.

Narasimham Committee has also proposed that well managed non-banking financial intermediates and merchant bank should also be allowed to operate in the money market. As and when implemented this will widen the scope of money market.

10. Permission to Foreign Institutional Investors (FII):

FII's are allowed to operate in all dated government securities. The policy for 1998-99 had allowed them to buy treasury Bills' within approved debt ceiling.

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Capital Market in India: Classification and Growth

(Taken from https://www.yourarticlelibrary.com/economics/market/indian-capital-market-classification-and-growth-of-indian-capital-market/23476)

The Indian capital market is the market for long term loanable funds as distinct from money market which deals in short-term funds. It refers to the facilities and institutional arrangements for borrowing and lending 'term funds', medium term and long term funds. In principal capital market loans are used by industries mainly for fixed investment. It does not deal in capital goods, but is concerned with raising money capital or purpose of investment.

Meaning and Classification:

The capital market in India includes the following institutions (i.e., supply of funds tor capital markets comes largely from these); (i) Commercial Banks; (ii) Insurance Companies (LIC and GIC); (iii) Specialised financial institutions like IFCI, IDBI, ICICI, SIDCS, SFCS, UTI etc.; (iv) Provident Fund Societies; (v) Merchant Banking Agencies; (vi) Credit Guarantee Corporations. Individuals who invest directly on their own in securities are also suppliers of fund to the capital market.

Thus, like all the markets the capital market is also composed of those who demand funds (borrowers) and those who supply funds (lenders). An ideal capital market at tempts to provide adequate capital at reasonable rate of return for any business, or industrial proposition which offers a prospective high yield to make borrowing worthwhile.

The Indian capital market is divided into gilt-edged market and the industrial securities market. The gilt-edged market refers to the market for government and semi-government securities, backed by the RBI. The securities traded in this market are stable in value and are much sought after by banks and other institutions.

The industrial securities market refers to the market for shares and debentures of old and new companies. This market is further divided into the new issues market and old capital market meaning the stock exchange.

The new issue market refers to the raising of new capital in the form of shares and debentures, whereas the old capital market deals with securities already issued by companies.

The capital market is also divided in primary capital market and secondary capital market. The primary market refers to the new issue market, which relates to the issue of shares, preference shares, and debentures of non-government public limited companies and also to the realising of fresh capital by government companies, and the issue of public sector bonds.

Growth of Indian Capital Market:

Indian Capital Market before Independence:

Indian capital market was hardly existent in the pre-independence times. Agriculture was the mainstay of economy but there was hardly any long term lending to agricultural sector. Similarly the growth of industrial securities market was very much hampered since there were

very few companies and the number of securities traded in the stock exchanges was even smaller.

Indian capital market was dominated by gilt-edged market for government and semi-government securities. Individual investors were very few in numbers and that too were limited to the affluent classes in the urban and rural areas. Last but not the least, there were no specialised intermediaries and agencies to mobilise the savings of the public and channelise them to investment.

Indian Capital Market after Independence:

Since independence, the Indian capital market has made widespread growth in all the areas as reflected by increased volume of savings and investments. In 1951, the number of joint stock companies (which is a very important indicator of the growth of capital market) was 28,500 both public limited and private limited companies with a paid up capital of Rs. 775 crore, which in 1990 stood at 50,000 companies with a paid up capital of Rs. 20,000 crore. The rate of growth of investment has been phenomenal in recent years, in keeping with the accelerated tempo of development of the Indian economy under the impetus of the five year plans.

New Financial Intermediaries in Capital Market:

Since 1988 financial sector in India has been undergoing a process of structural transformation.

Some important new financial intermediaries introduced in Indian capital market are:

Merchant Banking:

Merchant bankers are financial intermediaries between entrepreneurs and investors. Merchant banks may be subsidiaries of commercial banks or may have been set up by private financial service companies or may have been set up by firms and individuals engaged in financial up by firms and individuals engaged in financial advisory business. Merchant banks in India manage and underwrite new issues, undertake syndication of credit, advice corporate clients on fund raising and other financial aspects.

Since 1993, merchant banking has been statutorily brought under the regulatory framework of the Securities Exchange Board of India (SEBI) to ensure greater transparency in the operation of merchant bankers and make them accountable. The RBI supervises those merchant banks which were subsidiaries, or are affiliates of commercial banks.

Leasing and Hire-Purchase Companies:

Leasing has proved a popular financing method for acquiring plant and machinery specially or small and medium sized enterprises. The growth of leasing companies has been due to advantages of speed, informality and flexibility to suit individual needs.

The Narasimhan Committee has recognised the importance of leasing and hire-purchase companies in financial intermediation process and has recommended that: (i) a minimum capital requirement should be stipulated; (ii) prudential norms and guidelines in respect of conduct of business should be laid down; and (iii) supervision should be based on periodic returns by a unified supervisory authority.

Mutual Funds:

It refers to the pooling of savings by a number of investors-small, medium and large. The corpus of fund thus collected becomes sizeable which is managed by a team of investment specialists backed by critical evaluation and supportive data.

A mutual fund makes up for the lack of investor's knowledge and awareness. It attempts to optimise high return, high safety and high liquidity trade off for maximum of investor's benefit. It thus aims at providing easy accessibility of media including stock market in country to one and all, especially small investors in rural and urban areas.

Mutual funds are most important among the newer capital market institutions. Several public sector banks and financial institutions set up mutual funds on a tax exempt basis virtually on same footing as the Unit Trust of India (UTI) and have been able to attract strong investor support and have shown significant progress.

Government has now decided to throw open the field to private sector and joint sector mutual funds. At present Securities and Exchange Board of India (SEBI) has authority to lay down guidelines and to supervise and regulate working of mutual funds.

The guidelines issued by the SEBI in January 1991, are related in advertisements and disclosure and reporting requirements etc. The investors have to be informed about the status of their investments in equity, debentures, government securities etc.

The Narasimhan Committee has made the following recommendations regarding mutual funds: (i) creation of an appropriate regulatory framework to promote sound, orderly and competitive growth of mutual fund business: (ii) creation of proper legal framework to govern the establishments and operation of mutual funds (the UTI is governed by a special statute), and (iii) equality of treatment between various mutual funds including the UTI in the area of tax concessions.

Global Depository Receipts (GDR):

Since 1992, the Government of India has allowed foreign investment in the Indian securities through the issue of Global Depository Receipts (GDRs) and Foreign Currency Convertible Bonds (FCCBs). Initially the Euro-issue proceeds were to be utilised for approved end uses within a period of one year from the date of issue.

Since there was continued accumulation of foreign exchange reserves with RBI and there were long gestation periods of new investment the government required the issuing companies to retain the Euro-issue proceeds abroad and repatriate only as and when expenditure for the approved end uses were incurred.

Venture Capital Companies (VCC):

The aim of venture capital companies is to give financial support to new ideas and to introduction and adaptation of new technologies. They are of a great importance to technocrat entrepreneurs who have technical competence and expertise but lack venture capital.

Financial institutions generally insist on greater contribution to the investment financing, in which technocrat entrepreneurs can depend on venture capital companies. Venture capital financing involves high risk.

According to the Narasimhan Committee the guidelines for setting up of venture capital companies are too restrictive and unrealistic and have impeded their growth. The committee has recommended a review and amendment of guidelines.

Knowing the high risk involved in venture capital financing, the committee has recommended a reduction in tax on capital gains made by these companies and equality of tax treatment between venture capital companies and mutual funds.

Other New Financial Intermediaries:

Besides the above given institutions, the government has established a number of new financial intermediaries to serve the increasing financial needs of commerce and industry is the area of venture Capital, credit rating and leasing etc.

(1) Technology Development and Information Company of India (TDICI) Ltd., a technology venture finance company, which sanctions project finance to new technology venture since 1989.

(ii) Risk Capital and Technology Finance Corporation (RCTFC) Ltd., which provides risk capital to new entrepreneurs and offers technology finance to technology-oriented ventures since 1988.

(iii) Infrastructure Leasing and Financial Services (IL&FS) Ltd., set up in 1988 focuses on leasing of equipment for infrastructure development.

(iv) The credit rating agencies namely credit rating information services of India (CRISIS) Ltd., setup in 1988; Investment and Credit Rating Agency (ICRA) setup in 1991, and Credit Analysis and Research (CARE) Ltd., setup in 1993 provide credit rating services to the corporate sector.

Credit rating promotes investors interests by providing them information on assessed comparative risk of investment in the listed securities of different companies. It also helps companies to raise funds more easily and at relatively cheaper rate if their credit rating is high.

(v) Stock Holding Corporation of India (SHCIL) Ltd., setup in 1988, with the objective of introducing a book entry system for transfer of shares and other type of scrips thereby avoiding the voluminous paper work involved and thus reducing delays in transfers.

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Capital Market Reforms in India

Here we detail about the five measures regarding capital market reforms.

The five measures are: (1) Establishment of SEBI, (2) Setting up of Private Mutual Funds, (3) Opening up to Foreign Capital, (4) Access to International Capital Markets, and (5) Banks and Capital Markets.

1. Establishment of SEBI:

An important measure regarding capital market reforms is the setting up of Securities and Exchange Board of India (SEBI) as the regulator of equity market in India.

Regulation of stock markets is important to ensure:

(a) That the equity markets operate in a fair and orderly manner,

(b) That the brokers and other professionals of the stock markets deal justly with their customers,

(c) That the corporate firms who raise funds through the market provide all information about themselves which the investors need to make intelligent investment decisions. Since its inception SEBI has been addressing itself to these tasks.

SEBI has introduced various guidelines and regulations for the functioning of capital markets and assuming and selling of shares in the primary market.

The following important regulatory measures have been introduced by SEBI:

(a) SEBI has introduced a code of advertisement for public issues by companies for making fair and truthful disclosures. The companies are now required to disclose all material facts and specific risk factors associated with their projects while making public issues.

(b) It has required the stock exchanges to amend their listing agreements to ensure that a listed company furnishes annual statements to them showing variations between financial projections and project utilisation of funds made in the offer documents and actual. This will enable shareholders to make comparisons between performance and promises made by of company.

(c) An important reform SEBI has introduced is that it has brought merchant banking also under its regulatory framework. The merchant bankers are required to follow the code of conduct issued by SEBI in respect of pricing and premium fixation of issues of shares a companies.

(d) The practice of making preferential allotment of shares at prices unrelated to the prevailing price has been stopped by SEBI. Besides, to ensure transparency insider trading has also been banned.

(e) As a part of the process of establishing transparent rules for trading in stock exchanges, a notorious BADLA system has been banned and in its place Rolling Settlement System has been introduced.

2. Setting up of Private Mutual Funds:

Another important reform is the permission granted to the private sector firms to start Mutual Funds. Many private sector companies such as Tata, Reliance, Birla have set up their mutual funds through which they raise money from the public. In this way monopoly position of UTI in Mutual Fund business has come to end. Mutual Funds raise money by selling units to the public and the funds so raised are invested in a number of equities and debentures of companies.

A mutual fund may be entirely equity-based or debt-based or a balanced one having a particular combination of investment in equities and debentures of a number of companies. Investment in mutual funds enables the investors to reduce risk. Mutual funds have also been allowed to open offshore funds to invest in equities abroad. UTI has also been brought within the regulatory framework of SEBI.

3. Opening up to Foreign Capital:

A significant reform has been that Indian capital market has been opened up for foreign institutional institutions (FII). That is, FII can now buy shares and debentures of private Indian companies in the Indian stock market and can also invest in government securities. This has been done to attract foreign capital. Foreign Institutional Investors (FII) have been permitted full capital convertibility

4. Access to International Capital Markets:

The Indian corporate sector has been allowed to raise funds in the international capital markets through American Depository Receipts (ADRs), Global Depository Receipts (GDR), Foreign Currency Convertible Bonds (FCCBs) and External Commercial Borrowings (ECBs). Similarly, Overseas Corporate Bodies (OCBs) and Non-resident Indians have been allowed to invest in the equity capital of the Indian companies. FIIs have been allowed to invest in equities of private corporate Indian companies as well as in Government securities.

5. Banks and Capital Markets:

Another important step to strengthen the Indian capital market is that banks have been allowed to lend against various capital market instruments such as corporate shares and debentures to individuals, investment companies, trusts and endowment share and stock brokers, industrial and corporate buyers and SEBI-approved market makers.

Lending by banks against various capital market instruments to individuals, share and stock brokers, market makers is made in accordance with certain norms regarding purpose, capital adequacy, transparent transactions, maximum possible amount or ceiling, or duration of the loan. Bank lending against shares and debentures, according to C. Rangarajan, will “enable partial liquidity to scrip’s, to help reduce volatility in price movement, encourage the presence of market makers so as to reduce market concentration and help in widening and deepening of trading in the secondary market.”

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Unit 3

Interest Rates

This Unit covers the following topics:

- Meaning and Determination of Interest Rates
- Sources of Interest rate Differentials and
- Theories of Term Structure of Interest rates

Meaning of Interest rate

The interest rate is the amount charged on top of the principal by a lender to a borrower for the use of assets for the specified period of time. It is a percentage term.

Most mortgages use simple interest where Simple Interest Rate = Principal × Interest Rate × Time. However, some loans use compound interest, which is applied to the principal but also to the accumulated interest of previous periods.

Compound Interest = Principal × $[(1 + \text{Interest Rate})^{n} - 1]$

A loan that is considered low risk by the lender will have a lower interest rate. A loan that is considered high risk will have a higher interest rate.

Consumer loans typically use an APR (Annual Percentage Rate) which does not use compound interest. APR is the rate of return that lenders demand for the ability to borrow their money. For example, the interest rate on credit cards is quoted as an APR. In our example above, 15% is the APR for the mortgagor or borrower.

On the other hand, the APY (Annual Percentage Yield) is the interest rate that is earned at a bank or credit union from a savings account or certificate of deposit (CD) by taking compounding into account.

Determination of Interest Rate

Here we are going to discuss the theories of determination of rate of interest. They are:

1. Classical Theory
2. Loanable Funds Theory
3. Theory of Liquidity Preference

1. The Classical Theory of Rate of Interest

(Taken from https://www.economicsdiscussion.net/theories-of-interest/the-classical-theory-of-interest-with-diagarm/7512)

The classical theory of interest also known as the demand and supply theory was propounded by the economists like Marshall and Fisher. Later on, Pigou, Cassel, Knight and Taussig worked to modify the theory.

According to this theory rate of interest is determined by the intersection of demand and supply of investment and savings. It is called the real theory of interest in the sense that it explains the determination of interest by analyzing the real factors like savings and investment. Therefore, classical economists maintained that interest is a price paid for the supply of savings.

Demand for Investment

Demand for investment comes from those who want to invest in business activities. Demand for investment is derived demand. Any factor of production is demanded for its productivity. The demand for the factor is high when there are higher expectations from it. Since all the factors are not equally productive, so, capital demand will be high for more productive uses first and then gradually with the increase in its supply, will shift to less productive uses.

Therefore, classical economists maintained that with the aid of capital facilities we turn out more goods per man-hour than when we produce with bare hands or with scant tools. Moreover, marginal productivity of the business goes on decreasing with more and more doses of investment of savings in his business venture. It is due to the operation of the law of diminishing returns.

Now a very important question arises is that how much capital a person will demand because when a person borrows money he has to pay interest on it. The answer according to this theory is that demand for capital can be raised to a point where marginal productivity of capital becomes equal to the interest paid on it. Thus, if marginal productivity of capital is more than the interest paid, then it is beneficial to borrow money and vice-versa. Equilibrium will prevail at a point where marginal productivity of capital equals the rate of interest. This shows that there exists inverse relationship between demand for capital or investment (I) and the interest rate as given by the downward sloping curve in Figure 1.

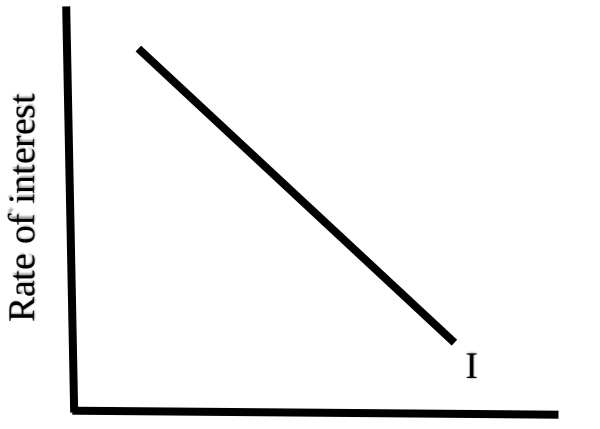


Figure 1

Supply of Savings: (S)

Supply of capital is the result of savings. It comes from those who have the excess of income over consumption. Thus, savings is the main source of capital which depends on the capacity to save, willingness to save, level of income and rate of interest etc. Capacity to save depends on the size of national income, size of personal income, size of family, price level and

purchasing power of money etc. Willingness to save depends on the family affection, further expectations etc.

To a large extent, willingness to save is affected by the rate of interest. On a higher rate of interest people save more to earn the benefits of high rate of interest. On the other hand, at the low rate of interest, people save less. Thus, we may say that there is a direct relationship between the supply of savings and the rate of interest as shown in Figure 2 which gives supply of savings (S) as upward sloping curve.

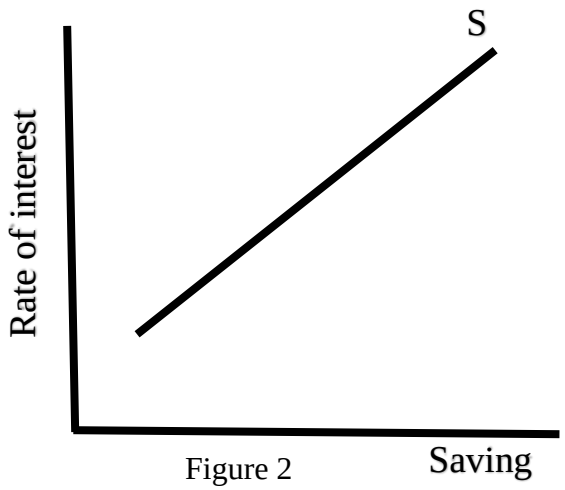


Figure 2

Equilibrium Rate of Interest in the Classical Theory

According to classical theory, equilibrium interest rate is restored at a point where demand for investment and supply savings are equal

In Fig. 3, rate of interest is determined by the intersection of demand and supply curves. Equilibrium is restored at point E which determines rate of interest as EM.

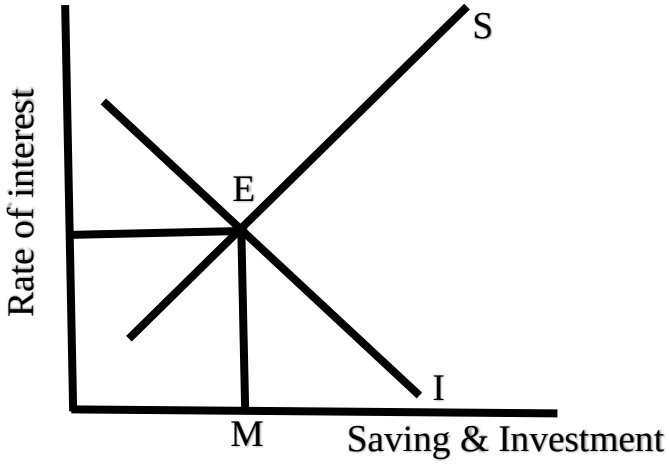


Figure 3

Criticism of Classical Theory of Interest

The classical theory of rate of interest has been criticized on the basis of the following shortcomings as discussed below:

1. Indeterminate Theory:

Keynes has maintained that the classical theory is indeterminate in the sense that it fails to determine the interest rate. In this theory, interest is determined by the equality of demand and supply. But the position of savings varies with the income level. Thus, unless we know the income, interest rate cannot be determined.

2. Fixed Level of Income:

Classical theory assumes that the level of income remains constant. But in actual practice income changes with a small change in investment. Thus, it is not correct to assume a fixed level of income.

3. Long Run:

Classical theory determines the interest rate through the interaction of demand and supply of capital in the long run. Keynes pointed out that in the long run we all are dead. Therefore, there was an urgent need of a theory which determines rate of interest in the short-run.

4. Full Employment:

This theory assumes that there is full employment of resources in the economy. But, in reality, unemployment or less than full employment is a general situation. Full employment is only an abnormal case… Thus, this theory does not apply to the present world.

5. Savings and Investment:

Classical economists assume that savings and investment are interring dependent. But actually investment changes, income also changes which leads to a change in savings. Thus, both are interdependent on each other.

6. Ignores Monetary Factors:

Classical theory takes into consideration only the real factors for determining the rate of interest and ignores the monetary factors.

2. Loanable Funds Theory of Interest

(https://www.economicsdiscussion.net/theories/loanable-funds-theory-with-diagram/7504)

The neo-classical theory of interest or loanable funds theory of interest owes its origin to the Swedish economist Knut Wicksell. Later on, economists like Ohlin, Myrdal, Lindahl, Robertson and J. Viner have considerably contributed to this theory.

According to this theory, rate of interest is determined by the demand for and supply of loanable funds. In this regard this theory is more realistic and broader than the classical theory of interest.

Demand for Loanable Funds: (LD)

According to this theory demand for loanable funds (LD) arises for the following three purposes viz.; Investment, hoarding and dissaving:

1. Investment (I):

The main source of demand for loanable funds is the demand for investment. Investment refers to the expenditure for the purchase of making of new capital goods including inventories. The price of obtaining such funds for the purpose of these investments depends on the rate of interest. An entrepreneur while deciding upon the investment is to compare the expected return from an investment with the rate of interest. If the rate of interest is low, the demand for loanable funds for investment purposes will be high and vice- versa. This shows that there is an inverse relationship between the demands for loanable funds for investment to the rate of interest.

2. Hoarding (H):

The demand for loanable funds is also made up by those people who want to hoard it as idle cash balances to satisfy their desire for liquidity. The demand for loanable funds for hoarding purpose is a decreasing function of the rate of interest. At low rate of interest demand for loanable funds for hoarding will be more and vice-versa.

3. Dissaving (DS):

Dissaving's is opposite to an act of savings. This demand comes from the people at that time when they want to spend beyond their current income. Like hoarding it is also a decreasing function of interest rate.

Diagram of Demand for Loanable Funds (LD) as given below in Figure 4 shows that LD is a downward sloping curve. At a higher rate of interest demand for loanable funds will be less than that at a lower rate of interest.

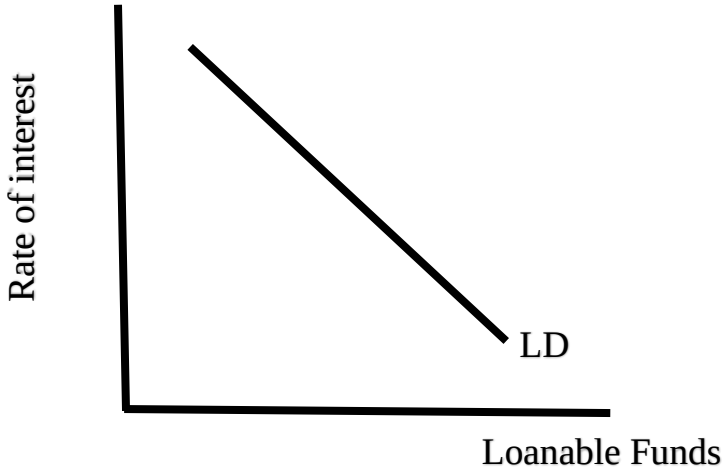


Figure 4

Supply of Loanable Funds: (LS)

The supply of loanable funds is derived from the basic four sources as savings, dishoarding, disinvestment and bank credit. They are explained as:

1. Savings (S):

Savings constitute the most important source of the supply of loanable funds. Savings is the difference between the income and expenditure. Since, income is assumed to remain unchanged, so the amount of savings varies with the rate of interest. Individuals as well as business firms will save more at a higher rate of interest and vice-versa.

2. Dishoarding (DH):

Dishoarding is another important source of the supply of loanable funds. Generally, individuals may dishoard money from the past hoardings at a higher rate of interest. Thus, at a higher interest rate, idle cash balances of the past become the active balances at present and become available for investment. If the rate of interest is low dishoarding would be negligible.

3. Disinvestment (DI):

Disinvestment occurs when the existing stock of capital is allowed to wear out without being replaced by new capital equipment. Disinvestment will be high when the present interest rate provides better returns in comparison to present earnings. Thus, high rate of interest leads to higher disinvestment and so on.

4. Bank Money (BM):

Banking system constitutes another source of the supply of loanable funds. The banks advance loans to the businessmen through the process of credit creation. The money created by the banks adds to the supply of loanable funds.

Supply of loanable funds (LS) is an upward sloping curve as given by Figure 5. At a higher rate of interest, supply of loanable funds will be higher and vice-versa.

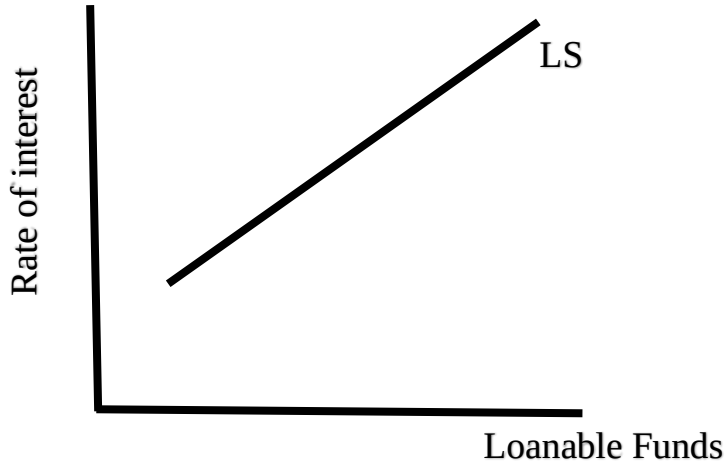


Figure 5

Determination of Rate of Interest in Loanable Funds Theory

According to loanable funds theory, equilibrium rate of interest is that which brings equality between the demand for and supply of loanable funds. In other words, equilibrium interest rate is determined at a point where the demand for loanable funds curve intersects the supply curve of loanable funds. It can be shown with the help of Figure 6.

The rate of interest is determined at the point of intersection of the two curves—the supply of loanable funds curve (LS) and the demand for loanable funds curve, LD. Figure 6 shows that the equilibrium rate of interest is EF; at this rate, the demand for loanable funds is equal to the supply of loanable funds i.e. OF.

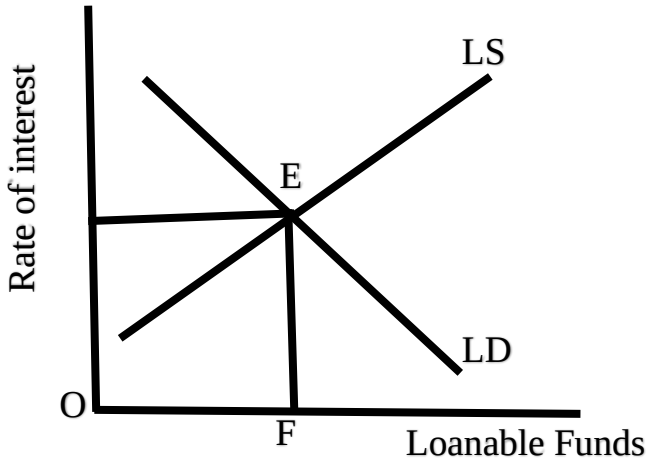


Figure 6

Criticism of Loanable Funds Theory

Although, loanable funds theory is superior to classical theory, yet, critics have criticised it on the following grounds:

1. Full Employment:

Keynes opined that loanable funds theory is based on the unrealistic assumption of full employment. As such, this theory also suffers from the defects as the classical theory does.

2. Indeterminate:

Like classical theory, loanable funds theory is also indeterminate. This theory assumes that savings and income both are independent. But savings depend on income. As the income changes savings also change and so does the supply of loanable funds.

3. Impracticable:

This theory assumes savings, hoarding, investment etc. to be related to interest rate. But in actual practice investment is not only affected by interest rate but also by the marginal efficiency of capital whose affect has been ignored.

4. Unsatisfactory Integration of Real and Monetary Factors:

This theory makes an attempt to integrate the monetary as well as real factors as the determinants of interest rate. But, the critics have maintained that these factors cannot be integrated in the form of the schedule as is evident from the frame work of this theory.

5. Constancy of National Income:

Loanable funds theory rests on the assumption that the level of national income remains unchanged. In reality, due to the change in investment, income level also changes accordingly.

Improvement over the Classical Theory

Loanable funds theory is considered to be an improvement over the classical theory on the following aspects:

1. Loanable funds theory recognizes the importance of hoarding as a factor affecting the interest rate which the classical theory has completely overlooked.

2. Loanable funds theory links together liquidity preference, quantity of money, savings and investment.

3. Loanable funds theory takes into consideration the role of bank credit which acts as a very important source of loanable funds.

3. Theory of Liquidity Preference

Liquidity Preference Theory of Interest Rate Determination was propounded by Keynes. The determinants of the equilibrium interest rate in the classical model are the 'real' factors of the supply of saving and the demand for investment. On the other hand, in the Keynesian analysis, determinants of the interest rate are the 'monetary' factors alone.

Keynes' analysis concentrates on the demand for and supply of money as the determinants of interest rate. According to Keynes, the rate of interest is purely "a monetary phenomenon." Interest is the price paid for borrowed funds. People like to keep cash with them rather than investing cash in assets. Thus, there is a preference for liquid cash.

People, out of their income, intend to save a part. How much of their resources will be held in the form of cash and how much will be spent depend upon what Keynes calls liquidity preference, Cash being the most liquid asset, people prefer cash. And interest is the reward for parting with liquidity. However, the rate of interest in the Keynesian theory is determined by the demand for money and supply of money.

Demand for Money:

Demand for money is not to be confused with the demand for a commodity that people 'consume'. But since money is not consumed, the demand for money is a demand to hold an asset. The desire for liquidity or demand for money arises because of three motives:

(a) Transaction motive

(b) Precautionary motive

(c) Speculative motive

(a) Transaction Demand for Money:

Money is needed for day-to-day transactions. As there is a gap between the receipt of income and spending, money is demanded. Incomes are earned usually at the end of each month or fortnight or week but individuals spend their incomes to meet day-to-day transactions.

Since payments or spending are made throughout a period and receipts or incomes are received after a period of time, an individual needs 'active balance' in the form of cash to finance his transactions. This is known as transaction demand for money or need- based money—which directly depends on the level of income of an individual and businesses.

People with higher incomes keep more liquid money at hand to meet their need-based transactions. In other words, transaction demand for money is an increasing function of money income.

Symbolically, MT = f (Y) where, MT stands for transaction demand for money and Y stands for money income.

(b) Precautionary Demand for Money:

Future is uncertain. That is why people hold cash balances to meet unforeseen contingencies, like sickness, death, accidents, danger of unemployment, etc. The amount of money held under this motive, called 'Idle balance', also depends on the level of money income of an individual.

People with higher incomes can afford to keep more liquid money to meet such emergencies. This means that this kind of demand for money is also an increasing function of money income. The relationship between precautionary demand for money (MP) and the volume of income is normally a direct one.

Thus, MP = f (Y) where Y is income.

(c) Speculative Demand for Money:

This sort of demand for money is really Keynes' contribution. The speculative motive refers to the desire to hold one's assets in liquid form to take advantages of market movements regarding the uncertainty and expectation of future changes in the rate of interest.

The cash held under this motive is used to make speculative gains by dealing in bonds and securities whose prices and rate of interest fluctuate inversely. If bond prices are expected to rise (or the rate of interest is expected to fall) people will now buy bonds and sell when their prices rise to have a capital gain. In such a situation, bond is more attractive than cash.

Contrarily, if bond prices are expected to fall (or the rate of interest is expected to rise) in future, people will now sell bonds to avoid capital loss. In such a situation, cash is more attractive than bond. Thus, at a low rate of interest, liquidity preference is high and, at a high rate of interest, securities are attractive. Now it is clear that the speculative demand for money (MSp) varies inversely with the rate of interest. Thus, MSp = f (r), where, r is the rate of interest.

Total Demand for Money:

The total demand for money (D_M) is the sum of all three types of demand for money. That is, D_M = MT + MP + MSp. In Figure 7 MT is shown as a vertical line which is perfectly inelastic to the rate of interest. MSp is inversely related to rate of interest. So, the demand for money

has a negative slope because of the inverse relationship between the speculative demand for money and the rate of interest.

However, the negative sloping liquidity preference curve becomes perfectly elastic at a low rate of interest. According to Keynes, there is a floor interest rate below which the rate of interest cannot fall. This minimum rate of interest indicates absolute liquidity preference of the people. This is what Keynes called 'liquidity trap'. In Figure 7, the total demand for money D_M is the liquidity preference curve comprising of vertical MT and downward sloping MSp. At minimum rate of interest, r-min, the MSp curve is perfectly elastic.

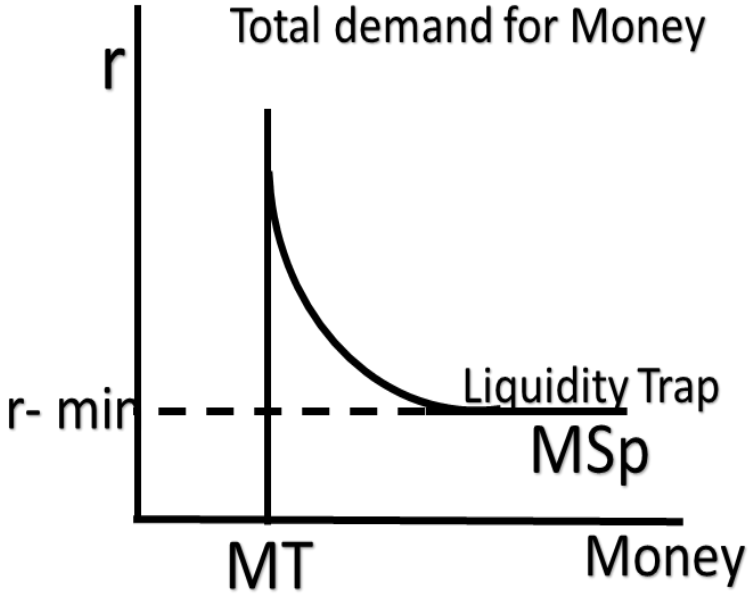


Figure 7

Money Supply:

The supply of money in a particular period depends upon the policy of the central bank of a country. Money supply curve, SM, has been drawn perfectly inelastic as it is institutionally given. (Figure 8).

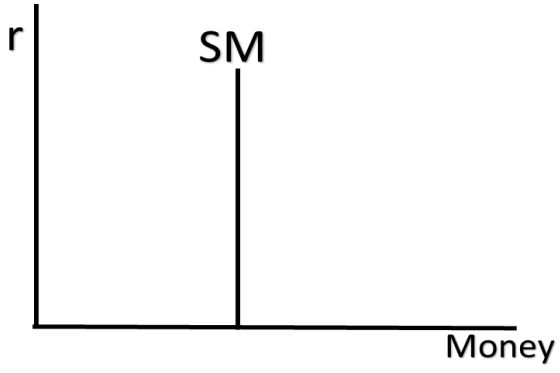


Figure 8

Determination of Interest Rate as per Liquidity Preference Theory:

According to Keynes, the rate of interest is determined by the demand for money and the supply of money as shown in Figure 9. OM is the total amount of money supplied by the central bank. At point E, demand for money becomes equal to the supply of money where the equilibrium interest rate is determined as r*. At a rate of interest is greater than this, supply of money will exceed the demand for money. People will purchase more securities. Consequently, its price will rise and interest rate will fall until demand for money becomes equal to the supply of money. On the other hand, if the rate of interest becomes less than or, demand for money will exceed supply of money, people will sell their securities. Price of securities will tumble and rate of interest will rise until we reach point E. Thus, the rate of interest is determined by the monetary variables only.

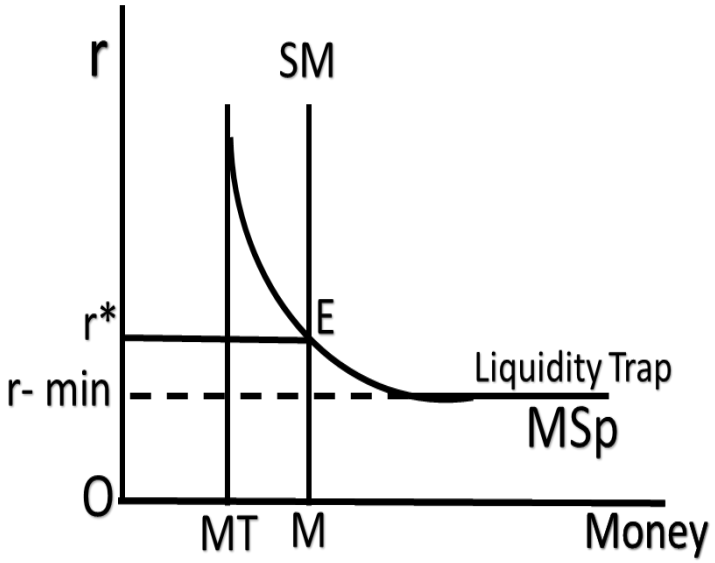


Figure 9

Limitations of the theory of Liquidity Preference

Even Keynes' liquidity preference theory is not free from criticisms:

Firstly, like the classical and neo-classical theories, Keynes' theory is an indeterminate one. Keynes charged the classical theory on the ground that it assumed the level of employment fixed.

Same criticism applies to the Keynesian theory since it assumes a given level of income. Keynes' theory suggests that Dm and SM determine the rate of interest. Without knowing the level of income we cannot know the transaction demand for money as well as the speculative demand for money. Obviously, as income changes, liquidity preference schedule changes—leading to a change in the interest rate.

Therefore, one cannot, determine the rate of interest until the level of income is known and the level of income cannot be determined until the rate of interest is known. Hence indeterminacy.

Hicks and Hansen solved this problem in their IS-LM analysis by determining simultaneously the rate of interest and the level of income.

It is indeed true also that the neo-classical authors or the pro-pounders of the loanable funds theory earlier made attempt to integrate both the real factors and the monetary factors in the interest rate determination but not with great successes. Such defects had been greatly removed by the neo-Keynesian economists—J.R. Hicks and A.H. Hansen.

Secondly, Keynes committed an error in rejecting real factors as the determinants of interest rate determination.

Thirdly, Keynes' theory gives a choice between holding risky bonds and riskless cash. An individual holds either bond or cash and never both. In the real world, it is the uncertainty or risk that induces an individual to hold both. This gap in Keynes' theory has been filled up by James Tobin. In fact, today people make a choice between a variety of assets.

Conclusion

Despite these criticisms, Keynes' liquidity preference theory tells a lot on income, output and employment of a country. His basic purpose was to demonstrate that a capitalist economy can never reach full employment due to the existence of liquidity trap.

Though the liquidity trap has been overemphasized by Keynes yet he demolished the classical conclusion the goal of full employment. Further, his theory has an important policy implication. A central bank is incapable of reviving a capitalistic economy during depression because of liquidity trap. In other words, monetary policy is useless during depressionary phase of an economy.

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Sources of Interest Rate Differentials

The major sources of interest rate differentials are: 1. Differences in Risk 2. Period of Loan 3. Volume of Loan 4. Nature of Security 5. Financial Standing of the Borrower 6. Market Imperfection 7. Variation in Demand and Supply of Money.

1. Differences in Risk:

Gross interest rates differ owing to the differences in risk and inconvenience involved, cost of maintaining accounts of borrowers, toil and trouble associated with the business of lending, etc. The greater the risk and inconvenience, the higher is the rate of interest. A Kabuliwala usually charges high rate of interest compared with banks because the former sanctions loan without asking any security.

2. Period of Loan:

Generally, longer the duration of loans the higher will be the interest rate. Loan sanctioned for a long period usually carries high interest rate since money or capital is locked for a long time. A short period loan carries a low interest rate.

3. Volume of Loan:

Rate of interest also depends on the amount of loan. Usually, a borrower pays low interest if he borrows larger amount of money. A small amount of loan may be available if a high rate of interest is paid.

4. Nature of Security:

Interest rate varies with the nature of securities offered by the borrowers. Loans against the security of gold or government bonds carry less interest since these securities can easily be converted into cash by the lenders. But loans are also sanctioned by giving securities of immovable properties, like land and house. Rate of interest against those securities will tend to be high since these are not easily converted into cash.

5. Financial Standing of the Borrower:

Rate of interest may vary due to the lender’s idea about the financial strength or creditworthiness.

A borrower with known integrity and reputation can get loans at low rates. The reverse will be the case if the credibility of the borrower is doubtful. Usually, whenever government takes loans from its own citizens, it pays low interest rate as no one casts any doubt on the ability of the government to pay back money in due time.

6. Market Imperfection:

Market imper-fections seem to be another reason for the variation in interest rates. There is variety of institutions that carry on money-lending business. For instance, banks, insurance companies, house building bank, etc., specialize in different kinds of loans charging different interest rates. A village moneylender enjoys some sort of monopoly power and, hence, charges a high rate of interest.

7. Variation in Demand and Supply of Money:

Finally, differences in interest rate may also be due to variation in the demand for money and supply of money in different markets. Nature of agricul-tural loan is different from industrial and commercial loan. Obviously, conditions of demand for and supply of money are different—thereby causing differences in interest rate.

Major Sources of Interest Rate Differentials as Observed in India

(Taken from: https://www.yourarticlelibrary.com/economics/major-sources-of-interest-rate-differentials-as-observed-in-india/40973)

Some of the major sources of interest rate differentials as observed in India are: 1. Differences in Risk of Default and Over dues, 2. Differences in the liquidity of debt, 3. Differences in term to maturity, 4. Differences in lender's cost of servicing loans, 5. Differences in lending practices and extra-loan services, 6. Differences in monopoly (or exploitative) gains and 7. Other reasons.

Wide differences in interest rates are quite common, and not the uniformity of them. What are these differences due to? As we shall see in this article, several factors are responsible for them. But monetary economists have devoted excessive attention to only one of them, namely the term to maturity of an otherwise-homogeneous debt, to the near-neglect of others.

The resulting discussion carried out under the caption of the 'term structure of interest rates' has, no doubt, offered economists an excellent opportunity for sophisticated analysis and empirical testability of competing hypotheses mainly with the help of the US data. But this excessive concentration on the term structure of interest rates has created a false impression as though the only difference in interest rates worth considering is that which may arise due to differences only in the term to maturity of debt, even though in fact differences arising from other sources may be relatively much more important.

Also, we can think of as many structures of interest rates as there are separate sources of differences in interest rates such as the default-risk-structure or the net- worth-structure of interest rates. The term-structure of interest rates is only one of these several structures which may coexist in an economy.

Further, a rate of interest can be a member of more than one structure. For example, a short-term rate is obviously a member of the term structure, but whether the debtor is the government or a private firm will determine its place in the risk structure, and the net worth of the private debtor will determine, its position in the net-worth structure, and so on.

We shall not go into any exhaustive theoretical discussion of several interest-rate structures. Instead, we offer-below a simple heuristic discussion of the major sources of interest-rate differentials as observed in India. Simultaneously we speak briefly of the associated interest-rate structure.

1. Differences in Risk of Default and Over dues:

The first consideration in loan-making is to ensure against the risk of default of the principal and the interest due. Overdues are also un-welcome, but relatively much less so than the total or partial loss of a sum due.

To guard against the risk of default some kind of primary collateral security is usually sought by the lender from the borrower. The additional point to note is that due to several reasons despite the security some amount of risk of default or over dues does survive in most cases. The security itself may turn out to be legally defective or inadequate. There are also unsecured loans.

Therefore, a part of the difference among recorded rates of interest is due to a sort of premium against risk of default and over dues. In theory, the economists talk of the pure rate of interest that is the rate of interest which does not include any risk premium. In practice, such a default-free rate is approximated by the government bond rate. It is the assumed total absence of default risk in the case of government securities that makes them gilt-edged or the securities of the highest quality.

All other securities or loans carry less or more risk of default. So, the rates of interest on them include a premium for this risk-taking. In the USA corporate bonds are given risk rating (in order of increasing risk) as AAA, AA, A, BBB, etc. by firms of security analysts. No such service exists in India.

For the government, it's 'promise to pay' is its security. The lender has full faith in the repaying capacity of the government, which is given by the government's capacity to raise resources through taxation, borrowing, and other sources. In the commercial sector, the risk of default is associated negatively with the net worth and expected profitability of the borrowing firm and positively with its debt-equity ratio.

In free (uncontrolled) credit markets this factor tends to yield an interest- rate structure which, other things being the same, makes the interest rate a declining function of the net worth of the borrower (subject to a certain minimum of the default-risk-free rate of interest). This can be called the net-worth-structure (or the default-risk-structure) of interest rates.

2. Differences in the liquidity of debt:

Liquidity of an asset refers to the degree of ease and certainty with which it can be converted into cash at short notice without any loss. Other things being the same, asset-holders or creditors generally prefer more liquid to less liquid assets. This means, that, other things being the same, a lender would be willing to charge a lower rate of interest on more liquid debt than on a less liquid debt.

For non-money financial assets or debts, the two main determi-nants of their liquidity are their (a) marketability and (b) term to maturity. There are some IOUs or financial instruments, like bonds, reference shares, treasury bills, etc. which are traded in well- organized markets. They can be easily bought and sold in these markets at short notice. In itself, this ready marketability of these (marketable) financial instruments increases their liquidity as com-pared to similar other financial instruments, which are not market-able. Such as loans and advances made by banks, moneylenders, etc.

The other factor affecting the liquidity of assets (debts) is their term to maturity. It is obvious in the case of non-marketable debts that the longer the term to maturity the less their liquidity, because a longer waiting period is required for their conversion into cash.

In the case of marketable debt what makes longer- term debt less liquid than shorter-term debt is the greater capital-uncertainly of the former than of the latter. For, it is theoretically well-

understood and empirically well-observed that the prices of long-term debt fluctuate over a wider range than do the prices of short-term debt.

3. Differences in term to maturity:

Other things being the same, rates of interest also differ according to term to maturity (time-length) of debt. The resulting structure is called the term structure of interest rates and the curve showing the relation between yield and term to maturity is called the 'yield curve'. Much theoretical and empirical work has been done on ' the term structure of government debt in the USA. This work is of little relevance to the Indian scene, because,in the Indian gilt-edged market, the treasury-bill rate is an administered rate and other rates in interest are effectively controlled by the RBI.

Corporate market debt even in the USA is considered sufficiently heterogeneous for the term-structure analysis, so that the observed differences in rates on such debt of different maturity cannot be attributed to differences in term to maturity alone.

In India, additionally, the market for this kind of debt is not well-developed; it is both narrow and shallow. Also, the institutional investors generally do not trade in securities and private dealers are small and unimportant, therefore, full importance of differences in term to maturity for market-determined interest-rate differentials in the Indian context cannot be easily assessed.

In the sphere of RBI-administered (or controlled) rates, the yield curve is upward-sloping, with the short rate (on treasury bills) pegged at 4.6% per year and the long rate averaging at about 6.6%. There is much, greater spread allowed on time deposits of banks, the rates currently varying from the low of 2.5% per year on fixed deposits of 15 to 45 days to the high of 10% per year on fixed deposits of more than 5 years.

The rates of interest offered on company deposits also have a rising structure as one move from one-year deposits to three-year deposits. Thus, in most cases, the rates of interest are higher on debts (assets) of longer-term maturity. Possibly, this is to encourage longer-term public savings or to attract longer-term funds.

The administered lending rates of financial institutions on loans and advances show a reverse term structure, as the rate of interest on term loans by development banks is generally lower than the rate of interest charged by commercial banks on their short-term commercial credit. Presumably, this policy has been adopted to encourage fixed-asset formation and to discourage excessive inventory-holding of goods in the general climate of shortages of several essential goods.

4. Differences in lender's cost of servicing loans:

Lenders incur costs on several counts in servicing loans. They have to appraise each loan application, which involves appraising income-earning prospects of the project plan for which loan is sought, various other sources of funds for the project, the adequacy of the financing arrangement, the value, marketability, and capital-certainty of the security against the loan offered by the borrower, the per-sonal integrity or honesty or the previous credit-record of the borrower, etc.

The use of the loan and the security against it have to be supervised during the currency of the loan; the loan accounts maintained; regular interest payments and loan instalments or

repayment of the principal at the maturity of the loan collected. In the case of accumulation of over dues or threatened default, special effort has to be made to recover maximum of the amount due. At that time, adequate security in the hands of the lender comes in handy as a safeguard.

Some of these costs are fixed per loan transaction, whatever the size of the loan. This gives a higher cost per rupee of loan for small loans than for large loans. Then, the problem of timely recovery of loans may be especially acute in the case of small borrowers. They may also be inconveniently located from the point of view of institutional lenders like banks, as is true of rural borrowers.

These factors increase further the cost per rupee of loan on small loans-, presumably, relatively much higher servicing cost per rupee of loan on small loans is one factor which discourages banks from going in for small loans to small borrowers in a much bigger way than they have done so far, despite the social need to expand their loan operations in favour of small borrowers.

5. Differences in lending practices and extra-loan services:

Finance companies in cities are in a position to charge higher rates of interest from their borrowers even when the latter are eligible to borrow from commercial banks at a much lower rate of interest. And yet finance companies are expanding their business. Why do such borrowers continue to borrow from finance companies than switch over to bank finance? Two reasons are generally given.

One is the informal lending practices of finance companies as compared to several formalities observed by banks in their loan-making, e.g., various kinds of disclosures about the state of business, loans outstanding from other sources, net worth, etc., strict adherence to margin requirements and length of the loan period, bank’s charge on goods purchased with the bank loan, maintenance of proper books of accounts, etc.

Some borrowers prefer to avoid all these formalities and restrictions even at the cost of paying a higher rate of interest. Then, finance companies provide fuller and more liberal finance. It is also more expeditious and assured. Rolling over from one term to another of debt is also easier. Some finance companies also provide additional trade services of various kinds to their customers. For example, they may have links with manufacturers or dealers of certain products and may offer to arrange delivery of goods from the manufacturer or dealer, arrange for insurance and other services to the customer at a small charge and also offer the necessary finance.

Thus, there is a package of special services and facilities which finance companies offer to their borrowers, which banks do not, and for these package finance companies are able to include an extra charge in the rate of interest. Besides, all the customers of finance companies do not have bankable assets with them, or are otherwise not eligible for borrowing from banks. For them, finance companies are the easiest or the cheapest source of funds. In their case, the factor discussed next (monopoly gain) also becomes a relevant factor.

6. Differences in monopoly (or exploitative) gains:

Rates of interest on loans and advances by private moneylenders, indigenous bankers, finance companies, etc. carry an element of Monopoly (or exploitative) gain as well. Usually, these rates are much higher than the rates charged on loans and advances by banks or the rates of

interest prevailing in the organized bond market. The excess of the former over the latter cannot all be easily explained in terms of the factors discussed above, namely, default risk, lower liquidity, and term to maturity and lender's cost of servicing the loan.

The small borrowers whom the private moneylenders serve usually have no alternative source of funds to turn to. In a village, a few local moneylenders may be the only source of credit. Thus, local money-lenders enjoy a position of monopoly or oligopoly. And being aware of it, they take full advantage of it.

Most small borrowers in villages and even in cities borrow under duress. Most often, with them profit calculus involving a comparison between the expected rate of return and the rate of interest on borrowed funds does not have much meaning. For, being self- employed (in agriculture, handicrafts, cottage industry, small trade, etc.), even the earning of labour income is entirely dependent on the availability of sufficient capital to carry on in their hereditary occupations. Therefore, such borrowers are in extremely weak bor-rowing position and often times submit to extortionist tactics of moneylenders.

The position of medium borrowers is slightly better, because they can borrow funds from institutional sources, and can also afford to reject potential loan offers if the rate of interest is too high in comparison with the expected rate of profit from their planned investment. Under such conditions, the only way to protect the small borrower from the high-cost finance of the moneylender is to provide him institutional finance through banks and cooperative credit societies. On the one hand, such institutions would, then, meet a growing proportion of the credit needs of the small borrower, and on the other, help lower the stranglehold of moneylenders on small borrowers.

7. Other reasons:

The lending rates of banks (whether commercial, co-operative, or other) are not uniform for all borrowers. Instead, under the policy directives of the RBI, credit is given to borrowers of priority sectors and weaker sections (under the Differential Interest Rate Scheme) at concessional rates. The same is also true of credit extended by other official credit agencies.

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Term Structure of Interest Rates

(Taken from https://investinganswers.com/dictionary/t/term-structure-interest-rates)

What is the Term Structure of Interest Rates?

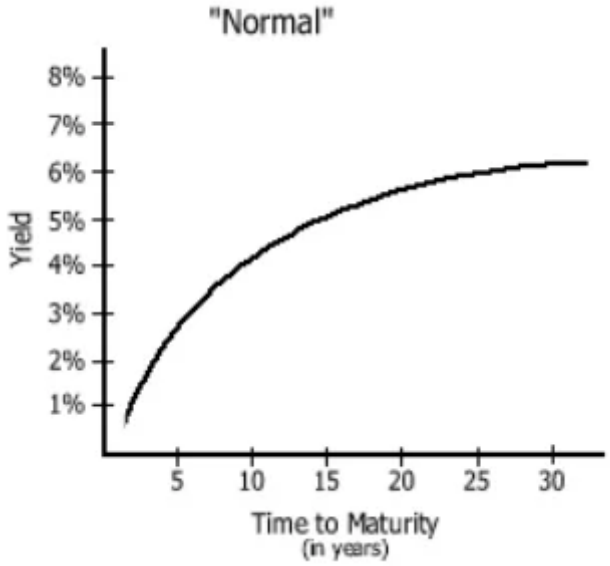
The term structure of interest rates, also called the yield curve, is a graph that plots the yields of similar-quality bonds against their maturities, from shortest to longest.

How Does the Term Structure of Interest Rates Work?

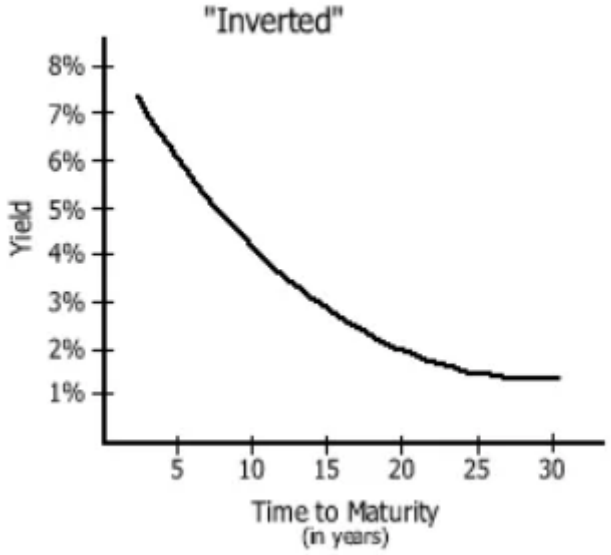
The term structure of interest rates shows the various yields that are currently being offered on bonds of different maturities. It enables investors to quickly compare the yields offered on short-term, medium-term and long-term bonds.

Note that the chart does not plot coupon rates against a range of maturities -- that graph is called the spot curve.

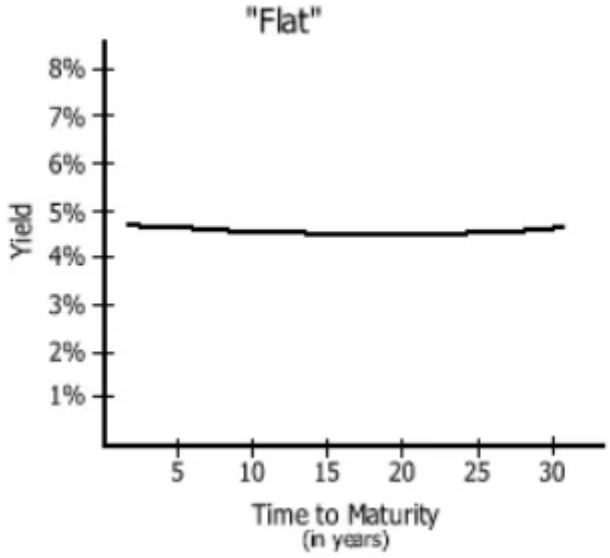
The term structure of interest rates takes three primary shapes. If short-term yields are lower than long-term yields, the curve slopes upwards and the curve is called a positive (or "normal") yield curve. Below is an example of a normal yield curve:



If short-term yields are higher than long-term yields, the curve slopes downwards and the curve is called a negative (or "inverted") yield curve. Below is example of an inverted yield curve:



Finally, a flat term structure of interest rates exists when there is little or no variation between short and long-term yield rates. Below is an example of a flat yield curve:



It is important that only bonds of similar risk are plotted on the same yield curve. The most common type of yield curve plots Treasury securities because they are considered risk-free and are thus a benchmark for determining the yield on other types of debt.

The shape of the curve changes over time. Investors who are able to predict how term structure of interest rates will change can invest accordingly and take advantage of the corresponding changes in bond prices.

Term structure of interest rates are calculated and published by The Wall Street Journal, the Federal Reserve, and a variety of other financial institutions.

Why Does the Term Structure of Interest Rates Matter?

In general, when the term structure of interest rates curve is positive, this indicates that investors desire a higher rate of return for taking the increased risk of lending their money for a longer time period.

Many economists also believe that a steep positive curve means that investors expect strong future economic growth with higher future inflation (and thus higher interest rates), and that a sharply inverted curve means that investors expect sluggish economic growth with lower future inflation (and thus lower interest rates). A flat curve generally indicates that investors are unsure about future economic growth and inflation.

Theories of Term Structure of Interest Rates

There are three central theories that attempt to explain why yield curves are shaped the way they are

1. The "expectations theory" says that expectations of increasing short-term interest rates are what create a normal curve (and vice versa).

2. The "liquidity preference hypothesis" says that investors always prefer the higher liquidity of short-term debt and therefore any deviance from a normal curve will only prove to be a temporary phenomenon.

3. The "segmented market hypothesis" says that different investors adhere to specific maturity segments. This means that the term structure of interest rates is a reflection of prevailing investment policies.

Because the term structure of interest rates is generally indicative of future interest rates, which are indicative of an economy's expansion or contraction, yield curves and changes in these curves can provide a great deal of information. In the 1990s, Duke University professor Campbell Harvey found that inverted yield curves have preceded the last five U.S. recessions.

Changes in the shape of the term structure of interest rates can also have an impact on portfolio returns by making some bonds relatively more or less valuable compared to other bonds. These concepts are part of what motivate analysts and investors to study the term structure of interest rates carefully.

Expectations Theory

What is Expectations Theory?

Expectations theory suggests that the forward rates in current long-term bonds are closely related to the bond market's expectation about future short-term interest rates.

How Does Expectations Theory Work?

Expectations theory attempts to explain the term structure of interest rates. There are three main types of expectations theories: pure expectations theory, liquidity preference theory and preferred habitat theory.

Expectations theories are predicated upon the idea that investors believe forward rates, as reflected (and some would say predicted) by future contracts are indicative of future short-term interest rates.

For example, if 3 months from today you want to buy a 6-month T-bill, you would look at the forward rate on the 6-month T-bill to see what its expected yield is projected to be in 3 months. Let's assume the forward rate is 1% for that specific T-bill. In this case, expectations theory would suggest that the 6-month interest rate 3 months from today will be 1%.

Why Does Expectations Theory Matter?

Investors make decisions partially based upon where they foresee the future level of interest rates. Expectations theory implies that long-term investors will choose to purchase or not to purchase debt instruments based on whether forward interest rates are more or less favorable than current short-term interest rates.

Liquidity Preference Hypotheses

What is Liquidity?

Liquidity is the ability to sell an investment at or near its value in a relatively short period of time.

How Does Liquidity Work?

Let’s say you take an old painting from the attic to the local filming of Antiques Roadshow. The expert says your painting is worth $50,000. Surprise!

That’s great news, except that it could take months to find a buyer, and the buyer may only want to pay $35,000 or $40,000. Your painting, while valuable, isn’t very liquid. That is, you can’t convert it to $50,000 very quickly or easily. Houses aren’t very liquid, either. They too can take months to sell, and buyers often don’t pay the sticker price.

Why Does Liquidity Matter?

Liquidity is a factor of supply and demand for a security. But it is also affected by the size of the original issue and the time since the original issue -- the smaller the number of securities out there or the longer the securities have been out there, the less liquid they tend to be.

Most people consider the size of the bid/ask spread as indicative of a security's liquidity -- the larger the spread, the less liquid (and thus the riskier) the security is. For example, let’s assume you are watching Company XYZ stock. If the bid price is $50 and the ask price is $51.50, then the bid-ask spread is $1.50. This spread may be high or low depending on what the spread typically is for Company XYZ stock. An increasing spread denotes increasing liquidity risk, and vice versa. In the worst-case scenario, liquidity risk makes it possible that the investor could take a loss if he or she has to sell the investment quickly.

All investments have liquidity risk. This is important to understand, because liquidity risk can compound other problems for investors. For example, if the investor is unable to liquidate his or her position, this may keep him from meeting debt obligations (that is, the liquidity risk increases the investor's credit risk). Buy-and-hold investors face less liquidity risk because they

are generally not interested in buying and selling securities quickly. This is particularly true for buy-and-hold bond investors, who are simply waiting for their bonds to mature and are not concerned with interim price movements.

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Unit 4

Banking System

This Unit covers the following topics:

- Balance Sheet of Commercial Bank
- Portfolio Management by a Commercial Bank
- Changing Role of Indian Banking System
- Structure of Indian Banking System
- Banking Sector Reforms in India

4.1: Balance Sheet and Portfolio Management

(A) Balance Sheet of Commercial Bank

(Taken from https://www.yourarticlelibrary.com/banking/commercial-bank-the-balance-sheet-of-a-commercial-bank-banking/10984)

The balance sheet of a commercial bank provides a picture of its functioning. It is a statement which shows its assets and liabilities on a particular date at the end of one year.

The assets are shown on the right- hand side and the liabilities on the left-hand side of the balance sheet. As in the case of a company, the assets and liabilities of a bank must balance. The balance sheet which every commercial bank in India is required to publish once in a year is shown as under:

We analyse the distribution of assets and liabilities of a commercial bank on the basis of the division given in the above Table.

The Distribution of Assets:

The assets of a bank are those items from which it receives income and profit. The first item on the assets side is the cash in liquid form consisting of coins and currency notes lying in reserve with it and in its branches. This is a certain percentage of its total liabilities which it is required to keep by law. Cash reserves do not yield income to the bank but are essential to satisfy the claims of its depositors.

The second item is in the form of balances with the central bank and other banks. The commercial banks are required to keep a certain percentage of their time and demand deposits with the central bank. They are the assets of the bank because it can withdraw from them in cash in case of emergency or when the seasonal demand for cash is high.

The third item, money at call and short notice, relates to very short-term loans advanced to bill brokers, discount houses and acceptance houses. They are repayable on demand within fifteen days. The banks charge low rate of interest on these loans. The fourth item of assets relates to bills discounted and purchased.

The bank earns profit by discounting bills of exchange and treasury bills of 90 days duration. Some bills of exchange are accepted by a commercial bank on behalf of its customers which i ultimately purchases. They are a liability but they are included under assets because the bank can get them rediscounted from the central bank in case of need.

The fifth item, investments by the bank in government securities, state bonds and industrial shares, yields a fixed income to the banks. The bank can sell its securities when there is need for more cash. The sixth item relating to loans and advances is the most profitable source of bank assets as the bank changes interest at a rate higher than the bank rate.

The bank makes advances on the basis of cash credits and overdrafts and loans on the basis of recognised securities. In the seventh item are included liabilities of the bank's customers which the bank has accepted and endorsed on their behalf. They are the assets of the bank because the liabilities of customers remain in the custody of the bank.

The bank charges a nominal commission for all acceptances and endorsements which is a source of income. The eighth item relates to the value of permanent assets of the bank in the form of property, furniture, fixtures, etc. They are shown in the balance sheet after allowing for depreciation every year. The last item includes profits retained by the bank after paying corporation tax and profits to shareholders.

The Distribution of Liabilities:

The liabilities of commercial banks are claims on it. These are the items which form the sources of its funds. Of the liabilities, the share capital of the bank is the first item which is contributed by its shareholders and is a liability to them. The second item is the reserve fund. It consists of accumulated resources which are meant to meet contingencies such as losses in any year.

The bank is required to keep a certain percentage of its annual profits in the reserve fund. The reserve fund is also a liability to the shareholders. The third item compresses both the time and demand deposits. Deposits are the debts of the bank to its customers.

They are the main source from which the bank gets funds for investment and are indirectly the source of its income. By keeping a certain percentage of its time and demand deposits in cash the bank lends the remaining amount on interest. Borrowings from other banks are the fourth item.

The bank usually borrows secured and unsecured loans from the central bank. Secured loans are on the basis of some recognised securities, and unsecured loans out of its reserve funds lying with the central bank. The fifth item bills payable refer to the bills which the bank pays out of its resources. The sixth items relates to bills for collection.

These are the bills of exchange which the bank collects on behalf of its customers and credits the amount to their accounts. Hence it is a liability to the bank. The seventh item is the acceptance and endorsement of bills of exchange by the bank on behalf of its customers. These are the claims on the bank which it has to meet when the bills mature.

The eighth item contingents liabilities relate to those claims on the bank which are unforeseen such as outstanding forward exchange contracts, claims on acknowledge debts, etc. In the last item, profit and loss, are shown profits payable to the shareholders which are a liability on the bank.

The various items of the balance sheet shown in Table 1 are a rough indicator of the assets and liabilities of commercial banks. The balance sheet of a particular bank showed its financial soundness. By studying the balance sheets of the major commercial banks of a country, one can also know the trend of the monetary market. "The bank balance sheet reflects bank credit

extension on its asset side in loans and investments, and on the liabilities side reflects the bank's operations as an intermediary in time deposits and its role as an element in the nation's monetary system in demand deposits."

Table 1. Form of Balance Sheet:

Liabilities	Assets
1. Share Capital	1. Cash
2. Reserve Fund	2. Balances with the Central Bank and other banks.
3. Deposits	3. Money at call and Short-Notice
4. Borrowings from other banks	4. Bills Discounted and Purchased.
5. Bills Payable	5. Investments
6. Bills for collection	6. Loans, Advances, Cash Credits and Overdrafts.
7. Acceptances. Endorsements and other obligations	7. Liabilities of Customer for Acceptances. Endorsements and other Obligations.
8. Contingent Liabilities	8. Property, Furniture, Fixtures less Depreciation
9. Profit and Loss	9. Profit and Loss.

(B) Portfolio Management of a Commercial Bank

The main aim of a commercial bank is to seek profit like any other institution. Its capacity to earn profit depends upon its investment policy. Its investment policy, in turn, depends on the manner in which it manages its investment portfolio.

Thus "commercial bank investment policy emerges from a straight forward application of the theory of portfolio management to the particular circumstances of commercial bank." Portfolio management refers to the prudent management of a bank's assets and liabilities in order to seek some optimum combination of income or profit, liquidity, and safety.

When a bank operates, it acquires and disposes of income-earning assets. These assets plus the bank's cash make up what is known as its portfolio. A bank's earning assets consist of (a) securities issued by the central and state governments, local bodies and government institutions, and (b) financial obligations, such as promissory notes, bills of exchange, etc. issues by firms. There earning assets constitute between one- fourth and one-third of a commercial bank's total assets. Thus a bank's earning assets are an important source of its income.

The manner in which banks manage their portfolios, that is acquiring and disposing of their earning assets, can have important affects on the financial markets, on the borrowing and spending practices of households and businesses, and on the economy as a whole.

Objectives of Portfolio Management:

There are three main objectives of portfolio management which a wise bank follows: liquidity, safety and income. The three objectives are opposed to each other. To achieve on the bank will have to sacrifice the other objectives. For example, if the banks seek high profit, it may have to sacrifice some safety and liquidity. If it seeks more safety and liquidity it may have to give up some income. We analyse these objectives one by one in relation to the other objectives.

1. Liquidity:

A commercial bank needs a higher degree of liquidity in its assets. The liquidity of assets refers to the ease and certainty with which it can be turned into cash. The liabilities of a bank are large in relation to its assets because it holds a small proportion of its assets in cash. But its liabilities are payable on demand at a short notice.

Therefore, the bank must hold a sufficiently large proportion of its assets in the form of cash and liquid assets for the purpose of profitability. If the bank keeps liquidity the uppermost, its profit will below. On the other hands, if it ignores liquidity and aims at earning more, It will be disastrous for it. Thus in managing its investment portfolio a bank must strike a balance between the objectives of liquidity and profitability. The balance must be achieved with a relatively high degree of safety. This is because banks are subject to a number of restrictions that limit the size of earning assets they can acquire.

The nature of conflict between liquidity and profitability is illustrated in earning assets are taken on the horizontal axis and cash on the vertical axis. CF is the investment possibility line which shows all combinations of cash and earning assets.

For instance, point A denotes a combination of OM of cash and OS of earning assets; and point B shows ON of cash and OT of earning assets. Each bank seeks to obtain its optimum point along the line CE which will be a combination of cash and earning assets so as to achieve the highest possible level of earnings consistent with its liquidity and safety.

Many types of assets are available to a commercial bank with varying degrees of liquidity. The most liquid of assets is money in cash. The next most liquid assets are deposits with the central bank, treasury bills and other short-term bills issues by the central and state governments and large firms, and call loans to other banks, firms, dealers and brokers in government securities.

The less liquid assets are the various types of loans to customers and investments in long term bonds and mortgages. Thus the principle sources of liquidity of a bank are its borrowings from the other banks and the central bank and from the sales of the assets.

But the amount of liquidity which the bank can have depends on the availability and cost of borrowings. If it can borrow large amounts at any time without difficulty at a low cost (interest rate), it willhod very little liquid assets. But if it is uncertain to borrow funds or the cost of borrowing is high, the bank will keep more liquid assets in its portfolio.

2. Safety:

A commercial bank always operates under conditions of uncertainty and risk. It is uncertain about the amount and cost of funds it can acquire and about its income in the future. Moreover, it face two types of risks. The first is the market risk which results from the decline in the prices of debt obligations when the market rate of interest rises. The second is the risk by default

where the bank fears that the debtors are not likely to repay the principle and pay the interest in time. “This risk is largely concentrated in customer loans, where banks have a special function to perform, and bank loans to businesses and bank mortgage loans are among the high grade loans of these types.”

In the light of these risks, a commercial bank has to maintain the safety of its assets. It is also prohibited by law to assume large risks because it is required to keep a high ration of its fixed liabilities to its total assets with itself and also with the central bank in the form of cash. But if the bank follows the safety principle strictly by holding only the safest assets it will not be able to create more credit.

It will thus lose customers to other banks and its income will also be very low. One the other hand, if the bank takes too much risk, it may be highly harmful for it. Therefore, a commercial bank “must estimate the amount of risks attached to the various types of available assets, compare estimated risk differentials, consider both long-turn and short- run consequences, and strike a balance.”

3. Profitability:

One of the principle objectives of a bank is to earn more profit. It is essential for the purpose of paying interest to depositors, wage to the staff, dividend to shareholders and meeting other expenses. It cannot afford to hold a large amount of funds in cash for that will mean forgoing income. But the conflict between profitability and liquidity is not very sharp. Liquidity and safety are primary considerations while profitability is subsidiary for the very existence of a bank depends on the first two.

Conclusion:

The three conflicting objectives of portfolio management lead to the conclusion that for a bank to earn more profit, it must strike a judicious balance between liquidity and safety.

Theories of Portfolio Management:

There are apparent conflicts between the objectives of liquidity, safety and profitability relating to a commercial bank. Economists have tried to resolve these conflicts by laying down certain theories from time to time. These principles or theories, in fact, govern the distribution of assets keeping in views these objectives. They have also come to be known as the theories of liquidity management which are discussed as under.

1. The Real Bills Doctrine:

The real bills doctrine or the commercial loan theory states that a commercial bank should advance only short-term self-liquidating productive loans to business firms. Self-liquidating loans are those which are meant to finance the production, and movement of goods through the successive stages of production, storage, transportation, and distribution.

When such goods are ultimately sold, the loans are considered to liquidate themselves automatically. For instance, a loan give by the bank to a businessman to finance inventories would be repaid out of the receipts from the sale of those very inventories, and the loan would be automatically self-liquidated.

The theory states that when commercial banks make only short term self-liquidating productive loans, the central bank, in turn, should only land to the banks on the security of such short-term loans. This principle would ensure the proper degree of liquidity for each bank and the proper money, supply for the whole economy.

The central bank was expected to increase or diminish bank reserves by rediscounting approved loans. When business expanded and the needs of trade increased, banks were able to acquire additional reserves by rediscounting bills with the central banks. When business fell and the needs of trade declined, the volume of rediscounting of bills would fall, the supply of bank reserves and the amount of bank credit and money would also contract.

Merits:

Such short-term self-liquidating productive loans possess three advantages. First, they possess liquidity that is why they liquidate themselves automatically. Second, since they mature in the short run and are for productive purposes, there is no risk of their running to bad debts. Third, being productive such loans earn income for the banks.

Demerits:

Despite these merits, the real bills doctrine suffers from certain defects.

First, if a bank refuses to grant a fresh loan till the old loan is repaid, the disappointed borrower will have to reduce production which will adversely affect business activity. If all the banks follow the same rule, this may lead to reduction in the money supply and price in the community. This may, in turn, make it impossible for existing debtors to repay their loans in time.

Second, the doctrine assumes that loans are self-liquidating under normal economic conditions. If there is depression, production and trade suffer and the debtor will not be able to repay the debt at maturity.

Third, this doctrine neglects the fact that the liquidity of a bank depends on the saleability of its liquid assets and not on real trade bills. If a bank possesses a variety of assets like bills and securities which can be readily should in the money and capital markets, it can ensure safety, liquidity and profitability. Then the bank need not rely on maturitis in time of trouble.

Fourth, the basic defect of the theory is that no loan is in itself automatically self-liquidating. A loan to a retailer to purchase inventor is not self-liquidating if the inventories are not sold to consumers and remain with the retailer. Thus a loan to be successful involves a third party, the consumers in this case, besides the lender and the borrower.

Fifth, this theory is based on the “needs of trade” which is no longer accepted as an adequate criterion for regulating this type of bank credit. If bank credit and money supply fluctuate on the basis of the needs of trade, the central bank cannot prevent either spiralling recession or inflation.

2. The Shiftability Theory:

The shiftability theory of bank liquidity was propounded by H.G. Moulton who asserted that if the commercial banks maintain a substantial amount of assets that can be shifted on to the other

banks for cash without material loss in case of necessity, then there is no need to rely on maturities.

According to this view, an asset to be perfectly shiftable must be immediately transferable without capital loss when the need for liquidity arises. This is particularly applicable to short term market investments, such as treasury bills and bills of exchange which can be immediately sold whenever it is necessary to raise funds by banks. But in a general crisis when all banks are in need of liquidity, the shiftability theory requires that all banks should possess such assets which can be shifted on to the central bank which is the lender of the last resort.

This theory has certain elements of truth. Banks now accept sound assets which can be shifted on to other banks. Shares and debentures of large companies are accepted as liquid assets along with treasury bills and bills of exchange. This has encouraged term lending by banks.

Demerits:

But it has its weaknesses. First, mere shiftability of assets does not provide liquidity to the banking system. It entirely depends upon the economic circumstances. Second, the shiftability theory ignores the fact that in times of acute depression, the shares and debentures cannot be shifted on to others by the banks. In such a situation, there are no buyers and all who possess them want to sell them. Third, a single bank may have shiftable assets in sufficient quantities but if it tries to sell them when there is a run on the bank, it may adversely affect the entire banking system, fourth, If all the banks simultaneously start shifting their assets, it would have disastrous effects on both the lenders and borrowers.

3. The Anticipated Income Theory:

The anticipated income theory was developed by H.V. Prochanow in 1944 on the basis of the practice of extending term loans by the US commercial banks. According to this theory, regardless of the nature and character of a borrower's business, the bank plans the liquidation of the term-loan from the anticipated income of the borrower. A term-loan is for a period exceeding one year and extending to less than five years.

It is granted against the hypothecation of machinery, stock and even immovable property. The bank puts restrictions on the financial activities of the borrower while granting this loan. At the time of granting a loan, the bank takes into consideration not only the security but the anticipated earnings of the borrower. Thus a loan by the bank gets repaid out of the future income of the borrower in instalments, instead of in a lump sum at the maturity of the loan.

Merits:

This theory is superior to the real bills doctrine and the shiftability theory because it fulfills the three objectives of liquidity, safety and profitability. Liquidity is assured to the bank when the borrower saves and repays the loan regularly in instalments. It satisfies the safety principle because the bank grants a loan not only on the basis of a good security but also on the ability of the borrower to repay the loan. The bank can utilise its excess reserves in granting term-loan and is assured of a regular income. Lastly, the term-loan is highly beneficial for the business community which gets funds for medium-terms.

Demerits:

The theory of anticipated income is not free from a few defects.

i. Analyses Creditworthiness:

It is not a theory but simply a method to analyse a borrower's creditworthiness. It gives the bank criteria for evaluating the potential of a borrower to successfully repay a loan on- time.

ii. Fails to Meet Emergency Cash Needs:

Repayment of loans in instalments to the bank no doubt provides a regular stream of liquidity, but they fail to meet emergency cash needs of the lender bank.

4. The Liabilities Management Theory:

This theory was developed in the 1960s. According to this theory, there is no need for banks to grant self-liquidating loans and keep liquid assets because they can borrow reserve money in the money market in case of need. A bank can acquire reserves by creating additional liabilities against itself from different sources. These sources include the issuing of time certificates of deposit, borrowing from other commercial banks, borrowing from the central banks, raising of capital funds by issuing shares, and by ploughing back of profits. We discuss these sources of bank funds briefly.

(a) Time Certificates of Deposits:

These are the principal sources of reserve money for a commercial bank in the USA. Time certificates of deposits are of different maturities ranging from 90 days to less than 12 months. They are negotiable in the money market. So a bank can have access to liquidity by selling them in the money market. But there are two limitations.

First, if during a boom, the interest rate structure in the money market is higher than the ceiling rate set by the central bank, time deposit certificates cannot be sold in the market. Second, they are not a dependable source of funds for the commercial banks. Bigger commercial banks are at an advantage in selling these certificates because they have large certificates which they can afford to sell at even low interest rates. So the smaller banks are at a disadvantage in this respect.

(b) Borrowing from other Commercial Banks:

A bank may create additional liabilities by borrowing from other banks having excess reserves. But such borrowings are only for a very short duration, for a day or week at the most. The interest rate of such borrowings depends upon the prevailing rate in the money market. But borrowings from other banks are only possible during normal economic conditions. In abnormal times, no bank can afford to lend to others.

(c) Borrowing from the Central Bank:

Banks also create liabilities on themselves by borrowing from the central bank of the country. They borrow to meet their liquidity needs for short term and by discounting bills from the central bank. But such borrowings are relatively costlier than borrowings from other sources.

(d) Raising Capital Funds:

Commercial banks acquire funds by issuing fresh shares or debentures. But the availability of funds through these sources depends on the amount of dividend or interest rate which the bank is prepared to pay. Usually, the banks are not in a position to pay rates higher than paid by

manufacturing and trading companies. So they are not able to get sufficient funds from this sources.

(e) Ploughing Back Profits:

Another source of liquid funds for a commercial bank is the ploughing back of its profits. But how much it can get from this source will depend upon its rate of profit and its dividend policy. It is the larger banks that can depend on this source rather than the smaller banks.

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4.2: Indian Banking System

(A) Changing Role and Structure of Indian Banking System

(Taken from https://kalyan-city.blogspot.com/2012/04/changing-role-of-banks-in-india-since.html)

Changing Role of Banks in India

The role of banks in India has changed a lot since economic reforms of 1991. These changes came due to LPG, i.e. liberalization, privatization and globalization policy being followed by GOI. Since then most traditional and outdated concepts, practices, procedures and methods of banking have changed significantly. Today, banks in India have become more customer-focused and service-oriented than they were before 1991. They now also give a lot of importance to their rural customers. They are even willing ready to help them and serve regularly the banking needs of country-side India.

The changing role of banks in India can be glanced in points depicted below.

The following points briefly highlight the changing role of banks in India.

1. Better customer service,
2. Mobile banking facility,
3. Bank on wheels scheme,
4. Portfolio management,
5. Issue of electro-magnetic cards,
6. Universal banking,
7. Automated teller machine (ATM),
8. Internet banking,
9. Encouragement to bank amalgamation,
10. Encouragement to personal loans,
11. Marketing of mutual funds,
12. Social banking, etc.

The above-mentioned points indicate the role of banks in India is changing. Now let's discuss how banking in India is getting much better day after day.

1. Better Customer Service

Before 1991, the overall service of banks in India was very poor. There were very long queues (lines) to receive payment for cheques and to deposit money. In those days, some bank staffs were very rude to their customers. However, all this changed remarkably after Indian economic reforms of 1991.

Banks in India have now become very customer and service focus. Their service has become quick, efficient and customer-friendly. This positive change is mostly due to rising competition from new private banks and initiation of Ombudsman Scheme by RBI.

2. Mobile Banking

Under mobile banking service, customers can easily carry out major banking transactions by simply using their cell phones or mobiles.

Here, first a customer needs to activate this service by contacting his bank. Generally, bank officer asks the customer to fill a simple form to register (authorize) his mobile number. After registration, this service is activated, and the customer is provided with a username and password. Using secret credentials and registered phone, customer can now comfortably and securely, find his bank balance, transfer money from his account to another, ask for a cheque book, stop payment of a cheque, etc.

Today, almost all banks in India provide a mobile-banking service.

3. Bank on Wheels

The 'Bank on Wheels' scheme was introduced in the North-East Region of India. Under this scheme, banking services are made accessible to people staying in the far-flung (remote) areas of India. This scheme is a generous attempt to serve banking needs of rural India.

4. Portfolio Management

In portfolio management, banks do all the investments work of their clients.

Banks invest their clients' money in shares, debentures, fixed deposits, etc. They first enter a contract with their clients and charge them a fee for this service. Then they have the full power to invest or disinvest their clients' money. However, they have to give safety and profit to their clients.

5. Issue of Electro-Magnetic Cards

Banks in India have already started issuing Electro-Magnetic Cards to their customers. These cards help to carry out cash-less transactions, make an online purchase, avail ATM facility, book a railway ticket, etc.

Banks issue many types of electro-magnetic cards, which are as follows:

Credit cards help customers to spend money (loaned up to a certain limit as previously settled by the bank) which they don't have in hand. They get a monthly statement of their purchases and withdrawals. Along with the transacted amount, this statement also includes the interest and service fee. The entire amount (as reflected in the statement of credit card) must be paid back to the bank either fully or in installments, but before due date.

Debit cards help customers to spend that money which they have saved (credited) in their individual bank accounts. They need not carry cash but instead can use a debit card to make a purchase (for shopping) and/or withdraw money (get cash) from an ATM. No interest is charged on the usage of debit cards.

Charge cards are used to spend money up to a certain limit for a month. At the end of the month, customer gets a statement. If he has a sufficient balance, then he only had to pay a small fee. However, if he doesn't have a necessary balance, he is given a grace period (which is generally of 25 to 50 days) to repay the money.

Smart cards are currently being used as an alternative to avail public transport services. In India, this covers Railways, State Transport and City (Local) Buses. Smart card has an integrated circuit (IC) embedded in its plastic body. It is made as per norms specified by ISO.

Kisan credit cards are used for the benefit of the rural population of India. The Indian farmers (kisans) can use this card to buy agricultural inputs and goods for self-consumption. These cards are issued by both Commercial and Co-operative banks.

6. Universal Banking

In India, the concept of universal banking has gained recognition after year 2000. The customers can get all banking and non-banking services under one roof. Universal bank is like a super store. It offers a wide range of services, including banking and other financial services like insurance, merchant banking, etc.

7. Automated Teller Machine (ATM)

There are many advantages of ATM. As a result, many banks have opened up ATM centres to offer convenience to their customers. Now banks are operating ATM centres not only in their branches but also at public places like airports, railway stations, hotels, etc. Some banks have joined together and agreed upon to set up common ATM centres all over India.

8. Internet Banking

Internet banking is also called as an E-banking or net banking. Here, the customer can do banking transactions through the medium of the internet or world wide web (WWW). The customer need not visit the bank's branch. Through this facility, the customer can easily inquiry about bank balance, transfer funds, request for a cheque book, etc. Most large banks offer this service to their tech-savvy customers.

9. Encouragement to Bank Amalgamation

Failure of banks is well-protected with the facility of amalgamation. So depositors need not worry about their deposits. When weaker banks are absorbed by stronger banks, it is called amalgamation of banks.

10. Encouragement to Personal Loans

Today, the purchasing power of Indian consumers has increased dramatically because banks give them easy personal loans. Generally, interest charged by the banks on such loans is very high. Interest is calculated on reducing balance. Large banks offer loans up to a huge amount like one crore. Some banks even organise Loan Mela (Fair) where a loan is sanctioned on the spot to deserving candidates after they submit proper documents.

11. Marketing of Mutual Funds

A mutual fund collects money from many investors and invests the money in shares, bonds, short-term money market instruments, gold assets; etc. Mutual funds earn income by interest and dividend or both from its investments. It pays a dividend to subscribers. The rate of dividend fluctuates with the income on mutual fund investments. Now banks have started selling these funds in their own names. These funds are not insured like other bank deposits. There are different types of funds such as open-ended funds, closed-ended funds, growth funds, balanced funds, income funds, etc.

12. Social Banking

The government uses the banking system to alleviate poverty and unemployment. Many social development programmes are initiated by the banks from time to time. The success of these programmes depends on financial support provided by the banks. Banks supply a lot of finance to farmers, artisans, scheduled castes (SC) and scheduled trib**e** (ST) families, unemployed youth and people living below the poverty line (BPL).

(B) Structure of Indian Banking System

(Taken from https://cdn.yourarticlelibrary.com/wp-content/uploads/2014/01/image118.png)

Bank is an institution that accepts deposits of money from the public. Anybody who has account in the bank can withdraw money. Bank also lends money. India has banks operating in both unorganised and organised sectors.

Banking in Unorganised Sector: This is also called Indigenous Banking.

Indigenous Banking:

The exact date of existence of indigenous bank is not known. But, it is certain that the old banking system has been functioning for centuries. Some people trace the presence of indigenous banks to the Vedic times of 2000-1400 BC. It has admirably fulfilled the needs of the country in the past.

However, with the coming of the British, its decline started. Despite the fast growth of modern commercial banks, however, the indigenous banks continue to hold a prominent position in the Indian money market even in the present times. It includes shroffs, seths, mahajans, chettis, etc. The indigenous bankers lend money; act as money changers and finance internal trade of India by means of hundis or internal bills of exchange.

Defects of Indigenous Banking

The main defects of indigenous banking are:

(i) They are unorganised and do not have any contact with other sections of the banking world.

(ii) They combine banking with trading and commission business and thus have introduced trade risks into their banking business.

(iii) They do not distinguish between short term and long term finance and also between the purpose of finance.

(iv) They follow vernacular methods of keeping accounts. They do not give receipts in most cases and interest which they charge is out of proportion to the rate of interest charged by other banking institutions in the country.

Suggestions for Improvements:

(i) The banking practices need to be upgraded.

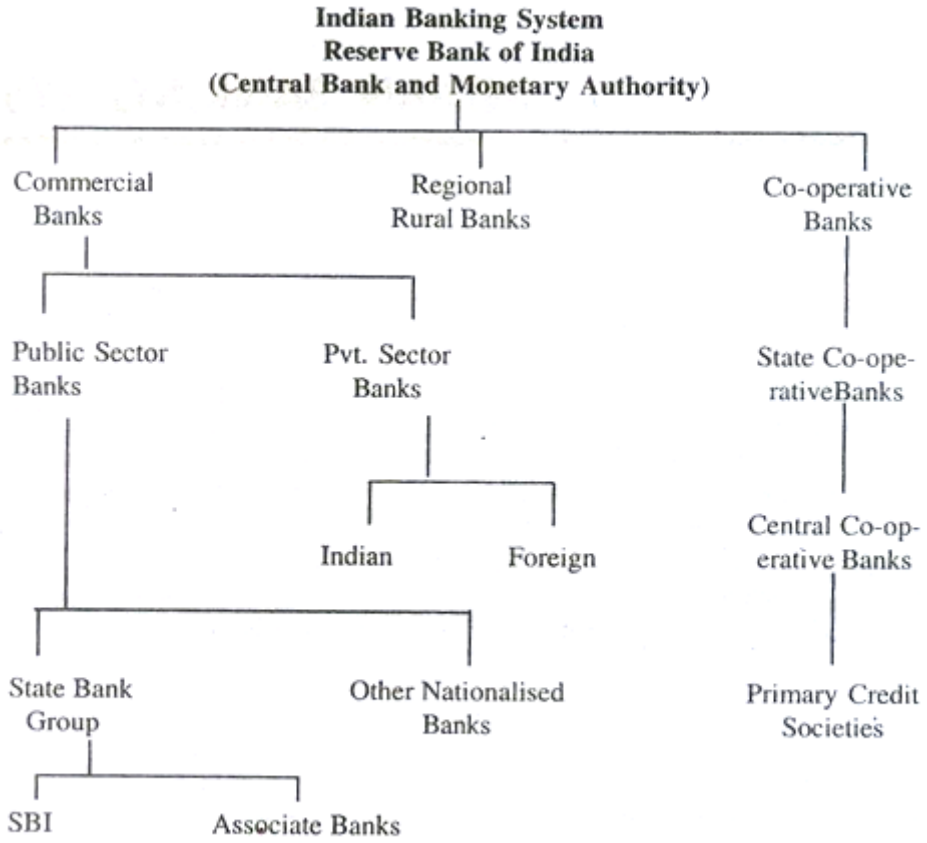
(ii) Encouraging them to avail of certain facilities from the banking system, including the RBI.

(iii) These banks should be linked with commercial banks on the basis of certain understanding in the respect of interest charged from the borrowers, the verification of the same by the commercial banks and the passing of the concessions to the priority sectors etc.

(iv) These banks should be encouraged to become corporate bodies rather than continuing as family based enterprises.

Structure of Organised Indian Banking System

The organised banking system in India can be classified as given below:



STRUCTURE OF INDIAN BANKING SYSTEM

Reserve Bank of India (RBI):

The country had no central bank prior to the establishment of the RBI. The RBI is the supreme monetary and banking authority in the country and controls the banking system in India. It is called the Reserve Bank' as it keeps the reserves of all commercial banks.

Commercial Banks:

Commercial banks mobilise savings of general public and make them available to large and small industrial and trading units mainly for working capital requirements.

Commercial banks in India are largely Indian-public sector and private sector with a few foreign banks. The public sector banks account for more than 92 percent of the entire banking business in India—occupying a dominant position in the commercial banking. The State Bank of India and its 7 associate banks along with another 19 banks are the public sector banks.

Scheduled and Non-Scheduled Banks:

The scheduled banks are those which are enshrined in the second schedule of the RBI Act, 1934. These banks have a paid-up capital and reserves of an aggregate value of not less than Rs. 5 lakhs, hey have to satisfy the RBI that their affairs are carried out in the interest of their depositors.

All commercial banks (Indian and foreign), regional rural banks, and state cooperative banks are scheduled banks. Non- scheduled banks are those which are not included in the second schedule of the RBI Act, 1934. At present these are only three such banks in the country.

Regional Rural Banks:

The Regional Rural Banks (RRBs) the newest form of banks, came into existence in the middle of 1970s (sponsored by individual nationalised commercial banks) with the objective of developing rural economy by providing credit and deposit facilities for agriculture and other productive activities of al kinds in rural areas.

The emphasis is on providing such facilities to small and marginal farmers, agricultural labourers, rural artisans and other small entrepreneurs in rural areas.

Other special features of these banks are:

(i) their area of operation is limited to a specified region, comprising one or more districts in any state; (ii) their lending rates cannot be higher than the prevailing lending rates of cooperative credit societies in any particular state; (iii) the paid-up capital of each rural bank is Rs. 25 lakh, 50 percent of which has been contributed by the Central Government, 15 percent by State Government and 35 percent by sponsoring public sector commercial banks which are also responsible for actual setting up of the RRBs.

These banks are helped by higher-level agencies: the sponsoring banks lend them funds and advise and train their senior staff, the NABARD (National Bank for Agriculture and Rural Development) gives them short-term and medium, term loans: the RBI has kept CRR (Cash Reserve Requirements) of them at 3% and SLR (Statutory Liquidity Requirement) at 25% of their total net liabilities, whereas for other commercial banks the required minimum ratios have been varied over time.

Cooperative Banks:

Cooperative banks are so-called because they are organised under the provisions of the Cooperative Credit Societies Act of the states. The major beneficiary of the Cooperative Banking is the agricultural sector in particular and the rural sector in general.

The cooperative credit institutions operating in the country are mainly of two kinds: agricultural (dominant) and non-agricultural. There are two separate cooperative agencies for the provision of agricultural credit: one for short and medium-term credit, and the other for long-term credit. The former has three tier and federal structure.

At the apex is the State Co-operative Bank (SCB) (cooperation being a state subject in India), at the intermediate (district) level are the Central Cooperative Banks (CCBs) and at the village level are Primary Agricultural Credit Societies (PACs).

Long-term agriculture credit is provided by the Land Development Banks. The funds of the RBI meant for the agriculture sector actually pass through SCBs and CCBs. Originally based in rural sector, the cooperative credit movement has now spread to urban areas also and there are many urban cooperative banks coming under SCBs.

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(C) Banking Sector Reforms in India

The banking sector is the heart of all the economic activity of a country and a small change in its regulation affects the entire economy. The banks are the institutions that impinge on the economy and affect their performance for better or worse. The banking system helps in

- Capital accumulation
- Growth by encouraging savings
- Mobilising the capital
- Allocating the capital for alternative uses, etc.

History of Banking Sector Reforms in India

Modern banking in India started way back to 1786, with the establishment of the General Bank of India. In 1806, the East India Company established the first Presidency Bank in Kolkata.

Two more banks were established in 1840 and 1843 named Bank of Bombay and Bank of Madras. Reserve Bank of India (RBI) came into formation on April 1, 1935, with the enactment 50 of the Reserve Bank of India Act, 1934.

The objective of establishing the Reserve Bank, as stated in the preamble to the RBI Act, was to "regulate the issue of banknotes and the keeping of reserves with a view of securing monetary stability in India".

Post- Independence

Even after independence, the banks were mainly urban-oriented and were beyond the reach of the rural population. A large section of the rural population still had to look upon the moneylenders as their resort for credit.

That's why the government decided to nationalize all the major banks in India. The first Nationalization took place in 1969 and the second one in 1985.

Reasons behind the Banking Reforms in India

The main reason behind the banking sector reforms in India is as follows.

The Indian economy witnessed a series of difficulties like uncertain political situation, persistent fiscal imbalance, double-digit inflation, the balance of payments crisis, etc.

The fiscal situation, which was under strain throughout the 1980s, reached a critical situation in 1990-91, the external payment crisis and the high rate of inflation both, reached their peak level in the middle of 1991.

Growth of real GDP decelerated partly because of lower industrial growth and partly because of the slowdown in agriculture.

The industries were affected because of lower government investment, non-availability of inputs due to import squeeze, recession prevailed in the industrial economy due to the collapse of demand in the markets of Kuwait and Iraq in the wake of Gulf crisis, and the collapse of the erstwhile Soviet Union.

Objectives of Banking Sector Reforms in India

Banking sector reforms started with the objective to improve the overall performance of the Indian banking sector. The various objectives of banking sector reforms in India are as follows.

- Reforms were aimed at bringing a transformation change in the structure, efficiency and stability of the banking system, and also integration with the international markets.
- To make Indian banks, internationally competitive and encourage them to play an effective role in accelerating the process of growth.
- The reforms in the banking sector in India intended to enhance the stability and efficiency of banks
- To remove the operational rigidities in the credit delivery system to ensure allocation efficiency and achievement of social objectives. To place the Indian banking system on par with international standards in respect of capital adequacy and other prudential norms.
- The strengthening measures aimed at reducing the vulnerability of banks in the face of fluctuations in the economic environment. These included capital adequacy, income recognition, asset classification, provisioning norms, exposure norms, improved levels of transparency, and disclosure standards.

First Phase of Reforms – The Narasimham Committee I

To rebuild the financial health of commercial banks and to make their functioning efficient and profitable, the Government of India appointed a High-Level Committee. The name of the committee was “The Committee on Financial System” (CFS) under the Chairmanship of M. Narasimham.

It was a committee of nine members along with Mr. M, Narasimham. The committee gave its recommendation in Nov 1991 which was the blueprint of the first-generation banking sector reforms in India. The objectives of the committee were given below

To make recommendations for improving and modernising the organisational systems and procedures as well as managerial policies

Make recommendations for infusing greater competitive viability into the system so as to enable the banks and financial institution to respond more effectively to the emerging credit needs of the economy

To examine the cost, composition and adequacy of the capital structure of the various financial institutions and to make suitable recommendations in this regard

To review the relative roles of different types of financial institutions in the financial system and to make recommendations for their balanced growth.

Many of the recommendations of the committee have been accepted and implemented by the Government of India.

Improvement in Financial Health: The first necessary step was to improve the financial health of banks. These measures aimed at reducing the vulnerability of banks in the face of

fluctuations in the economic environment. These included the introduction of prudential norms more or less in keeping with international thinking.

Transparency on Financial Statement: The Committee was of the view that the balance sheets of banks and financial institutions should be made transparent and full disclosures are made in the balance sheets as required by the International Accounting Standards Committee. In conformity with this recommendation, RBI modified the format of balance sheets of banks in 1992.

Institutional Strengthening: Institutional framework conducive to the development of banks needs to be developed so, an important aspect of banking sector reform was to strengthen the institutional base of the banking system. These included a variety of measures such as the licensing of new banks in private sector, enabling the public sector banks to go to the market and augment their capital base, creation of Debt Recovery Tribunals to deal with loans owed to the commercial banks.

Asset Reconstruction Fund: The Committee suggested the setting up of an ARF to take over bad and doubtful assets of the balance sheets of the banks and DFIs at a discount so that the banks could recycle the funds realised through this process into new productive assets. The rate of discount will be determined by independent auditors on the basis of clearly stipulated guidelines.

Second Phase of Reforms – Narasimham Committee-II

In 1998, the Government set up Committee on Banking Sector Reforms in India under the chairmanship of M. Narasimham in order to review the progress of banking sector reforms to date and chart a programme of financial sector reforms necessary to strengthen the Indian Financial System and make it internationally competitive.

The Committee recommended that by 2000 the entire portfolio of government securities should be marked to the market and there should be 5% weight for market risk for government and approved securities which was zero earlier. The report also mentioned that the risk weight for a government guaranteed advance should be the same as for other advances.

The benefits of the Second Phase of Banking Sector Reforms India by Narasimham committee recommendations are as follows.

Deregulation of Branch Licensing: With the Narasimham Committee's recommendations governing branch licensing restrictions, the RBI changed its licensing policy in 1992 in order to give banks the operational autonomy to rationalise their branch networks.

The Committee recommended that branch licensing be abolished and the matter of opening branches or closing of branches (other than branches for the present) be left to the commercial judgments of the individual banks. Banks were allowed to shift their existing branches within the same locality, open certain types of specialised branches, convert existing nonviable rural branches into satellite offices, a spin-off the business of a branch, and open extension counters and administrative units without prior approval of the RBI.

Prudential Norms and Disclosure Requirements: With regard to income recognition, the Committee recommended the introduction of the norm of 90 days i.e. income stops accruing when interest on or instalment of principal is not paid within 90 days, in a phased manner by

the year 2002, which was 180 days previously. It also suggested for a general provision on standard assets which was not there previously. As far as the future loans were concerned, prudential norms as income recognition asset classification and provisioning norms should be applied to government guaranteed advances in the same manner as for any other advances. The Banking Sector Reforms in India helped to transform the Indian economy.

Capital Adequacy: Taking the present financial scenario of the economy into account, the Committee recommended that market risk which is defined as the risk of losses with respect to on and off-balance sheet positions arising from movements in market prices, should be given greater attention.

The report suggested that RBI should work towards implementing the amendment to the Basel norms which is the standard measurement method which uses a building block approach in which specific risk and 86 general market risk arising from debt and equity positions are calculated separately.

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Unit 5

Central Banking and Monetary Policy

This Unit covers the following topics:

- Definition and Functions of Central Bank
- Balance Sheet of Central Bank
- Monetary Policy: Goals/Objectives and Instruments
- Current Monetary Policy in India
- Monetary Policy in the Open Economy

Definition and Functions of Central Bank

(https://www.economicsdiscussion.net/banks/central-bank-and-its-functions/4165)

A central bank plays an important role in monetary and banking system of a country. It is responsible for maintaining financial sovereignty and economic stability of a country, especially in underdeveloped countries.

"A Central Bank is the bank in any country to which has been entrusted the duty of regulating the volume of currency and credit in that country"-Bank of International Settlement. It issues currency, regulates money supply, and controls different interest rates in a country. Apart from this, the central bank controls and regulates the activities of all commercial banks in a country. According to Bank of International Settlement, "A Central Bank is the bank in any country to which has been entrusted the duty of regulating the volume of currency and credit in that country."

Bank of England was the world's first effective central bank that was established in 1694. As per the resolution passed in Brussels Financial Conference, 1920, all the countries should establish a central bank for interest of world cooperation. Thus, since 1920, central banks are formed in almost every country of the world. In India, RBI operates as a central bank.

Functions of Central Bank:

The central bank does not deal with the general public directly. It performs its functions with the help of commercial banks. The central bank is accountable for protecting the financial stability and economic development of a country.

Apart from this, the central bank also plays a significant part in avoiding the cyclical fluctuations by controlling money supply in the market. As per the view of Hawtrey, a central bank should primarily be the "lender of last resort." On the other hand, Kisch and Elkins believed that "the maintenance of the stability of the monetary standard" as the essential function of central bank. The functions of central bank are broadly divided into two parts, namely, traditional functions and developmental functions.

(a) Traditional Functions:

Refer to functions that are common to all central banks in the world. The traditional functions of the central bank include the following:

(i) Bank of issue:

Possesses an exclusive right to issue notes (currency) in every country of the world. In the initial years of banking, every bank enjoyed the right of issuing notes. However, this led to a number of problems, such as notes were over-issued and the currency system became disorganized. Therefore, the governments of different countries authorized central banks to issue notes. The issue of notes by one bank has led to uniformity in note circulation and balance in money supply.

(ii) Government's banker, agent, and advisor:

Implies that a central bank performs different functions for the government. As a banker, the central bank performs banking functions for the government as commercial banks performs for the public by accepting the government deposits and granting loans to the government. As an agent, the central bank manages the public debt, undertakes the payment of interest on this debt, and provides all other services related to the debt.

As an advisor, the central bank gives advice to the government regarding economic policy matters, money market, capital market, and government loans. Apart from this, the central bank formulates and implements fiscal and monetary policies to regulate the supply of money in the market and control inflation.

(iii) Custodian of cash reserves of commercial banks:

Implies that the central bank takes care of the cash reserves of commercial banks. Commercial banks are required to keep certain amount of public deposits as cash reserve, with the central bank, and other part is kept with commercial banks themselves.

The percentage of cash reserves is deeded by the central bank! A certain part of these reserves is kept with the central bank for the purpose of granting loans to commercial banks Therefore, the central bank is also called banker's bank.

(iv) Custodian of international currency:

Implies that the central bank maintains a minimum reserve of international currency. The main aim of this reserve is to meet emergency requirements of foreign exchange and overcome adverse requirements of deficit in balance of payments.

(v) Bank of rediscount:

Serve the cash requirements of individuals and businesses by rediscounting the bills of exchange through commercial banks. This is an indirect way of lending money to commercial banks by the central bank. Discounting a bill of exchange implies acquiring the bill by purchasing it for the sum less than its face value.

Rediscounting implies discounting a bill of exchange that was previously discounted. When owners of bill of exchange are in need of cash they approach the commercial bank to discount these bills. If commercial banks are themselves in need of cash they approach the central bank to rediscount the bills.

(vi) Lender of last resort:

Refer to the most crucial function of the central bank. The central bank also lends money to commercial banks. Instead of rediscounting of bills, the central bank provides loans against treasury bills, government securities, and bills of exchange.

(vii) Bank of central clearance, settlement, and transfer:

Implies that the central bank helps in settling mutual indebtness between commercial banks. Depositors of banks give checks and demand drafts drawn on other banks. In such a case, it is not possible for banks to approach each other for clearance, settlement, or transfer of deposits.

The central bank makes this process easy by setting a clearing house under it. The clearing house acts as an institution where mutual indebtedness between banks is settled. The representatives of different banks meet in the clearing house to settle inter-bank payments. This helps the central bank to know the liquidity state of the commercial banks.

(viii) Controller of Credit:

Implies that the central bank has power to regulate the credit creation by commercial banks. The credit creation depends upon the amount of deposits, cash reserves, and rate of interest given by commercial banks. All these are directly or indirectly controlled by the central bank. For instance, the central bank can influence the deposits of commercial banks by performing open market operations and making changes in CRR to control various economic conditions.

(b) Developmental Functions:

This Refers to the functions that are related to the promotion of banking system and economic development of the country. These are not compulsory functions of the central bank. These are discussed as follows:

(i) Developing specialized financial institutions:

Refer to the primary functions of the central bank for the economic development of a country. The central bank establishes institutions that serve credit requirements of the agriculture sector and other rural businesses.

Some of these financial institutions include Industrial Development Bank of India (IDBI) and National Bank for Agriculture and Rural Development (NABARD). These are called specialized institutions as they serve the specific sectors of the economy.

(ii) Influencing money market and capital market:

Implies that central bank helps in controlling the financial markets Money market deals in short term credit and capital market deals in long term credit. The central bank maintains the country’s economic growth by controlling the activities of these markets.

(iii) Collecting statistical data:

Gathers and analyzes data related to banking, currency, and foreign exchange position of a country. The data is quite helpful for researchers, policymakers, and economists. For instance, the Reserve Bank of India publishes a magazine called Reserve Bank of India

Bulletin, whose data is useful for formulating different policies and making macro-level decisions.

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Balance Sheet of Central Bank

A central bank’s balance sheet summarizes its financial position, and is made up of assets, liabilities and equity. Assets equal liabilities plus equity. In contrast to a corporation, currency in circulation (cash) is a liability for a central bank. Through the purchase of any asset, financial or not, a central bank may increase its balance sheet at no cost, simply by creating new money (increasing its currency in circulation liability by the exact amount it increases its assets).

Traditionally, money was created by either minting coins or printing currency. Nowadays, most money is stored electronically as account information, so money can be created or destroyed simply by changing the information in the accounts. Before 1900, sovereign governments were in charge of minting coins or printing currency — sometimes with disastrous results.

Today, the supply of money is managed by central banks, not to satisfy the whims of politicians, but to achieve specific well-established objectives, such as low inflation, maximum growth, or high employment. Money is usually created — or destroyed — electronically as information in accounts held by central banks. The creation or destruction of money is recorded in the central bank's balance sheet. Therefore, to understand the supply of money, one must understand how it is recorded in the bank's balance sheet.

A central bank's balance sheet, like most balance sheets, is divided into assets and liabilities. The central bank's balance sheet can also be divided further into assets and liabilities as the bankers' bank and assets and liabilities as the government's bank, as shown in the following table:

Central Bank Balance Sheet		
	Assets	Liabilities
Bankers' Bank	Loans	Bank Accounts
Government's Bank	Securities	Currency
	Foreign Exchange Reserves	Government's Account

To simplify this discussion, we will focus on the supply of money by the Federal Reserve (Fed) of the United States and its balance sheet. Although the Fed's balance sheet is rather complicated, only the main components are necessary to understand the money supply process. Central bank assets include:

securities, mainly in the form of Treasuries; foreign exchange reserves, which are mainly held in the form of foreign bonds issued by foreign governments; and loans to commercial banks.

Of these, the most important asset is securities, which the Fed uses to directly control the supply of money in the United States. In other countries, where exports are important, such as China, federal exchange reserves may be the dominant asset.

Central bank liabilities include: currency, which is held by the public, federal government's bank account, which the federal bank uses just as anyone would use their own checking account, depositing its revenues, mostly in the form of tax revenues, into its account, and paying its bills, mostly in electronic format; commercial bank accounts, otherwise known as reserves, where commercial banks keep their deposits with the Fed. Vault cash, which is cash held in the banks' vaults, is also part of the commercial banks' reserves, because the cash is used to service its customers.

Reserves can be further classified as either required reserves or excess reserves. Required reserves are reserves banks must hold as a legal minimum to ensure their financial soundness while excess reserves is the amount exceeding the required reserves, which banks keep to conduct their daily business or because they failed to lend it.

Because changes in the supply of money are revealed in the central bank's balance sheet, the balance sheet is the most important item that the central bank discloses. The Federal Reserve and most other central banks publish their balance sheets weekly as a way to maintain transparency. When a central bank fails to publish its balance sheet, it often indicates trouble, usually in the form of increasing the supply of money at the behest of politicians.

Monetary Base

The quantity of money in any economy is determined by the monetary base, which are the banking reserves and currency held by the public. In other words, the monetary base consists of the actual quantity of money. However, because money also has velocity, in that the same dollar is used in multiple transactions over time, the monetary base is often called high-powered money because the total value of all financial transactions is a multiple of the monetary base.

Only a central bank can control its balance sheet at will, since only a central bank can create or destroy money. Because of the relationship between the supply of money and the bank's balance sheet, the creation of money is sometimes called expanding the central bank's balance sheet, because both its assets and liabilities increase; likewise, the destruction of money causes the contraction of the central bank's balance sheet.

Example: Balance Sheet of Reserve Bank of India

RESERVE BANK OF INDIA BALANCE SHEET AS ON JUNE 30, 2019 (Amount in Rs. Billion)	
Liabilities	Assets
Capital	Assets of Banking Department (BD)
Reserve Fund	Notes, Rupee Coin, Small Coin
Other Reserves	Gold Coin and Bullion
Deposits	Investments-Foreign-BD
Other Liabilities and Provisions	Investments-Domestic-BD
	Bills Purchased and Discounted
	Loans and Advances
	Investment in Subsidiaries
	Other Assets
Liabilities of Issue Department	Assets of Issue Department (ID)
Notes issued	Gold Coin and Bullion (as backing for Note issue)
	Rupee Coin
	Investments-Foreign-ID
	Investments-Domestic-ID
	Domestic Bills of Exchange and other Commercial Papers
Total Liabilities	Total Assets

Monetary Control or Monetary Policy

What Is Monetary Policy?

Monetary policy, the demand side of economic policy, refers to the actions undertaken by a nation's central bank to control money supply and achieve macroeconomic goals that promote sustainable economic growth. Monetary policy is the process of drafting, announcing, and implementing the plan of actions taken by the central bank, currency board, or other competent monetary authority of a country that controls the quantity of money in an economy and the channels by which new money is supplied.

Monetary policy consists of the management of money supply and interest rates, aimed at meeting macroeconomic objectives such as controlling inflation, consumption, growth, and liquidity. This is achieved by actions such as modifying the interest rate, buying or selling government bonds, regulating foreign exchange (forex) rates, and changing the amount of money banks are required to maintain as reserves.

Economists, analysts, investors, and financial experts across the globe eagerly await monetary policy reports and the outcome of meetings involving monetary policy decision-makers. Such developments have a long-lasting impact on the overall economy, as well as on specific industry sectors or markets.

Monetary policy is formulated based on inputs gathered from a variety of sources. For instance, the monetary authority may look at macroeconomic numbers such as gross domestic product (GDP) and inflation, industry/sector-specific growth rates and associated figures, as well as geopolitical developments in international markets—including oil embargos or trade tariffs. These entities may also ponder concerns raised by groups representing industries and businesses, survey results from organizations of repute, and inputs from the government and other credible sources.

Monetary policy can be broadly classified as either expansionary or contractionary.

Expansionary

If a country is facing a high unemployment rate during a slowdown or a recession, the monetary authority can opt for an expansionary policy aimed at increasing economic growth and expanding economic activity. As a part of expansionary monetary policy, the monetary authority often lowers the interest rates through various measures, serving to promote spending and make money-saving relatively unfavorable.

Increased money supply in the market aims to boost investment and consumer spending. Lower interest rates mean that businesses and individuals can secure loans on convenient terms to expand productive activities and spend more on big-ticket consumer goods. An example of this expansionary approach is the low to zero interest rates maintained by many leading economies across the globe since the 2008 financial crisis.

Contractionary

Increased money supply can lead to higher inflation, raising the cost of living and cost of doing business. Contractionary monetary policy, increasing interest rates, and slowing the growth of

the money supply, aims to bring down inflation. This can slow economic growth and increase unemployment, but is often necessary to cool down the economy and keep it in check.

In the early 1980s when inflation hit record highs and was hovering in the double-digit range of around 15%, the Fed raised its benchmark interest rate to a record 20%. Though the high rates resulted in a recession, it managed to bring back inflation to the desired range of 3% to 4% over the next few years.

Goals or Objectives of Monetary Policy

the objective of monetary policy varies from country to country and from time to time, a brief description of the same has been as following:

(i) Neutrality of money

(ii) Stability of exchange rates

(iii) Price stability

(iv) Full Employment

(v) Economic Growth

(vi) Equilibrium in the Balance of Payments.

1. Neutrality of Money:

Economists like Wicksteed, Hayek and Robertson are the chief exponents of neutral money. They hold the view that monetary authority should aim at neutrality of money in the economy. Any monetary change is the root cause of all economic fluctuations. According to neutralists, the monetary change causes distortion and disturbances in the proper operation of the economic system of the country.

They are of the confirmed view that if somehow neutral monetary policy is followed, there will be no cyclical fluctuations, no trade cycle, no inflation and no deflation in the economy. Under this system, money is kept stable by the monetary authority. Thus the main aim of the monetary authority is not to deviate from the neutrality of money. It means that quantity of money should be perfectly stable. It is not expected to influence or discourage consumption and production in the economy.

2. Exchange Stability:

Exchange stability was the traditional objective of monetary authority. This was the main objective under Gold Standard among different countries. When there was disequilibrium in the balance of payments of the country, it was automatically corrected by movements. It was popularly known, “Expand Currency and Credit when gold is coming in; contract currency and credit when gold is going out.” This system will correct the disequilibrium in the balance of payments and exchange stability will be maintained.

It must be noted that if there is instability in the exchange rates, it would result in outflow or inflow of gold resulting in unfavorable balance of payments. Therefore, stable exchange rates play a key role in international trade. Thus, it is clear from this fact that: the main objective of monetary policy is to maintain stability in the external equilibrium of the country. In other words, they should try to eliminate those adverse forces which tend to bring instability in exchange rates.

(i) It leads to violent fluctuations resulting in encouragement to speculative activities in the market.

(ii) Heavy fluctuations lead to loss of confidence on the part of domestic and foreign capitalists resulting in adverse impact in capital outflow which may also result in capital formation and growth.

(iii) Fluctuations in exchange rates bring repercussions in the internal price level.

3. Price Stability:

The objective of price stability has been highlighted during the twenties and thirties of the present century. In fact, economists like Crustar Cassels and Keynes suggested price stabilization as a main objective of monetary policy. Price stability is considered the most genuine objective of monetary policy. Stable prices repose public confidence because cyclical fluctuations are totally eliminated.

It promotes business activity and ensures equitable distribution of income and wealth. As a consequence, there is general wave of prosperity and welfare in the community. Price stability also impedes economic progress as there is no incentive left with the business community to increase production of qualitative goods.

It discourages exports and encourages imports. But it is admitted that price stability does not mean ‘price rigidity’ or price stagnation’. A mild increase in the price level provides a tonic for economic growth. It keeps all virtues of a stable price.

4. Full Employment:

During world depression, the problem of unemployment had increased rapidly. It was regarded as socially dangerous, economically wasteful and morally deplorable. Thus, full employment assumed as the main goal of monetary policy. In recent times, it is argued that the achievement of full employment automatically includes prices and exchange stability.

However, with the publication of Keynes’ General Theory of Employment, Interest and Money in 1936, the objective of full employment gained full support as the chief objective of monetary policy. Prof. Crowther is of the view that the main objective of monetary policy of a country is to bring about equilibrium between saving and investment at full employment level.

Similarly, Prof. Halm has also favoured Keynes’ view. Prof. Gardner Ackley regards that the concept of full employment is ‘slippery’. Classical economists believed in the existence of full employment which is the normal feature of an economy. Full employment, thus, exists when all those who are ready to work at the existing wage rate get work. Voluntary, frictional and seasonal unemployed are also called employed.

According to their version, full employment means absence of involuntary unemployment. Therefore, it implies not only employment of all types of labourers but also includes the employment of all economic resources. It is not an end in itself rather a pre-condition for maximum social and economic welfare.

Keynes equation of income, Y = C + I throws light as to how full employment can be secured with monetary policy. He argues that to increase income, output and employment, it is necessary to increase consumption expenditure and investment expenditure simultaneously. This indirectly solves the problem of unemployment in the economy. Since the consumption function is more or less stable in the short period, the monetary policy should aim at raising investment expenditure.

As monetary policy is the government policy regarding currency and credit, in this way, government measures of currency and credit can easily overcome the problem of trade fluctuations in the economy. On the other side, when the economy is facing the problem of depression and unemployment, private investment can be stimulated by adopting 'cheap money policy' by the monetary authority.

Therefore, this policy will serve as an effective and ideal stimulant to private investment as there is pessimism all round in the economy. Further, the objective of full-employment must be integrated with other objectives, like price and exchange stabilization.

The advanced countries like U.S.A. and U.K. are normally working at full employment level as their main concern is how to maintain full employment and avoid fluctuations in the level of employment and production. While, on the contrary, the main problem in underdeveloped country is as to how to achieve full employment.

Therefore, in such economies, monetary policy can be designed to meet with the problem of under employment and disguised unemployment and by further creating new opportunities for employment. The most suitable and favourable monetary policy should be followed to promote full-employment through increased investment, which in turn having multiplier and acceleration effects.

After achieving the objective of full-employment, monetary policy should aim at exchange and price stability. In short, the policy of full employment has the far-reaching beneficial effects.

(a) Keeping in view the present situation of unemployment and disguised unemployment particularly in more growing populated countries, the said objective of monetary policy is most suitable.

(b) On humanitarian grounds, the policy can go a long way to solve the acute problem of unemployment.

(c) It is useful tool to provide economic and social welfare of the community.

(d) To a greater extent, this policy solves the problem of business fluctuations.

5. Economic Growth:

In recent years, economic growth is the basic issue to be discussed among economists and statesmen throughout the world. Prof. Meier defined "Economic growth as the process whereby the real per capita income of a country increases over a long period of time." It implies an

increase in the total physical or real output, production of goods for the satisfaction of human wants.

In other words, it means utilization of all the productive natural, human and capital resources in such a manner as to ensure a sustained increase in national and per capita income over time.

Therefore, monetary policy promotes sustained and continuous economic growth by maintaining equilibrium between the total demand for money and total production capacity and further creating favourable conditions for saving and investment. For bringing equality between demand and supply, flexible monetary policy is the best course.

In other words, monetary authority should follow an easy or tight monetary policy to suit the requirements of growth. Again, monetary policy in a growing economy, has to satisfy the growing demand for money. Thus, it is the responsibility of the monetary authority to circulate the proper quantity and quality of money.

6. Equilibrium in the Balance of Payments:

Equilibrium in the balance of payments is another objective of monetary policy which emerged significant in the post war years. This is simply due to the problem of international liquidity on account of the growth of world trade at a more faster speed than the world liquidity.

It was felt that increasing of deficit in the balance of payments reduces, the ability of an economy to achieve other objectives. As a result, many less developed countries have to curtail their imports which adversely effects development activities. Therefore, monetary authority makes efforts that equilibrium should be maintained in the balance of payments.

Instruments of Monetary Policy:

The instruments of monetary policy are of two types: first, quantitative, general or indirect; and second, qualitative, selective or direct. They affect the level of aggregate demand through the supply of money, cost of money and availability of credit. Of the two types of instruments, the first category includes bank rate variations, open market operations and changing reserve requirements. They are meant to regulate the overall level of credit in the economy through commercial banks. The selective credit controls aim at controlling specific types of credit. They include changing margin requirements and regulation of consumer credit. We discuss them as under:

Bank Rate Policy:

The bank rate is the minimum lending rate of the central bank at which it rediscounts first class bills of exchange and government securities held by the commercial banks. When the central bank finds that inflationary pressures have started emerging within the economy, it raises the bank rate. Borrowing from the central bank becomes costly and commercial banks borrow less from it.

The commercial banks, in turn, raise their lending rates to the business community and borrowers borrow less from the commercial banks. There is contraction of credit and prices are checked from rising further. On the contrary, when prices are depressed, the central bank lowers the bank rate.

It is cheap to borrow from the central bank on the part of commercial banks. The latter also lower their lending rates. Businessmen are encouraged to borrow more. Investment is encouraged. Output, employment, income and demand start rising and the downward movement of prices is checked.

Open Market Operations:

Open market operations refer to sale and purchase of securities in the money market by the central bank. When prices are rising and there is need to control them, the central bank sells securities. The reserves of commercial banks are reduced and they are not in a position to lend more to the business community.

Further investment is discouraged and the rise in prices is checked. Contrariwise, when recessionary forces start in the economy, the central bank buys securities. The reserves of commercial banks are raised. They lend more. Investment, output, employment, income and demand rise and fall in price is checked.

Changes in Reserve Ratios:

This weapon was suggested by Keynes in his Treatise on Money and the USA was the first to adopt it as a monetary device. Every bank is required by law to keep a certain percentage of its total deposits in the form of a reserve fund in its vaults and also a certain percentage with the central bank.

When prices are rising, the central bank raises the reserve ratio. Banks are required to keep more with the central bank. Their reserves are reduced and they lend less. The volume of investment, output and employment are adversely affected. In the opposite case, when the reserve ratio is lowered, the reserves of commercial banks are raised. They lend more and the economic activity is favourably affected.

Selective Credit Controls:

Selective credit controls are used to influence specific types of credit for particular purposes. They usually take the form of changing margin requirements to control speculative activities within the economy. When there is brisk speculative activity in the economy or in particular sectors in certain commodities and prices start rising, the central bank raises the margin requirement on them.

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Current Monetary Policy in India

(Taken from https://indianexpress.com/article/business/economy/rbi-reserve-bank-of-india-monetary-policy-committee-february-2021-meeting-outcome-key-announcements-7175322/)

RBI Monetary Policy 2021 announcements: The six-member Monetary Policy Committee (MPC) headed by Reserve Bank of India (RBI) Governor Shaktikanta Das kept the repo rate unchanged at 4 per cent, while the reverse repo rate also was kept unchanged at 3.35 per cent. Here's what the Indian central bank announced.

RBI Monetary Policy 2021: The Reserve Bank of India's (RBI) Monetary Policy Committee (MPC) kept its repo rate unchanged at 4 per cent while maintaining an 'accommodative stance' as long as necessary at least through the current financial year to the next year, RBI Governor Shaktikanta Das announced on Friday.

The RBI governor announced that the decision was taken unanimously and added that the reverse repo rate too was kept unchanged at 3.35 per cent.

The central bank had slashed the repo rate by 115 basis points since late March 2020 to support growth. This is the fourth time in a row that the MPC decided to keep the policy rate unchanged. The RBI had last revised its policy rate on May 22 in an off-policy cycle to perk up demand by cutting interest rates to a historic low. The central bank also sees FY22 GDP growth at 10.5 per cent. The RBI governor said that the inflation has eased below the level of 6 per cent. The outlook on growth has also improved significantly. He also said that the MPC judged that need for the hour is to continue supporting the growth. He added that the signs of recovery have strengthened further and list of normalising sectors is expanding.

This is the first MPC meeting after the presentation of the Union Budget 2021-22. The six-member MPC headed by RBI Governor Shaktikanta Das meets every two months to analyze the state of the Indian economy and inflation and address the monetary issues in the country. This month, it began the 3-day bi-monthly meeting on Wednesday, February 3. Mr. Das said that the CPI projection is revised to 5.2 per cent for Q4 FY21 and CPI inflation is pegged at 5-5.2 per cent in H1 FY22.

The RBI Governor said "capacity utilisation in the manufacturing sector improved to 63.3 per cent in Q2 vs 47.3 per cent in Q1. FDI and FPI investments have surged in recent months, reposing faith in the Indian economy." Speaking on non-banking financial companies (NBFCs), Das said that the funds from banks though the TLTRO scheme will now available to NBFCs. He also said that the cash reserve ratio (CRR) will be restored in two phases to 3.5 per cent from March 27 and 4 per cent from May 22, 2021. The RBI governor also said that the CRR normalisation will open up space for a variety of market operations. The Indian central bank chief announced that the retail investors can now open gilt accounts with the RBI. Das also said that the retail investors can now access the primary and secondary government bond market. He also said that the resident individuals will be able to make remittances to IFSCs for the NRIs.

Shaktikanta Das announced an integrated ombudsman scheme for customer grievance redressal, which will be rolled out by June 2021. Concluding his address, Das said that the Indian economy is poised to move only in one direction, that is upward. Separately, in a press

conference post the monetary policy announcement, Shaktikanta Das said that the central bank is awaiting a formal proposal from the government on the proposed Asset Reconstruction Company (ARC) for the management of NPAs. The government had proposed the creation of an ARC during the recent Budget 2021-22. “Government and RBI have discussed the idea of a bad bank. We will examine the formal proposal on the ARC once it is made,” he said.

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Monetary Policy in the Open Economy

Monetary policy has international implications as well. Changes in interest rates lead to changes in supply and demand in the foreign exchange market. In turn, changes in exchange rates affect exports and imports and influence the overall demand for goods and services. Among other things, this means that the monetary policy of other countries will have an effect on your own country. So, if you live in India, you are not immune to Federal Open Market Committee (FOMC) actions. And if you live in the United States, you are not immune to the actions of the European Central Bank (ECB).

The Monetary Transmission Mechanism in the Open Economy

The key element in the monetary transmission mechanism is the ability of the central bank to influence the real interest rate. Changes in real interest rates lead to changes in spending on durable goods, which are a component of aggregate expenditures. But there is also another channel of influence. If the RBI cuts interest rates, for example, then the demand for INR to invest in Indian asset markets will be reduced. This will reduce the foreign currency price of INR. The weaker INR means that goods produced in India are cheaper, so Indian exports will increase, and Indian imports will decrease. Thus, changes in interest rates lead to changes in exchange rates, which in turn lead to changes in net exports. Net exports are also a component of aggregate expenditures. Here is an additional channel of the monetary transmission mechanism that operates through the exchange rate. Changes in interest rates lead to changes in exchange rates, which in turn lead to changes in net exports. This channel reinforces the effect operating through interest rates.

Even when we include this channel, it is just as easy to understand the monetary transmission mechanism as it was before. When interest rates are cut, there is an increase both in spending on durables and net exports. Both channels lead to higher aggregate spending and thus higher output.

Monetary Policy in the Rest of the World

The United States or India do not exist alone in the world economy. US and Indian financial markets are influenced by events in other countries, such as the actions of the ECB. Likewise, citizens in Europe are influenced by monetary policy in the United States.

Suppose the ECB cuts interest rates in Europe. As in the United States, the typical mechanism for this would be a purchase of debt issued by European governments. An increase in the price of this debt is equivalent to a decrease in interest rates. If nothing else happens, this decrease in European interest rates gives rise to an arbitrage opportunity. Investors want to move funds to the United States to take advantage of the higher interest rates. There is an increased demand for US assets and hence an increased demand for dollars. Interest rates in the United States decrease, which tends to increase durable goods spending and stimulate the US economy. Against that, the higher value of the dollar leads to fewer exports from the United States and more imports into the United States, so US net exports will decrease.

Completely analogously, monetary policy in the United States influences interest rates in other countries. If the Fed undertakes an open market sale of US government debt, for example, interest rates will increase in other countries as well as in the United States.

The US Federal Reserve and the ECB are big players in world financial markets. Their actions move world interest rates and world currency markets. There are other countries that are relatively small in the world economy. For example, suppose the Central Bank of Iceland increases interest rates in that country. The mechanisms that we have explained still apply: investors will find Icelandic assets more attractive, and there will be an increased demand for the Icelandic krona. However, the flows of capital into Iceland will be negligible in terms of the world economy. They will not have any noticeable effect on interest rates in Europe or the United States.

KEY TAKEAWAYS

In an open economy, interest rate changes induced by monetary policy influence exchange rates and thus net exports.

Actions by monetary authorities in other countries influence the net exports of the United States through exchange rate changes and through the level of aggregate spending on the United States by households in other countries.

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